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Microcontroller Infineon Software Architecture

User's Manual

GTM Driver

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Date	Version	Change Description
2017-02-09	4.0	Updated for the review comment REV 11633
2017-01-30	3.9	Updated the date format in the version history which was not aligned correctly based on JIRA 0000054991-85 Updated user manual based on review comment from REV_011542.Updated section 8.1
2016-08-10	3.8	Updated based on review comments RE_010594.
2016-08-04	3.7	Updated limitation section for Mcfg configuration.
2016-06-09	3.6	Updated based on review comments Rev_010071
2016-05-11	3.5	Updated based on JIRA 0000051261-231 - Updated File structure in section 3 - Updated I/O mode in section 5.2 - Updated the source for API's in section 5.2
2015-11-20	3.4	Modified section 4.2.10, 4.2.11, 5.2.54, 5.2.55, 5.2.66 for 0000051100-202. Modified the multiplicity for TC29x and added ModuleNumber input parameter for all FIFO APIs.
2015-10-08	3.3	Added recommended mode of interrupt to all sub peripheral interrupt mode. Modified IO mode for APIs for user-1 mode/ supervisory mode implementation. Modified Service ID and SAFE MCAL ERROR for Gtm_IsrTomModule, Gtm_IsrAtomModule and Gtm_IsrTimModule.
2015-09-07	3.2	Modified return type of Gtm_Init.
2015-05-28	3.1	Updated caveats for Gtm_FlushFifo, Gtm_InitDpllRam, Gtm_ClearMcsRam and Gtm_Init.
2015-05-18	3.0	Added limitation for GtmConfiguration container name.
2015-04-22	2.9	Supported features for Gtm_DeInit and McuDeInitApi.
2014-11-06	2.8	Inserted I/O Mode for all the API functions.
2014-08-26	2.7	Changes as per REV_006124.
2014-08-12	2.6	Added changes specific to AI00238938. (Separate configurability supported for ATOM ccu0 and ccu1)
2014-07-14	2.5	Removed reference for software safety information document.
2014-03-26	2.4	Modified reference for software safety information document.
2014-02-28	2.3	Added EP changes.
2014-02-07	2.2	Modified TIM/TOM/ATOM notifications (added 4 th param). UTP (AI00237990)
2013-11-14	2.1	Added modified SFR's to the configuration parameters.
2013-09-25	2.0	Modified ISR's, InitCheck and Changed assumptions and deviations sections. Changes for UTP [AI00231910]
2013-08-23	1.9	Added ISR's to the section 5.3
2013-08-22	1.8	Changed assumptions and deviations sections.
2013-08-20	1.7	changes as per AI00172352 Added safety changes.
2013-06-27	1.6	Updated for TC26x changes.
2013-04-02	1.5	Modified as per review [REV_004404]
2013-03-19	1.4	Added B-step changes

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Date	Version	Change Description
2012-04-07	1.3	Errata information updated
2012-12-06	1.2	Minor formal changes
2012-14-05	1.1	Updated for TC27x New API's included Few API's removed ISR concept modified. Channel Notification concept introduced.
2011-13-10	1.0	Document Refined
2011-03-22	0.1	Initial Version

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1 Introduction

This User Manual provides information regarding the functionality and API implementation for the GTM Driver. The GTM Driver forms the basis for ICU / PWM / GPT Drivers. The user should necessarily configure GTM peripheral first before configuring ICU / PWM / GPT Driver.

This User Manual is intended to help the user to get acquainted with the GTM Driver implementation for the AURIX Family Hardware Platform. It is the documentation that describes how to configure and use GTM Driver. ICU, PWM and GPT Drivers heavily depend on GTM Driver configuration with respect to resource and interrupts usage. WDG and ADC also depend on this driver for its functioning.

The User Manual covers the hardware features of the AURIX Family Platform used in the GTM Driver implementation, Driver configuration and the software API implementation.

1.1 Scope

This document addresses the following features of the GTM Driver implementation,

- AURIX HW peripheral used to realize the Driver
- File structure of GTM Driver
- Configuration parameters of the Driver
- API specification
- Code snippets illustrating the common use cases.
- Limitations and assumptions
- This documentation applies to the GTM Driver Variant: Variant PB.

1.2 Abbreviations

Table 1 Abbreviations used in this document

Abbreviation	Explanation
AEI	AE- Interface. GTM Generic Bus Interface.
API	Application Programming Interface
ATOM	Aru-Connected Timer Output Module
AUTOSAR	AUTomotive Open System Architecture
DTM	Dead Time Module
EP	Efficiency Platform
GTM	Generic Timer Module
HE	High End
ISR	Interrupt Service Routine
IFX	Infineon Technologies
MC-ISAR	Microcontroller Infineon Software Architecture
SW	Software
TGC	TOM Global Channel Control
TIM	Timer Input Module
TOM	Timer Output Module
VariantPB	This variant allows a mix of pre-compile time-, post build-time configuration parameters. The intention of this variant is to optimize the parameters configuration for a re-loadable binary.
VariantPC	This variant is limited to pre-compile-configuration parameters only. The intention of this variant is to optimize the parameters configuration for a source code delivery.

1.3 References

This section lists down all relevant documents for this User Manual.

Some of the artifacts listed below will not contain the version number. For those artifacts, kindly refer the latest version.

- [1] TC29x User Manual
tc29xB_um_v1.3.pdf
- [2] TC27x User Manual
tc27xC_um_v2.2.pdf
tc27xD_um_v2.2.pdf
- [3] TC26x User Manual
tc27xC_um_v2.2.pdf
- [4] TC23x User Manual
tc23x_tc22x_um_v1.1.pdf
- [5] HW Errata Analysis
GTM_IP-ES_IFX_20130123.pdf
- [6] MCU User Manual
MC-ISAR_AURIX_UM_McuDriver.doc
- [7] Port User Manual
MC-ISAR_AURIX_UM_PortDriver.doc
- [8] Irq User Manual
MC-ISAR_AURIX_UM_IrqDriver.doc

2 GTM Driver Overview

GTM driver is responsible for initializing the clock, the Input/output multiplexer registers and providing GTM interface to ICU, PWM and GPT modules. The initialization of GTM driver must be called before executing ICU / PWM / GPT functionality or any other module which is dependent on GTM. The GTM driver is also responsible for calling the respective interrupt service routines as per the configuration done by the user. Besides, GTM driver provides basic interfaces for all the sub-modules that are available in GTM peripheral.

2.1 Hardware Software Mapping

This section introduces the user to the HW features used for implementation. The [Figure 1](#) illustrates the Hardware peripherals and their interaction with the Software drivers.

From MCAL layer, Mcu Driver calls Gtm Initialization function (Gtm_Init through Mcu_Init) and Gtm Driver initializes the GTM clock and Input Output multiplexer. GTM Driver also initializes the resource registers that are used specifically by GTM Driver. ICU, PWM and GPT Driver also use the GTM peripheral (not shown in figure).

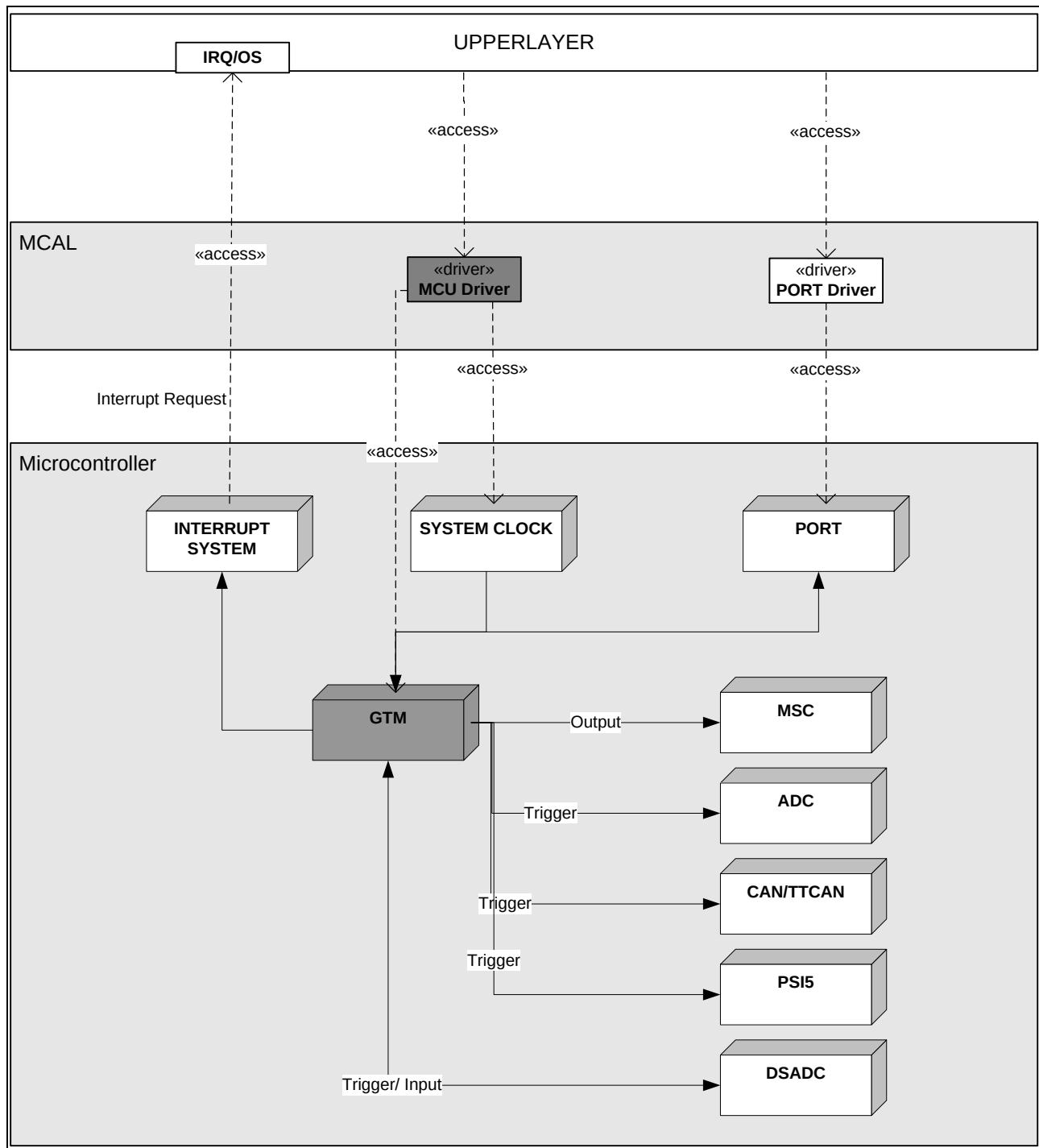


Figure 1 Software Hardware mapping of GTM

2.1.1 GTM

The GTM Driver is used in order to realize the GTM hardware functionality. GTM peripheral can be used independently; however the Driver support is of very basic in nature hence the user needs to be well versed with the features in order to avail the complete functionality.

The units of GTM that are used for implementation of GTM Driver are as given in [Table 2](#). (The module availability in EP and HE are mentioned.)

Table 2 Resource Usage

GTM Sub-Unit	Purpose	HE Applicability	EP Applicability
TOM	3 TOM Modules. Each with 16 Channels. Used for GPT, PWM and Complex usage	YES	YES
ATOM	5 ATOM Modules. Each with 8 Channels. Used for GPT, PWM and Complex usage	YES	NO
TIM	4 TIM Modules. Each with 8 Channels. Used for ICU and Complex usage	YES	YES
ARU	Advance Routing Unit. Used in Complex Driver.	YES	NO
BRC	Broadcast Module. Used in Complex Driver.	YES	NO
FIFO	First In First Out. Used in Complex Driver.	YES	NO
AFD	AEI to FIFO Data Interface. Used in Complex Driver.	YES	NO
F2A	FIFO to ARU Unit. Used in Complex Driver.	YES	NO
CMU	Clock Management Unit. Used in Complex Driver.	YES	YES
TBU	Timebase Unit. Used in Complex Driver. Also used in ICU for Timebase capture.	YES	YES
MCS	Multi Channel Sequencer. Used in Complex Driver.	YES	NO
MCFG	Memory Configuration. Used in Complex Driver.	YES	NO
MAP	TIM0 Input Mapping Module. Used in Complex Driver.	YES	NO
DPLL	Digital phase Locked Loop. Used in Complex Driver.	YES	NO
SPE	Sensor Pattern Evaluation. Used in Complex Driver.	YES	NO
ICM	Interrupt Concentrator Module. Used in Complex Driver.	YES	YES
CMP	Output Compare Unit. Used in Complex Driver.	YES	NO
MON	Monitor Unit. Used in Complex Driver.	YES	NO

For detailed study on GTM hardware configuration refer the user manuals of respective controller.

The Architecture figure in [Figure 2](#) gives an overview of GTM hardware features.

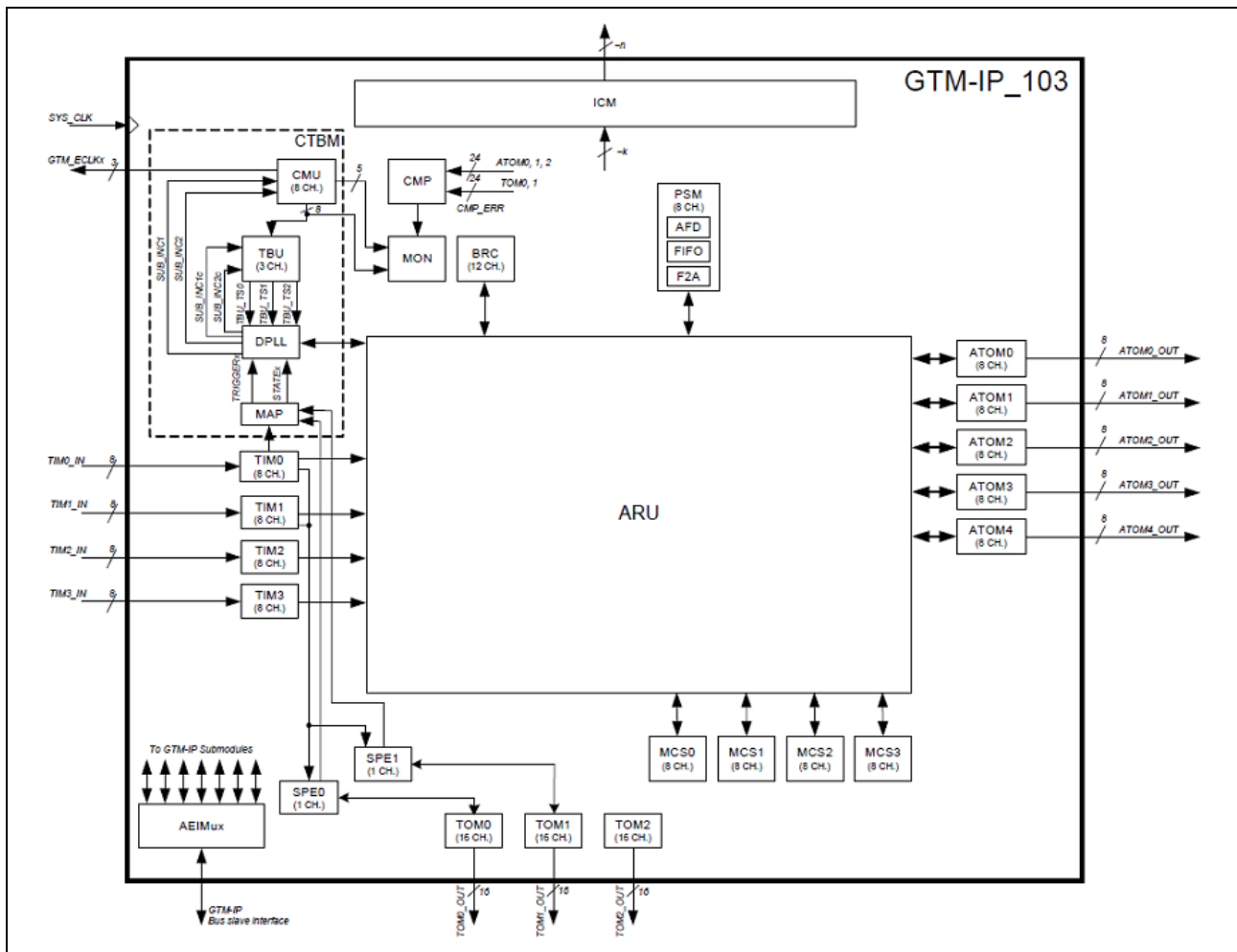


Figure 2 GTM Hardware Block Diagram

2.1.2 Multiplexer Programming

Port pins are interfaced to GTM through TIM, TOM and ATOM interface. The input ports are connected to TIM channels and Output ports are connected to TOM and ATOM channels. A multiplexer is present to choose the port pins that will be connected to these Sub-Modules. The Input Multiplexer can be configured using TIMINSEL register where 4 bits are allotted per TIM channel to choose the corresponding Port Pin. The Output Multiplexer can be configured using TOUTSEL register where 2 bits are allotted for each port pin to choose which TOM/ATOM channel need to be interfaced to it.

Please refer [1] for information on Port Pin interfaces to GTM.

2.1.2.1 Multiplexing for On-Chip Trigger and Inter Module connections

The GTM resources can be used as a trigger resource and also for supplying Input Signal or analyzing Outputs Signals. These resources are used by the other peripheral devices like MSC, DSADC, ADC, SENT, CAN, TTCAN, QSPI, CCU6x, PSI5 and PSI5S.

2.1.3 System Clock

The system clock is set up via the MCU Driver. This is mandatory for the user to set up an appropriate system clock. Note that the GTM peripheral clock is basically derived from the system clock configured. MCU related information is available in the MCU user manual. Refer [\[6\]](#)

2.1.4 Port

The PORT Driver controls all access to the pins required by GTM for input and output configuration.

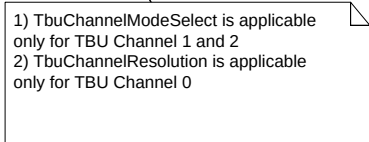
The required configuration for each pin in order to be used by GTM is explained in the PORT user manual. Refer [\[7\]](#).

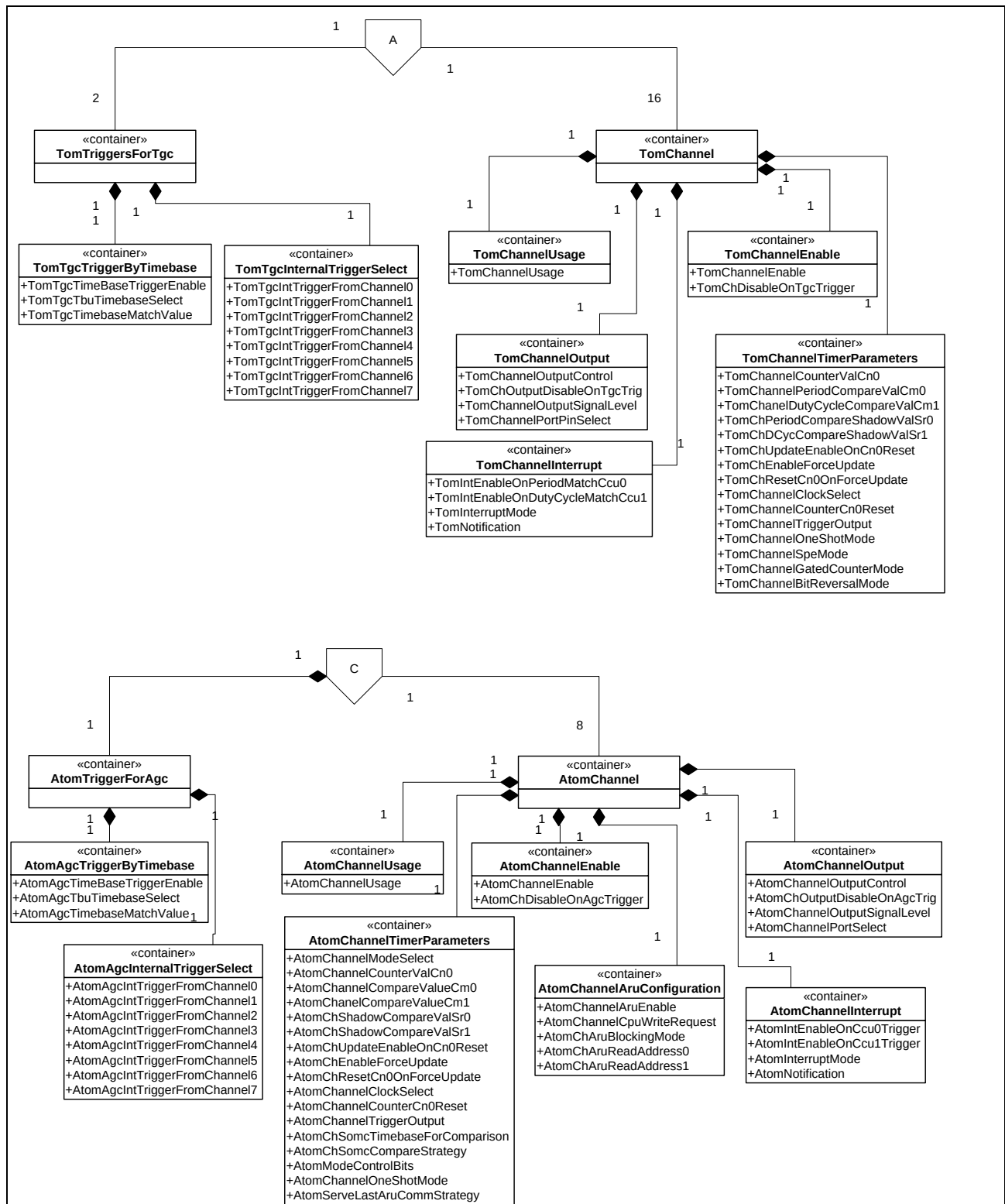
2.1.5 Irq

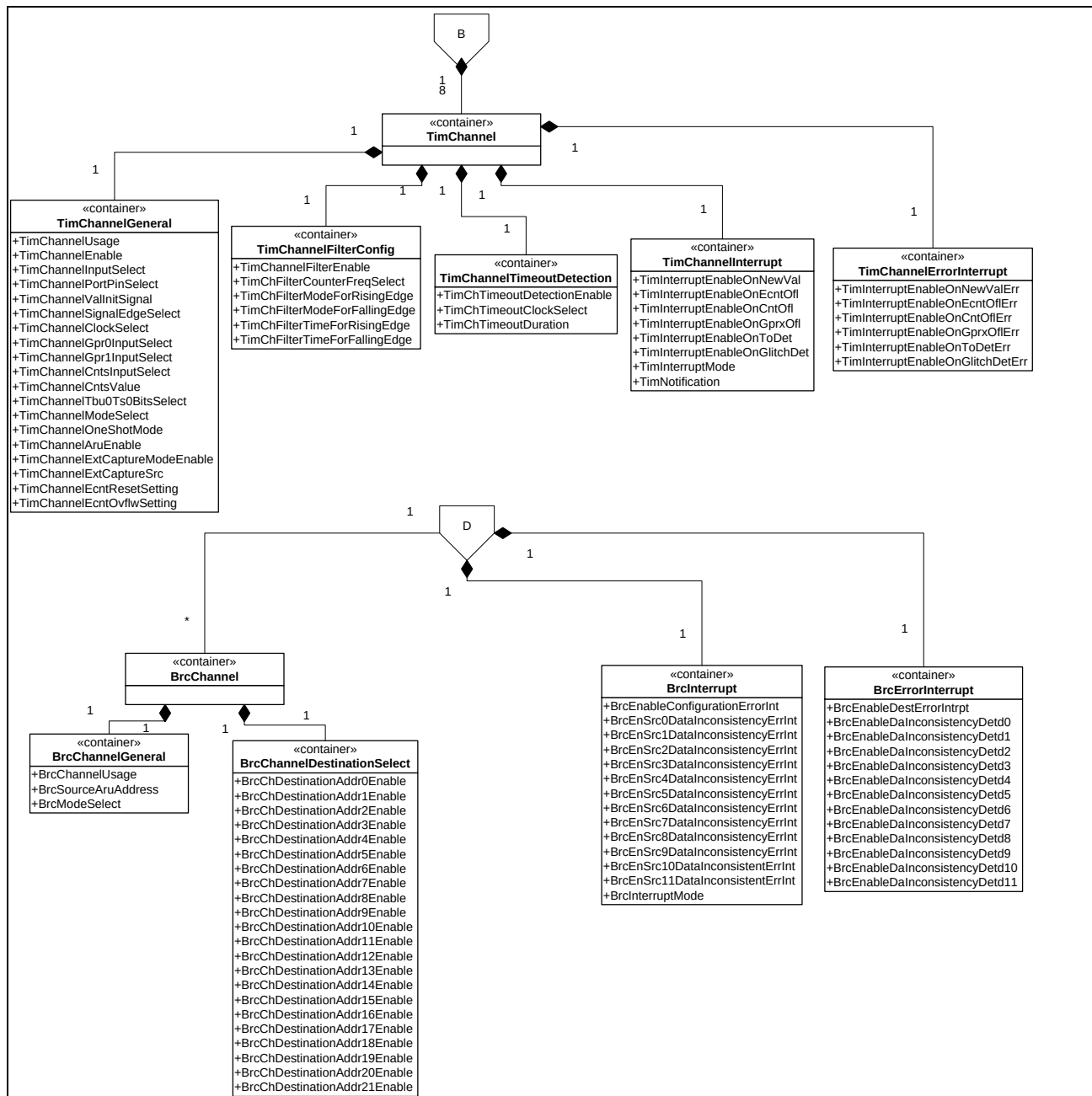
The GTM Peripheral can trigger interrupts upon multitudes of events, varying for each sub-module. These interrupts are configured and handled using the IRQ Driver. Refer [\[8\]](#) Irq User Manual for details.

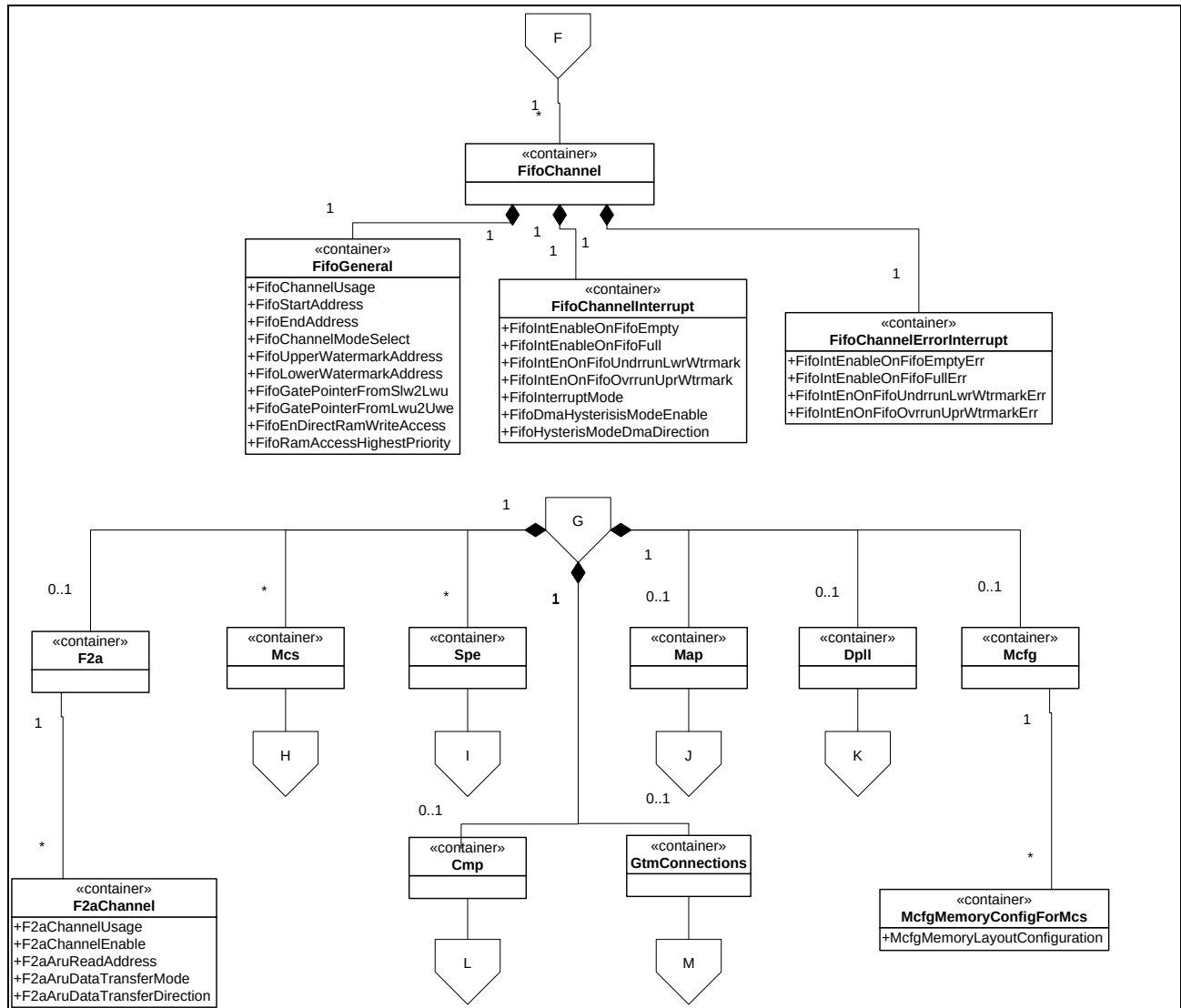
2.2 Software Driver Description

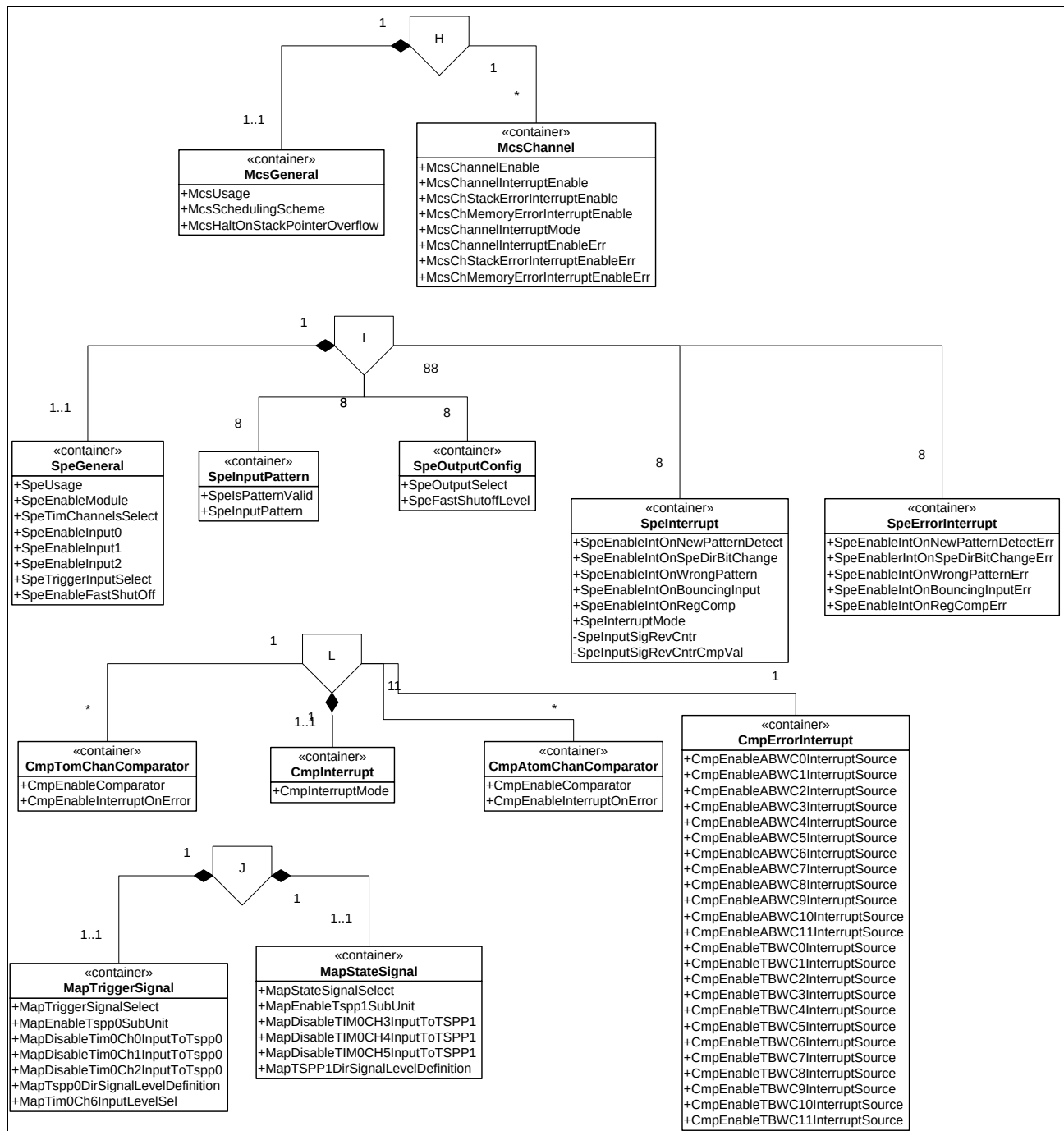
The containers and interfaces of GTM Driver and API specification are as given in Figure 3. The interfaces for complex functionality are very specific to a resource in order to keep it highly optimized.

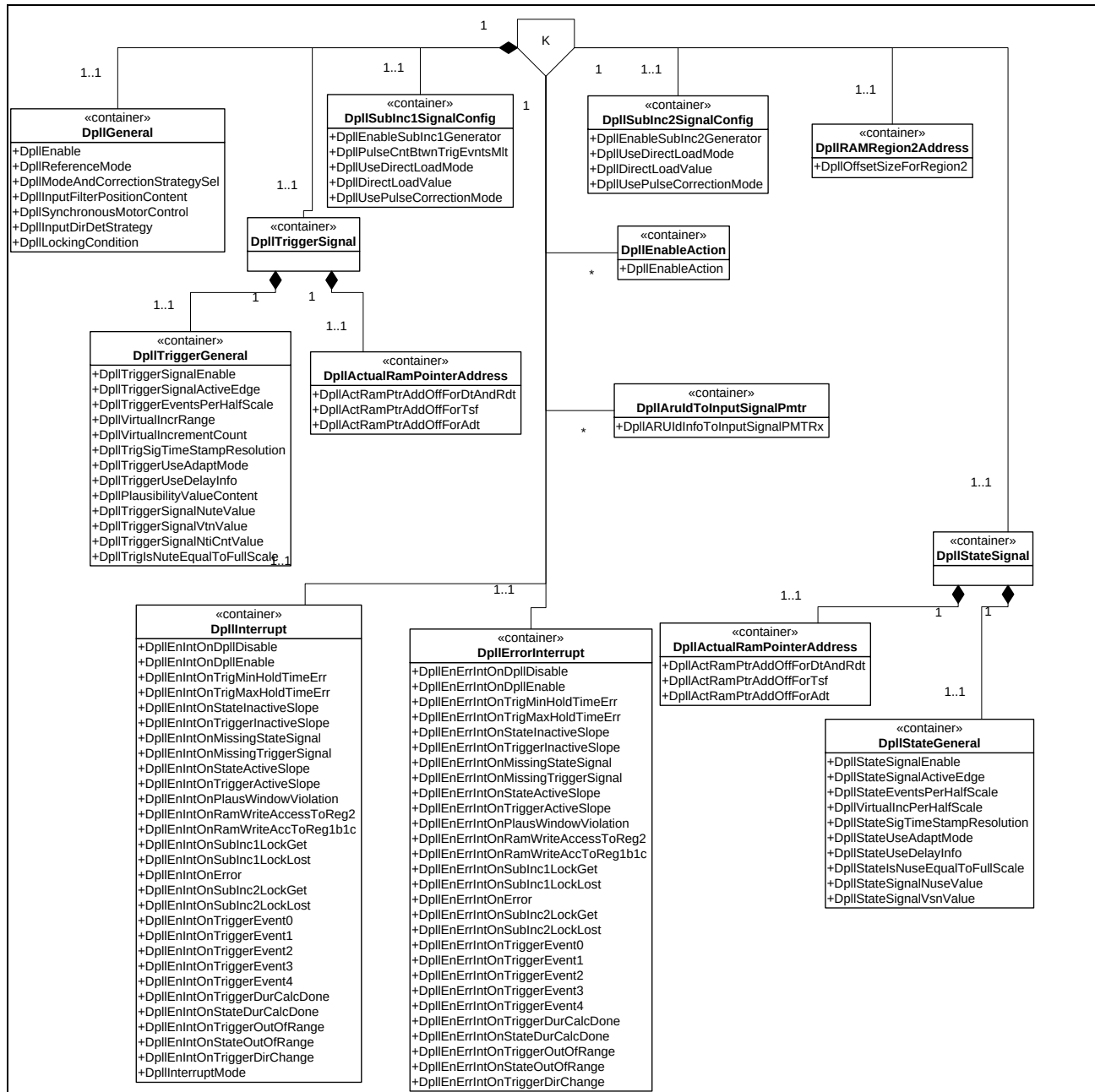


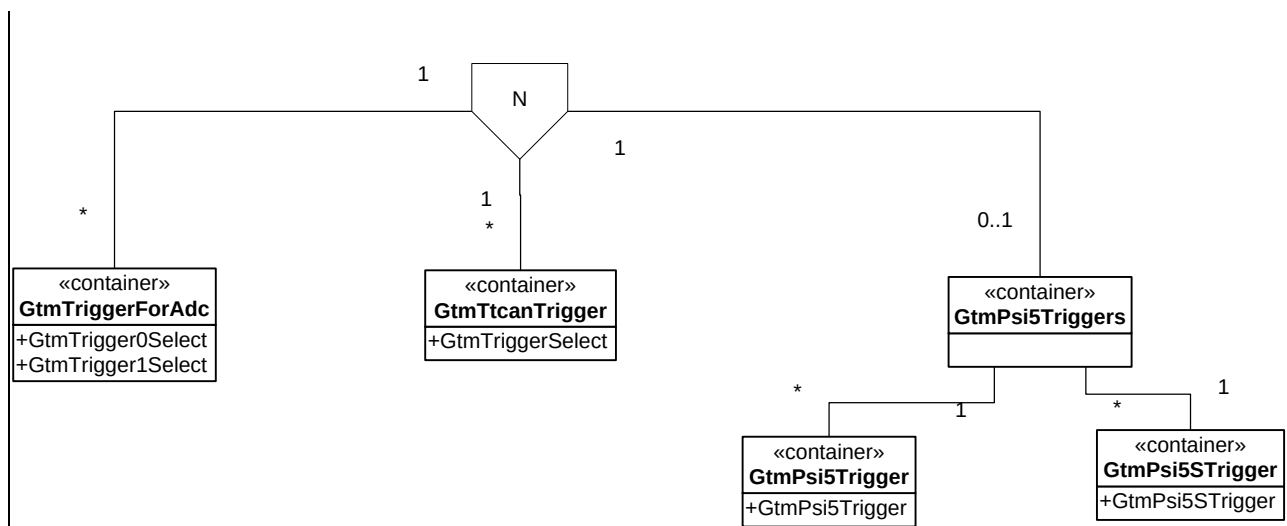
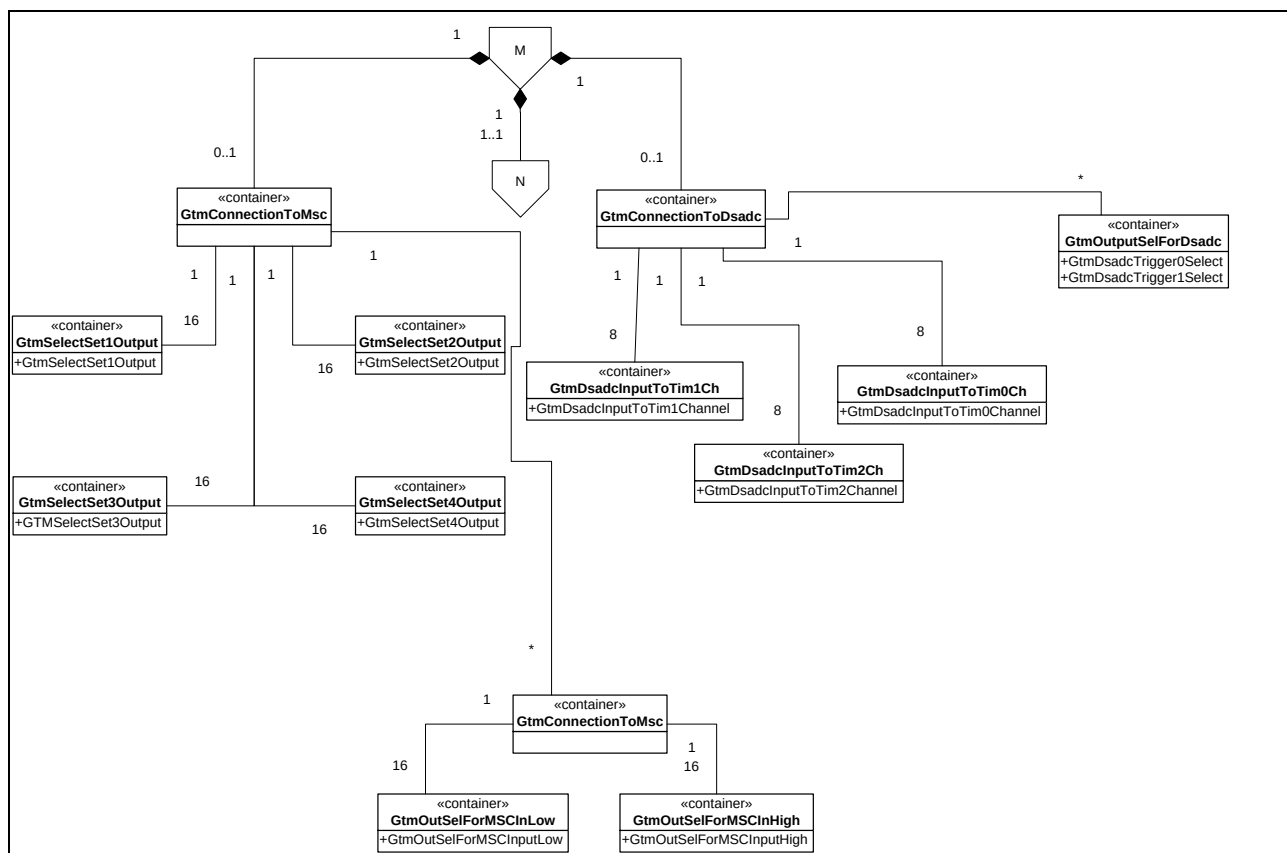












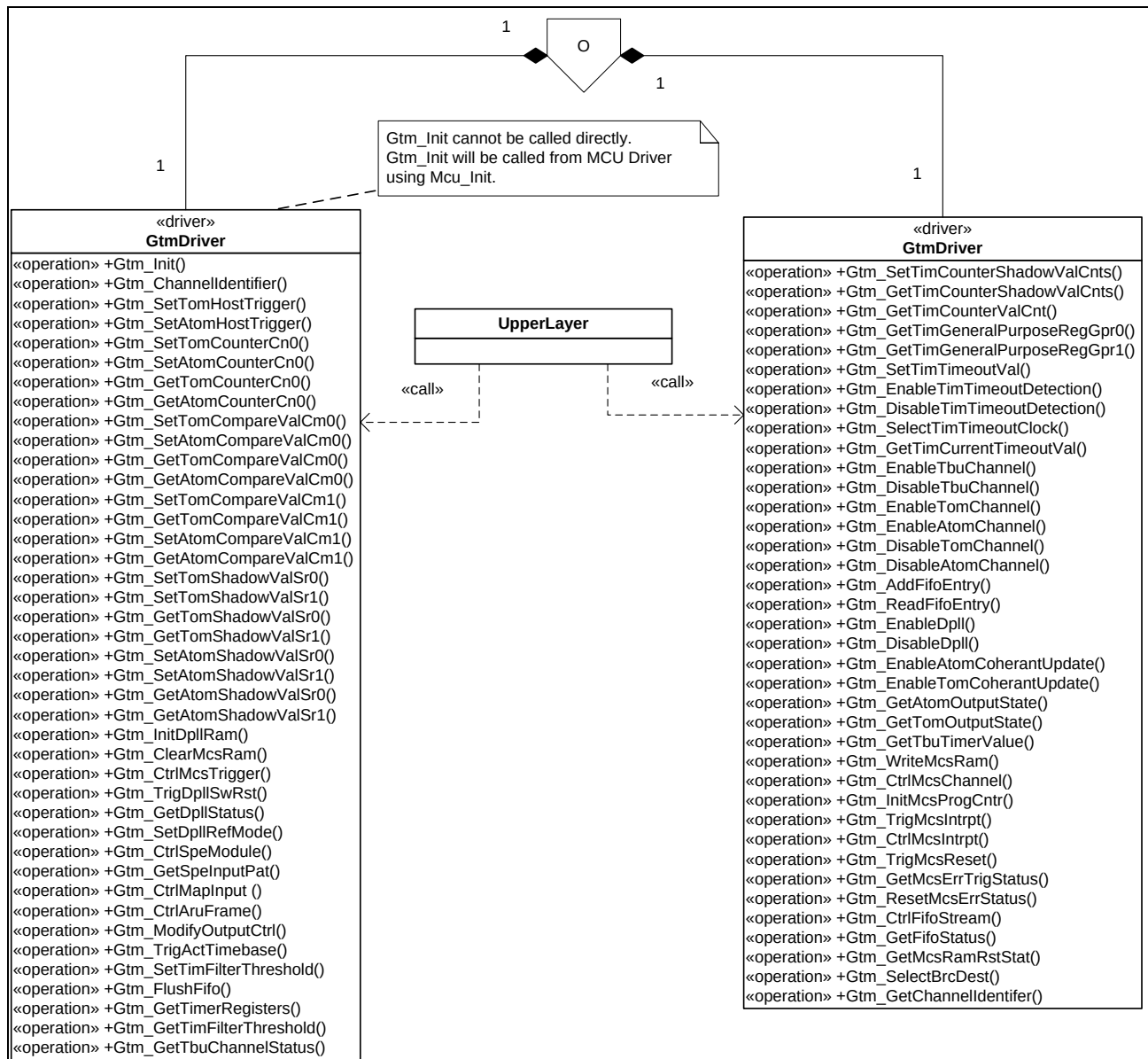


Figure 3 Interfaces of GTM Driver

3 File Structure

This section provides details about the GTM Driver files and other related files.

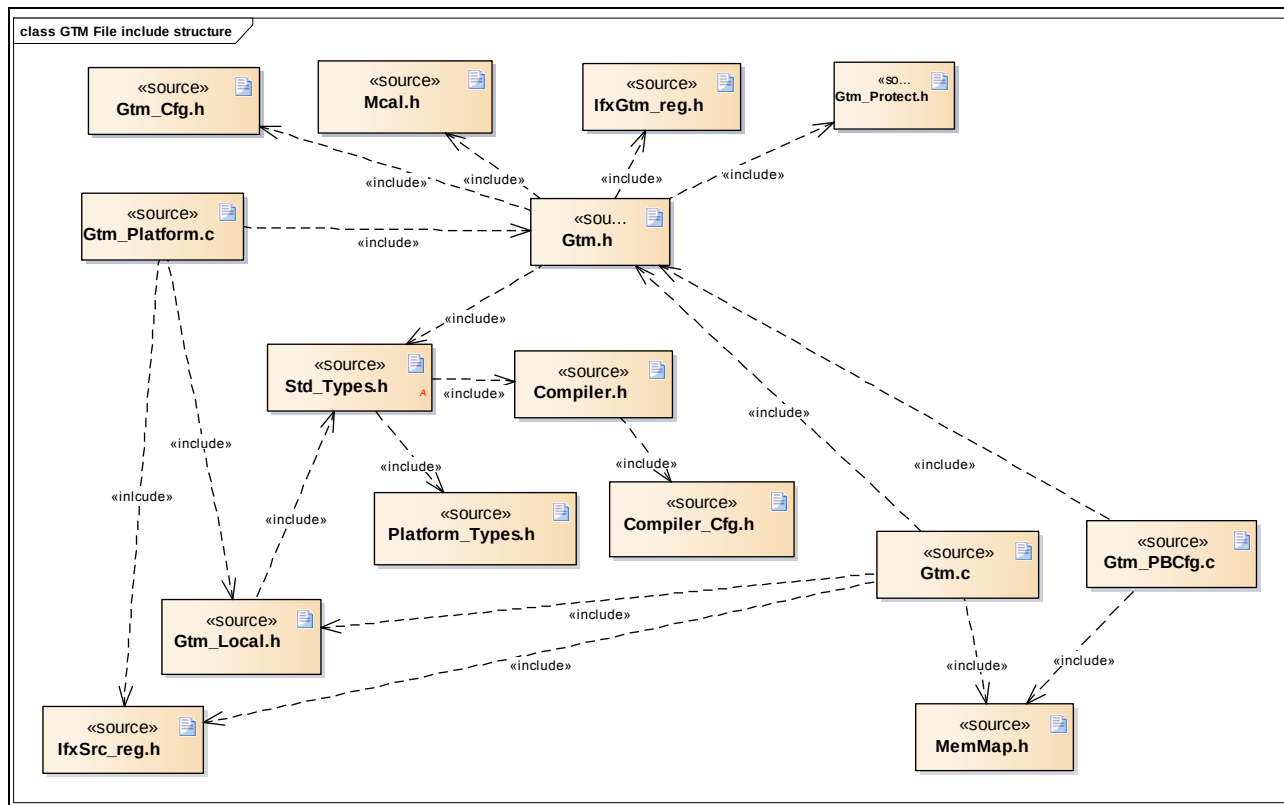


Figure 4 File structure of GTM Driver

Table 3 Code Structure

Filename	File Contents
Platform_Types.h	Platform dependencies (Aurix) are declared here.
Std_Types.h	Standard data types to be used are declared here.
Compiler.h	Compiler dependencies are abstracted here.
Compiler_Cfg.h	Contains abstraction of compiler specific keywords used for addressing data and code within declaration and definition.
MemMap.h	Mapping of code and data (variables, constant variables) to specific memory sections
IfxGtm_reg.h	Includes the register definition file for GTM.
IfxSrc_reg.h	Includes the register definition file for GTM.
Mcal.h	Provides library functions for MCAL layer.
Gtm.h	Driver configuration structure and API functions are exported to the user.
Gtm_Local.c	This file contains include functionality for platform specific file.
Gtm.c	Implementation of GTM Driver functionality.
Gtm_Platform.c	This file contains derivate specific code.
Gtm_Cfg.h	The pre-compile configuration macros required for GTM driver implementation is present in this file.
Gtm_PBCfg.c	Contains Driver post build configuration parameters.
Gtm_Protect.h	Included to define all the first level indirection macros for Protected mode support for supervisory access via OS calls

4 Configuration Documentation

This chapter summarizes

- Configuration concept used for structuring the Driver
- Configuration parameter variance w.r.t supported HW
- All configuration parameters and their description

4.1 Configuration Concept

4.1.1 Configuration Class

The development of Basic Software Drivers involves the following development cycles:

- Compiling,
- Linking and
- Downloading of the executable to ECU memory.

Configuration of parameters can be done in any of these process-steps: pre-compile time, link time or even post-build time. According to the process-step that does the configuration of parameters, the configuration classes are categorized as below

Pre-compile time: The pre-compile time configuration is implemented by means of compiler switches or variables affecting the total build process. Pre-compiler configuration requires the Driver to be compiled, linked and located and hence the values of pre-compile parameters must be frozen before compilation and therefore can be used in source code distributions of the Driver only.

Link time: The Link time configuration doesn't affect compilation stage but impacts the linkage and locating stages. The Link time parameters can be used in binary (library) distributions of the Driver and the values of the Link time parameters are specified by means of external configuration source code that must be linked to the Driver during the build process.

Post-build time: The Post-build time configuration will not affect the build process and hence Post-build time configuration is applied to Driver at runtime. Since configuration location is statically independent of the Driver functionality itself but linked dynamically (e.g. at Driver initialization), post-build time configuration is used for re-programmable configuration data.

4.1.2 Configuration Variant

Based on the most suitable configuration class that may be needed for the configuration parameters of the Driver, a Driver can be delivered as the following variants

Variant PC: This variant is limited to pre-compile configuration parameters only. The intention of this variant is to optimize the parameters configuration for a source code delivery.

Variant LT: This variant allows a mix of pre-compile time, link time configuration parameters. The intention of this variant is to optimize the parameters configuration for an object code delivery.

Variant PB: This variant allows a mix of pre-compile time, post-build time configuration parameters. The intention of this variant is to optimize the parameters configuration for a re-loadable binary.

Note: The GTM Driver is delivered as a Variant PB.

4.1.3 How to read the Configuration Class field

The examples cited below help the user to identify the configuration class of the parameter for a given delivery type.

Note: Topics [4.1.1](#), [4.1.2](#) should be read for understanding the configuration class field

Table 4 Configuration Class Example 1

Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--

'x' indicates that the parameter is implemented as a pre-compile parameter in all types of deliveries i.e. Variant PB, Variant LT, and Variant PC.

Table 5 Configuration Class Example 2

Configuration Class:	Pre-compile	x	Variant PC
	Link Time	--	--
	Post build	x	Variant PB

'x' indicates that the parameter is implemented as a

1. pre-compile parameter if the delivery type is Variant PC delivery
2. post-build parameter if the delivery type is a Variant PB delivery

4.2 Container:GtmConfiguration

[Note: In configuration parameters following conventions are used.

i – Module number

j – Channel number

k – Group number

n – Module number for port and other output connections]

Note: The Container names within GtmConfiguration container cannot be changed by the user. This limitation is imposed because of the dependencies of other modules with GTM.

4.2.1 Container: GtmGeneral

This container configures the clock initialization and the required divider value. The GTM clock depends on the MCU system clock.

The GTM Peripheral clock frequency and Sleep control (Needed for Power down Feature from the MCU) can be set up here. The GTM peripheral clock is derived from the system clock after 2 levels of prescaling. First level of Prescaling happens in Mcu configuration. The Second level of Prescaling is done here.

4.2.1.1 GtmGeneral

Table 6 GTM Sleep mode

Name:	GtmSleepMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_SLEEP_ENABLE – GTM Module enter sleep mode upon request. GTM_SLEEP_DISABLE – GTM module does not enter sleep mode upon request. Default: GTM_SLEEP_ENABLE		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	Parameter to set the module sleep mode. If the ICU / GPT Driver consist of any wakeup capable channels then it is mandatory that the GTM clock should be configured with GTM_ SLEEP_DISABLE.		
SFR Modified	CLC		
Scope:	Driver		
Dependency:	None		

4.2.1.2 GtmGlobalClkDivNumerator

Table 7 GtmGlobalClkDivNumerator

Name:	GtmGlobalClkDivNumerator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 – 0xFFFFFFFF Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Numerator value for the Global Clock Divider. This modifies GTM_CMU_GCLK_NUM [GCLK_NUM]. Global Clock = System Clock * (GtmGlobalClkDivNumerator/ GtmGlobalClkDivDenominator).		
SFR Modified	GTM_CMU_GCLK_NUM		
Scope:	Driver		
Dependency:	None		

4.2.1.3 GtmGlobalClkDivDenominator

Table 8 GtmGlobalClkDivDenominator

Name:	GtmGlobalClkDivDenominator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 – 0xFFFFFFFF Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This parameter specifies the Denominator value for the Global Clock Divider. This modifies GTM_CMU_GCLK_NUM [GCLK_DEN]. Global Clock = System Clock * (GtmGlobalClkDivNumerator/ GtmGlobalClkDivDenominator). This value should be configured lesser than Numerator value.
SFR Modified	GTM_CMU_GCLK_NUM
Scope:	Driver
Dependency:	None

4.2.1.4 GtmEnRstAndForceIntFunctionality

Table 9 GtmEnRstAndForceIntFunctionality

Name:	GtmEnRstAndForceIntFunctionality		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables RST and Force Interrupt Functionality of all modules of GTM FALSE – Disables RST and Force Interrupt Functionality of all modules of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the RST and Force Interrupt Functionality of all modules of GTM. This Clears/ Sets GTM_CTRL[RF_PROT]		
SFR Modified	GTM_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.2 Container: GtmAeiInterface

This container contains the Configuration for GTM AEI Interface

4.2.2.1 GtmAeiTimeOutMode

Table 10 GtmAeiTimeOutMode

Name:	GtmAeiTimeOutMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: OBSERVE: Observe mode is chosen ABORT: Abort mode is chosen Default: OBSERVE		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	Specifies the AEI Timeout mode for GTM. OBSERVE: If timeout_counter=0 the address and rw signal in addition with timeout flag will be stored to the GTM_AEI_ADDR_XPT register. Following timeout_counter=0 accesses will not overwrite the first entry in the aei_addr_timeout register. Clearing the timeout flag/aei_status error_code will reenables the storing of a next faulty access. ABORT: In addition to observe mode the pending access will be aborted by signaling an illegal module access on aei_status and sending ready. In case of a read deliver as data 0 by serving of next AEI accesses.		
SFR Modified	GTM_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.2.2 GtmAeiTimeOutValue

Table 11 GtmAeiTimeOutValue

Name:	GtmAeiTimeOutValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 – 31 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the AEI Timeout Value for GTM. These bits define the number of cycles after which a timeout event occurs. When Timeout value equals zero (0) the AEI timeout functionality is disabled. This changes the contents of GTM_CTRL[TO_VAL]		
SFR Modified	GTM_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.2.3 GtmAeiTimeoutExceptionIntEn

Table 12 GtmAeiTimeoutExceptionIntEn

Name:	GtmAeiTimeoutExceptionIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on AEI Timeout Exception FALSE – Disables interrupt on AEI Timeout Exception Default: FALSE		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on AEI Timeout Exception. This enables GTM_IRQ_EN[AEI_TO_XPT_IRQ_EN]		
SFR Modified	GTM_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.2.4 GtmAeiUnsupportedAddressIntEn

Table 13 GtmAeiUnsupportedAddressIntEn

Name:	GtmAeiUnsupportedAddressIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on AEI Unsupported Address FALSE – Disables interrupt on AEI Unsupported Address Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on AEI Unsupported Address ie. AEI_USP_ADDR Interrupt was raised by the AEI interface. This enables GTM_IRQ_EN[AEI_USP_ADDR_IRQ_EN]		
SFR Modified	GTM_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.2.5 GtmAeiIllegalModuleAddIntEn

Table 14 GtmAeiIllegalModuleAddIntEn

Name:	GtmAeiIllegalModuleAddIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on AEI Illegal Module Address FALSE – Disables interrupt on AEI Illegal Module Address Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on AEI Illegal Module Address i.e. AEI_IM_ADDR interrupt was raised by the AEI interface. This enables GTM_IRQ_EN[AEI_IM_ADDR_IRQ_EN]		

SFR Modified	GTM_IRQ_EN
Scope:	Driver
Dependency:	None

4.2.2.6 GtmAeiUnsupportedByteEnIntEn

Table 15 GtmAeiUnsupportedByteEnIntEn

Name:	GtmAeiUnsupportedByteEnIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on AEI Unsupported byte enable FALSE – Disables interrupt on AEI Unsupported byte enable Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on AEI Unsupported byte enable i.e. AEI_USP_BE interrupt was raised by the AEI interface. This enables GTM_IRQ_EN[AEI_USP_BE_IRQ_EN]		
SFR Modified	GTM_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.2.6.1 GtmAeiUnsupportedByteEnIntEn

Table 16 GtmAeiUnsupportedByteEnIntEn

Name:	GtmAeiUnsupportedByteEnIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on AEI Unsupported byte enable FALSE – Disables interrupt on AEI Unsupported byte enable Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on AEI Unsupported byte enable i.e. AEI_USP_BE interrupt was raised by the AEI interface. This enables GTM_IRQ_EN[AEI_USP_BE_IRQ_EN]		
SFR Modified	GTM_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.2.6.2 GtmAeiTimeoutExceptionErrIntEn

Table 17 GtmAeiTimeoutExceptionErrIntEn

Name:	GtmAeiTimeoutExceptionErrIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables error interrupt on AEI Timeout Exception FALSE – Disables interrupt on AEI Timeout Exception Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
SFR Modified	GTM_EIRQ_EN		
Description:	Enables/Disables error interrupt on AEI Timeout Exception. This enables GTM_EIRQ_EN[AEI_TO_XPT_EIRQ_EN]		
Scope:	Driver		
Dependency:	None		

4.2.2.6.3 GtmAeiUnsupportedAddressErrIntEn

Table 18 GtmAeiUnsupportedAddressErrIntEn

Name:	GtmAeiUnsupportedAddressErrIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables error interrupt on AEI Unsupported Address FALSE – Disables error interrupt on AEI Unsupported Address Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables error interrupt on AEI Unsupported Address i.e. AEI_USP_ADDR Interrupt was raised by the AEI interface. This enables GTM_EIRQ_EN[AEI_USP_ADDR_EIRQ_EN]		
SFR Modified			
Scope:	Driver		
Dependency:	None		

4.2.2.6.4 GtmAeiIllegalModuleAddErrIntEn

Table 19 GtmAeiIllegalModuleAddErrIntEn

Name:	GtmAeiIllegalModuleAddErrIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables error interrupt on AEI Illegal Module Address FALSE – Disables error interrupt on AEI Illegal Module Address Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on AEI Illegal Module Address i.e. AEI_IM_ADDR interrupt was raised by the AEI interface. This enables GTM_EIRQ_EN[AEI_IM_ADDR_EIRQ_EN]		
SFR Modified	GTM_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.2.6.5 GtmAeiUnsupportedByteEnErrIntEn

Table 20 GtmAeiUnsupportedByteEnErrIntEn

Name:	GtmAeiUnsupportedByteEnErrIntEn		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables error interrupt on AEI Unsupported byte enable FALSE – Disables error interrupt on AEI Unsupported byte enable Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on AEI Unsupported byte enable i.e. AEI_USP_BE interrupt was raised by the AEI interface. This enables GTM_EIRQ_EN[AEI_USP_BE_EIRQ_EN]		
SFR Modified	GTM_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.2.7 GtmAeiInterruptMode

Table 21 GtmAeiInterruptMode

Name:	GtmAeiInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: GTM_INTERRUPT_LEVEL_MODE – Level Mode chosen GTM_INTERRUPT_PULSE_MODE - Pulse Mode chosen GTM_INTERRUPT_PULSE_NOTIFY_MODE – Pulse-Notify Mode chosen GTM_INTERRUPT_SINGLE_PULSE_MODE – Single Pulse Mode chosen Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use.		
SFR Modified	GTM_IRQ_MODE		
Scope:	Driver		
Dependency:	None		

4.2.3 Container: Cmu

This container contains the Clock Configuration for GTM. The max multiplicity is 1. If this container is not included, then the registers values of CMU except for Global Clock divider registers are retained as is.

4.2.3.1 Container: CmuFixedClock

This container contains the Fixed Clock Configuration for GTM

4.2.3.1.1 CmuEnableAllFixedClocks

Table 22 CmuEnableAllFixedClocks

Name:	CmuEnableAllFixedClocks		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable all the fixed Clocks of GTM FALSE – Disable all the fixed Clocks of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables/Disables the Fixed Clocks of GTM. This modifies GTM_CMU_CLK_EN[EN_FXCLK]. CMU provides 5 Fixed Clocks with the following Prescale values respectively; 1, 16, 256, 4096, 65536. Hence the final clock frequency will be, Gtm frequency (after Global prescaler) /Fixed Clock Prescale value
SFR Modified	GTM_CMU_CLK_EN
Scope:	Driver
Dependency:	None

4.2.3.1.2 CmuFxdClkSourceSelect

Table 23 CmuFxdClkSourceSelect

Name:	CmuFxdClkSourceSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_GLOBAL_CLOCK: Use Global Clock as input for Fixed Clock. CMU_CLOCK_0_DIVIDER: Use Clock 0 as the clock source for Fixed Clock CMU_CLOCK_1_DIVIDER: Use Clock 1 as the clock source for Fixed Clock CMU_CLOCK_2_DIVIDER: Use Clock 2 as the clock source for Fixed Clock CMU_CLOCK_3_DIVIDER: Use Clock 3 as the clock source for Fixed Clock CMU_CLOCK_4_DIVIDER: Use Clock 4 as the clock source for Fixed Clock CMU_CLOCK_5_DIVIDER: Use Clock 5 as the clock source for Fixed Clock CMU_CLOCK_6_DIVIDER: Use Clock 6 as the clock source for Fixed Clock CMU_CLOCK_7_DIVIDER: Use Clock 7 as the clock source for Fixed Clock Default: GTM_GLOBAL_CLOCK		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB
Description:	Clock Source select for Fixed Clock. This modifies CMU_CLK_EN [FXCLK_SEL] GTM_GLOBAL_CLOCK: Use Global Clock as input for Fixed Clock. CMU_CLOCK_0_DIVIDER: Use Clock 0 as the clock source for Fixed Clock CMU_CLOCK_1_DIVIDER: Use Clock 1 as the clock source for Fixed Clock CMU_CLOCK_2_DIVIDER: Use Clock 2 as the clock source for Fixed Clock CMU_CLOCK_3_DIVIDER: Use Clock 3 as the clock source for Fixed Clock CMU_CLOCK_4_DIVIDER: Use Clock 4 as the clock source for Fixed Clock CMU_CLOCK_5_DIVIDER: Use Clock 5 as the clock source for Fixed Clock CMU_CLOCK_6_DIVIDER: Use Clock 6 as the clock source for Fixed Clock CMU_CLOCK_7_DIVIDER: Use Clock 7 as the clock source for Fixed Clock Default: GTM_GLOBAL_CLOCK		
SFR Modified	CMU_CLK_EN		
Scope:	Driver		
Dependency:	None		

4.2.3.2 Container: CmuConfigurableClock

This container contains the Configurable Clock Configuration for GTM

4.2.3.2.1 CmuEnableConfigurableClk0

Table 24 CmuEnableConfigurableClk0

Name:	CmuEnableConfigurableClk0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Configurable Clock 0 of GTM FALSE – Disable Configurable Clock 0 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Configurable Clock 0 of GTM. This modifies GTM_CMU_CLK_EN[EN_CLK0]		
SFR Modified	GTM_CMU_CLK_EN		
Scope:	Driver		
Dependency:	None		

4.2.3.2.2 CmuConfigurableClk0Div

Table 25 CmuConfigurableClk0Div

Name:	CmuConfigurableClk0Div		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Divider value for the configurable Clock 0. This modifies GTM_CMU_CLK_0_CTRL[CLK_CNT]. Hence the final clock frequency will be, Gtm frequency (after Global prescaler) / (Divider value configured here + 1)		
SFR Modified	GTM_CMU_CLK_0_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.3.2.3 CmuEnableConfigurableClk1

Table 26 CmuEnableConfigurableClk1

Name:	CmuEnableConfigurableClk1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Configurable Clock 1 of GTM FALSE – Disable Configurable Clock 1 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Configurable Clock 1 of GTM. This modifies GTM_CMU_CLK_EN[EN_CLK1].		
SFR Modified	GTM_CMU_CLK_EN[EN_CLK1].		
Scope:	Driver		
Dependency:	None		

4.2.3.2.4 CmuConfigurableClk1Div

Table 27 CmuConfigurableClk1Div

Name:	CmuConfigurableClk1Div		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Divider value for the configurable Clock 1. This modifies GTM_CMU_CLK_1_CTRL[CLK_CNT] Hence the final clock frequency will be, Gtm frequency (after Global prescaler) / (Divider value configured here + 1)		
SFR Modified	GTM_CMU_CLK_1_CTRL[CLK_CNT]		
Dependency:	None		
Scope:	Driver		

4.2.3.2.5 CmuEnableConfigurableClk2

Table 28 CmuEnableConfigurableClk2

Name:	CmuEnableConfigurableClk2		
Source:	This parameter is microcontroller specific.		

File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Configurable Clock 2 of GTM FALSE – Disable Configurable Clock 2 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Configurable Clock 2 of GTM. This modifies GTM_CMU_CLK_EN[EN_CLK2].		
SFR Modified	GTM_CMU_CLK_EN[EN_CLK2].		
Scope:	Driver		
Dependency:	None		

4.2.3.2.6 CmuConfigurableClk2Div

Table 29 CmuConfigurableClk2Div

Name:	CmuConfigurableClk2Div		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Divider value for the configurable Clock 2. This modifies GTM_CMU_CLK_2_CTRL[CLK_CNT] Hence the final clock frequency will be, Gtm frequency (after Global prescaler) /(Divider value configured here + 1)		
SFR Modified	GTM_CMU_CLK_2_CTRL[CLK_CNT]		
Scope:	Driver		
Dependency:	None		

4.2.3.2.7 CmuEnableConfigurableClk3

Table 30 CmuEnableConfigurableClk3

Name:	CmuEnableConfigurableClk3		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Configurable Clock 3 of GTM FALSE – Disable Configurable Clock 3 of GTM Default: FALSE		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Configurable Clock 3 of GTM. This modifies GTM_CMU_CLK_EN[EN_CLK3].		
SFR Modified	GTM_CMU_CLK_2_CTRL[CLK_CNT]		
Dependency:	None		

4.2.3.2.8 CmuConfigurableClk3Div

Table 31 CmuConfigurableClk3Div

Name:	CmuConfigurableClk3Div		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Divider value for the configurable Clock 3. This modifies GTM_CMU_CLK_3_CTRL[CLK_CNT]. Hence the final clock frequency will be, Gtm frequency (after Global prescaler) /(Divider value configured here + 1)		
Scope:	Driver		
Dependency:	None		

4.2.3.2.9 CmuEnableConfigurableClk4

Table 32 CmuEnableConfigurableClk4

Name:	CmuEnableConfigurableClk4		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Configurable Clock 4 of GTM FALSE – Disable Configurable Clock 4 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Configurable Clock 4 of GTM. This modifies GTM_CMU_CLK_EN[EN_CLK4].		
SFR Modified	GTM_CMU_CLK_2_CTRL[CLK_CNT]		
Dependency:	None		
Scope:	Driver		

4.2.3.2.10 CmuConfigurableClk4Div

Table 33 CmuConfigurableClk4Div

Name:	CmuConfigurableClk4Div		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Divider value for the configurable Clock 4. This modifies GTM_CMU_CLK_4_CTRL[CLK_CNT]. Hence the final clock frequency will be, $\text{Gtm frequency (after Global prescaler)} / (\text{Divider value configured here} + 1)$		
SFR Modified	GTM_CMU_CLK_4_CTRL[CLK_CNT]		
Scope:	Driver		
Dependency:	None		

4.2.3.2.11 CmuEnableConfigurableClk5

Table 34 CmuEnableConfigurableClk5

Name:	CmuEnableConfigurableClk5		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Configurable Clock 5 of GTM FALSE – Disable Configurable Clock 5 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Configurable Clock 5 of GTM. This modifies GTM_CMU_CLK_EN[EN_CLK5].		
SFR Modified	GTM_CMU_CLK_EN[EN_CLK5].		
Scope:	Driver		
Dependency:	None		

4.2.3.2.12 CmuConfigurableClk5Div

Table 35 CmuConfigurableClk5Div

Name:	CmuConfigurableClk5Div		
Source:	This parameter is microcontroller specific.		

File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: Z		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Divider value for the configurable Clock 5. This modifies GTM_CMU_CLK_5_CTRL[CLK_CNT]. Hence the final clock frequency will be, Gtm frequency (after Global prescaler) /(Divider value configured here + 1)		
SFR Modified	GTM_CMU_CLK_5_CTRL[CLK_CNT].		
Scope:	Driver		
Dependency:	None		

4.2.3.2.13 CmuEnableConfigurableClk6

Table 36 CmuEnableConfigurableClk6

Name:	CmuEnableConfigurableClk6		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Configurable Clock 6 of GTM FALSE – Disable Configurable Clock 6 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Configurable Clock 6 of GTM. This modifies GTM_CMU_CLK_EN[EN_CLK6].		
SFR Modified	GTM_CMU_CLK_EN[EN_CLK5].		
Scope:	Driver		
Dependency:	None		

4.2.3.2.14 CmuClk6SourceSelect

Table 37 CmuClk6SourceSelect

Name:	CmuClk6SourceSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: CMU_CLK_6_DIVIDER: Use Global Clock prescaled using Clock 6 Divider as CMU Clock 6. CMU_SUB_INC2_SIGNAL: Use Sub_Inc2 as the CMU Clock 6 Default: CMU_CLK_6_DIVIDER		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Clock Source select for CMU Clock6. This modifies GTM_CMU_CLK_6_CTRL[CLK6_SEL] CMU_CLK_6_DIVIDER: Use Global Clock prescaled using Clock 6 Divider as CMU Clock 6 CMU_SUB_INC2_SIGNAL: Use Sub_Inc2 as the CMU Clock 6		
SFR Modified	GTM_CMU_CLK_6_CTRL[CLK6_SEL]		
Scope:	Driver		
Dependency:	None		

4.2.3.2.15 CmuConfigurableClk6Div

Table 38 CmuConfigurableClk6Div

Name:	CmuConfigurableClk6Div		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Divider value for the configurable Clock 6. This modifies GTM_CMU_CLK_6_CTRL[CLK_CNT]. Hence the final clock frequency will be, Signal chosen by CmuClk6SourceSelect /(Divider value configured here + 1)		
SFR Modified	GTM_CMU_CLK_6_CTRL[CLK_CNT]		
Scope:	Driver		
Dependency:	None		

4.2.3.2.16 CmuEnableConfigurableClk7

Table 39 CmuEnableConfigurableClk7

Name:	CmuEnableConfigurableClk7		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enable Configurable Clock 7 of GTM FALSE – Disable Configurable Clock 7 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables the Configurable Clock 7 of GTM. This modifies GTM_CMU_CLK_EN[EN_CLK7].		
SFR Modified	GTM_CMU_CLK_EN[EN_CLK7].		
Scope:	Driver		
Dependency:	None		

4.2.3.2.17 CmuClk7SourceSelect

Table 40 CmuClk7SourceSelect

Name:	CmuClk7SourceSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: CMU_CLK_7_DIVIDER: Use Global Clock prescaled using Clock 7 Divider as CMU Clock 7 CMU_SUB_INC1_SIGNAL: Use Sub_Inc2 as the CMU Clock 7 Default: CMU_CLK_7_DIVIDER		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Clock Source select for CMU Clock7. This modifies GTM_CMU_CLK_7_CTRL[CLK7_SEL] CMU_CLK_7_DIVIDER: Use Global Clock prescaled using Clock 7 Divider as CMU Clock 7 CMU_SUB_INC2_SIGNAL: Use Sub_Inc2 as the CMU Clock 7.		
SFR Modified	GTM_CMU_CLK_7_CTRL[CLK7_SEL]		
Scope:	Driver		
Dependency:	None		

4.2.3.2.18 CmuConfigurableClk7Div

Table 41 CmuConfigurableClk7Div

Name:	CmuConfigurableClk7Div		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Divider value for the configurable Clock 7. This modifies GTM_CMU_CLK_7_CTRL[CLK_CNT]. Hence the final clock frequency will be, Signal chosen by CmuClk6SourceSelect /(Divider value configured here + 1)		
SFR Modified	GTM_CMU_CLK_7_CTRL[CLK_CNT].		
Scope:	Driver		
Dependency:	None		

4.2.3.3 Container: CmuExternalClock

This container contains the External Clock Configuration for GTM

4.2.3.3.1 CmuEnableExternalClk0

Table 42 CmuEnableExternalClk0

Name:	CmuEnableExternalClk0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable External Clock 0 of GTM FALSE – Disable External Clock 0 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the External Clock 0 of GTM. This sets GTM_CMU_CLK_EN[EN_ECLK0].		
SFR Modified	GTM_CMU_CLK_EN		
Scope:	Driver		
Dependency:	None		

4.2.3.3.2 CmuExternalClk0Numerator

Table 43 CmuExternalClk0Numerator

Name:	CmuExternalClk0Numerator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	X	Variant PB
Description:	This parameter specifies the Numerator value for the External Clock Divider0. This modifies GTM_CMU_ECLK_0_NUM[ECLK_NUM]. Final Clock value = System clock Frequency/(2 *(CmuExternalClk0Numerator/ CmuExternalClk0Denominator))		
SFR Modified	GTM_CMU_ECLK_0_NUM[ECLK_NUM].		
Scope:	Driver		
Dependency:	None		

4.2.3.3.3 CmuExternalClk0Denominator

Table 44 CmuExternalClk0Denominator

Name:	CmuExternalClk0Denominator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Denominator value for the External Clock Divider0. This modifies GTM_CMU_ECLK_0_DEN[ECLK_DEN]. Final Clock value = System clock Frequency/(2 *(CmuExternalClk0Numerator/ CmuExternalClk0Denominator))		
SFR Modified	GTM_CMU_ECLK_0_DEN		
Scope:	Driver		
Dependency:	CmuExternalClk0Numerator		

4.2.3.3.4 CmuEnableExternalClk1

Table 45 CmuEnableExternalClk1

Name:	CmuEnableExternalClk1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable External Clock 1 of GTM FALSE – Disable External Clock 1 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the External Clock 1 of GTM. This sets GTM_CMU_CLK_EN[EN_ECLK1].		

SFR Modified	GTM_CMU_CLK_EN.
Scope:	Driver
Dependency:	None

4.2.3.3.5 CmuExternalClk1Numerator

Table 46 CmuExternalClk1Numerator

Name:	CmuExternalClk1Numerator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Numerator value for the External Clock Divider1. This modifies GTM_CMU_ECLK_1_NUM[ECLK_NUM]. $\text{Final Clock value} = \text{System clock Frequency} / (2 * (\text{CmuExternalClk1Numerator} / \text{CmuExternalClk1Denominator}))$		
SFR Modified	GTM_CMU_ECLK_1_NUM.		
Scope:	Driver		
Dependency:	None		

4.2.3.3.6 CmuExternalClk1Denominator

Table 47 CmuExternalClk1Denominator

Name:	CmuExternalClk1Denominator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Denominator value for the External Clock Divider1. This modifies GTM_CMU_ECLK_1_DEN[ECLK_DEN]. $\text{Final Clock value} = \text{System clock Frequency} / (2 * (\text{CmuExternalClk1Numerator} / \text{CmuExternalClk1Denominator}))$		
SFR Modified	GTM_CMU_ECLK_1_DEN.		
Scope:	Driver		
Dependency:	CmuExternalClk1Numerator		

4.2.3.3.7 CmuEnableExternalClk2

Table 48 CmuEnableExternalClk2

Name:	CmuEnableExternalClk2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable External Clock 2 of GTM FALSE – Disable External Clock 2 of GTM Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the External Clock 2 of GTM. This sets GTM_CMU_CLK_EN.		
SFR Modified	GTM_CMU_CLK_EN		
Scope:	Driver		
Dependency:	None		

4.2.3.3.8 CmuExternalClk2Numerator

Table 49 CmuExternalClk2Numerator

Name:	CmuExternalClk2Numerator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB
Description:	This parameter specifies the Numerator value for the External Clock Divider2. This modifies GTM_CMU_ECLK_2_NUM[ECLK_NUM]. Final Clock value = System clock Frequency/(2 *(CmuExternalClk2Numerator/ CmuExternalClk2Denominator))		
SFR Modified	GTM_CMU_ECLK_2_NUM		
Scope:	Driver		
Dependency:	None		

4.2.3.3.9 CmuExternalClk2Denominator

Table 50 CmuExternalClk2Denominator

Name:	CmuExternalClk2Denominator
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Denominator value for the External Clock Divider2. This modifies GTM_CMU_ECLK_2_DEN[ECLK_DEN]. Final Clock value = System clock Frequency/(2 *(CmuExternalClk2Numerator/ CmuExternalClk2Denominator))		
SFR Modified	GTM_CMU_ECLK_2_DEN		
Scope:	Driver		
Dependency:	CmuExternalClk2Numerator		

4.2.4 Container: Tbu

This container provides the TBU configuration. Maximum multiplicity is 1. If this container is not included, then the registers values of TBU are retained as is.

4.2.4.1 Container: TbuChannel

This container provides the TBU Channel configuration. Multiplicity depends on the Number of TBU Channels present in the controller.

4.2.4.1.1 TbuChannelUsage

Table 51 TbuChannelUsage

Name:	TbuChannelUsage		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: USED_BY_OTHER_DRIVERS_OR_UNUSED: Channel is either not used or will be initialized elsewhere. By setting this, Gtm will not initialize the register of this channel USED_BY_GTM_DRIVER: Channel will used by Gtm Driver and will be initialized in Gtm_Init Default: USED_BY_OTHER_DRIVERS_OR_UNUSED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB

Description:	This decides if the channel will be used by GTM Driver or will be initialized elsewhere. USED_BY_OTHER_DRIVERS_OR_UNUSED: Channel is either not used or will be initialized elsewhere. By setting to this, Gtm will not initialize the register of this channel USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init
SFR Modified	N/A
Scope:	Driver
Dependency:	None

4.2.4.1.2 TbuEnableChannel

Table 52 TbuEnableChannel

Name:	TbuEnableChannel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable the TBU Channel FALSE – Disable the TBU Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBU Channel.		
SFR Modified	Ifx_GTM_TBU		
Scope:	Driver		
Dependency:	None		

4.2.4.1.3 TbuChannelModeSelect

Table 53 TbuChannelModeSelect

Name:	TbuChannelModeSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: FREE_RUNNING_COUNTER_MODE: Free Running Mode chosen FORWARD_BACKWARD_COUNTER_MODE: Forward Backward Counter mode chosen Default: FREE_RUNNING_COUNTER_MODE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Mode for the TBU Channel. This is only applicable for TBU Channels 1 and 2		
SFR Modified	Ifx_GTM_TBU_CHj_CTRL		

Scope:	Driver
Dependency:	None

4.2.4.1.4 TbuChannelClockSource

Table 54 TbuChannelClockSource

Name:	TbuChannelClockSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_CONFIGURABLE_CLOCK_0: Configurable Clock0 is chosen GTM_CONFIGURABLE_CLOCK_1: Configurable Clock1 is chosen GTM_CONFIGURABLE_CLOCK_2: Configurable Clock2 is chosen GTM_CONFIGURABLE_CLOCK_3: Configurable Clock3 is chosen GTM_CONFIGURABLE_CLOCK_4: Configurable Clock4 is chosen GTM_CONFIGURABLE_CLOCK_5: Configurable Clock5 is chosen GTM_CONFIGURABLE_CLOCK_6: Configurable Clock6 is chosen GTM_CONFIGURABLE_CLOCK_7: Configurable Clock7 is chosen Default: GTM_CONFIGURABLE_CLOCK_0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB
Description:	This decides the Clock Source for the TBU Channel.		
SFR Modified	Ifx_GTM_TBU_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.4.1.5 TbuChannelResolution

Table 55 TbuChannelResolution

Name:	TbuChannelResolution		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: BITS_0_TO_23: Choose Lower 24 Bits of TBU BITS_3_TO_26: Choose Higher 24 Bits of TBU Default: BITS_0_TO_23		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the resolution for the TBU Channel. This is only applicable for TBU Channel 0 BITS_0_TO_23: Choose Lower 24 Bits of TBU BITS_3_TO_26: Choose Higher 24 Bits of TBU		
SFR Modified	Ifx_GTM_TBU_CHj_CTRL		

Scope:	Driver
Dependency:	None

4.2.4.1.6 TbuTimebaseValue

Table 56 TbuTimebaseValue

Name:	TbuTimebaseValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0x7FFFFFFF for Tbu Channel 0 0 to 0xFFFFFFFF for other channels Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
SFR Modified	Ifx_GTM_TBU_CHj_BASE		
Description:	This parameter specifies the Timebase value for the TBU Channel		
Scope:	Driver		
Dependency:	None		

4.2.5 Container: Tom

This container contains the TOM Configuration for GTM. Maximum Multiplicity depends on the number of TOM modules present in the Microcontroller.

4.2.5.1 Container: TomTriggersForTgc

This container contains the Trigger configuration for the TGC's

4.2.5.1.1 Container: TomTgcTriggerByTimebase

This container contains the TGC Timebase trigger configuration for TOM

4.2.5.1.1.1 TomTgcTimeBaseTriggerEnable

Table 57 TomTgcTimeBaseTriggerEnable

Name:	TomTgcTimeBaseTriggerEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Timebase Trigger for the TGC FALSE - Disables the Timebase Trigger for the TGC Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	Enables/Disables the Timebase Trigger for the TGC. This sets GTM_TOMi_TGC0_ACT_TB[TB_TRIG] where i depends on the TOM module number.		
SFR Modified	GTM_TOMi_TGC0_ACT_TB		
Scope:	Driver		
Dependency:	None		

4.2.5.1.1.2 TomTgcTbuTimebaseSelect

Table 58 TomTgcTbuTimebaseSelect

Name:	TomTgcTbuTimebaseSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TBU_TS0_TIMEBASE: TBU_TS0 will be used to trigger the TGC TBU_TS1_TIMEBASE: TBU_TS1 will be used to trigger the TGC TBU_TS2_TIMEBASE: TBU_TS2 will be used to trigger the TGC Default: TBU_TS0_TIMEBASE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter selects the timebase which generates the Trigger for TGC. This modifies GTM_TOMi_TGC0_ACT_TB[TBU_SEL] where i depends on the TOM module number. TBU_TS0_TIMEBASE: TBU_TS0 will be used to trigger the TGC. TBU_TS1_TIMEBASE: TBU_TS1 will be used to trigger the TGC. TBU_TS2_TIMEBASE: TBU_TS2 will be used to trigger the TGC.		
SFR Modified	GTM_TOMi_TGC0_ACT_TB		
Scope:	Driver		
Dependency:	None		

4.2.5.1.1.3 TomTgcTimebaseMatchValue

Table 59 TomTgcTimebaseMatchValue

Name:	TomTgcTimebaseMatchValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	This parameter specifies the match value for the Timebase Trigger. This modifies GTM_TOMi_TGC0_ACT_TB[TB_VAL] where i depends on the TOM module number. Range 0 to 0xFFFFFFFF		
SFR Modified	GTM_TOMi_TGC0_ACT_TB		
Scope:	Driver		
Dependency:	None		

4.2.5.1.2 Container: TomTgcIntTriggerSelect

This container contains the TGC Timebase trigger configuration for TOM

4.2.5.1.2.1 TomTgcIntTriggerFromChannel0

Table 60 TomTgcIntTriggerFromChannel0

Name:	TomTgcIntTriggerFromChannel0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from TOM Channel 0 FALSE - Disables the Internal Trigger from TOM Channel 0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/disables the Internal Trigger from TOM Channel 0. This sets GTM_TOMi_TGC0_INT_TRIG[INT_TRIG0] where i depends on the TOM module number.		
SFR Modified	GTM_TOMi_TGC0_INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.5.1.2.2 TomTgcIntTriggerFromChannel1

Table 61 TomTgcIntTriggerFromChannel1

Name:	TomTgcIntTriggerFromChannel1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from TOM Channel 1 FALSE - Disables the Internal Trigger from TOM Channel 1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB

Description:	Enables/disables the Internal Trigger from TOM Channel 1. This sets GTM_TOMi_TGC0_INT_TRIG[INT_TRIG1] where i depends on the TOM module number.
SFR Modified	GTM_TOMi_TGC0_INT_TRIG
Scope:	Driver
Dependency:	None

4.2.5.1.2.3 TomTgcIntTriggerFromChannel2

Table 62 TomTgcIntTriggerFromChannel2

Name:	TomTgcIntTriggerFromChannel2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from TOM Channel 2 FALSE - Disables the Internal Trigger from TOM Channel 2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/disables the Internal Trigger from TOM Channel 2. This sets GTM_TOMi_TGC0_INT_TRIG[INT_TRIG2] where i depends on the TOM module number.		
SFR Modified	GTM_TOMi_TGC0_INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.5.1.2.4 TomTgcIntTriggerFromChannel3

Table 63 TomTgcIntTriggerFromChannel3

Name:	TomTgcIntTriggerFromChannel3		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from TOM Channel 3 FALSE - Disables the Internal Trigger from TOM Channel 3 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/disables the Internal Trigger from TOM Channel 3. This sets GTM_TOMi_TGC0_INT_TRIG[INT_TRIG3] where i depends on the TOM module number.		
SFR Modified	GTM_TOMi_TGC0_INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.5.1.2.5 TomTgcIntTriggerFromChannel4

Table 64 TomTgcIntTriggerFromChannel4

Name:	TomTgcIntTriggerFromChannel4		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from TOM Channel 4 FALSE - Disables the Internal Trigger from TOM Channel 4 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB
Description:	Enables/disables the Internal Trigger from TOM Channel 4. This sets GTM_TOMi_TGC0_INT_TRIG[INT_TRIG4] where i depends on the TOM module number.		
SFR Modified	GTM_TOMi_TGC0_INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.5.1.2.6 TomTgcIntTriggerFromChannel5

Table 65 TomTgcIntTriggerFromChannel5

Name:	TomTgcIntTriggerFromChannel5		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from TOM Channel 5 FALSE - Disables the Internal Trigger from TOM Channel 5 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/disables the Internal Trigger from TOM Channel 5. This sets GTM_TOMi_TGC0_INT_TRIG[INT_TRIG5] where i depends on the TOM module number.		
SFR Modified	GTM_TOMi_TGC0_INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.5.1.2.7 TomTgcIntTriggerFromChannel6

Table 66 TomTgcIntTriggerFromChannel6

Name:	TomTgcIntTriggerFromChannel6		
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from TOM Channel 6 FALSE - Disables the Internal Trigger from TOM Channel 6 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/disables the Internal Trigger from TOM Channel 6. This sets GTM_TOMi_TGC0_INT_TRIG[INT_TRIG6] where i depends on the TOM module number.		
SFR Modified	GTM_TOMi_TGC0_INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.5.1.2.8 TomTgcIntTriggerFromChannel7

Table 67 TomTgcIntTriggerFromChannel7

Name:	TomTgcIntTriggerFromChannel7		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from TOM Channel 7 FALSE - Disables the Internal Trigger from TOM Channel 7 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/disables the Internal Trigger from TOM Channel 7. This sets GTM_TOMi_TGC0_INT_TRIG[INT_TRIG7] where i depends on the TOM module number.		
SFR Modified	GTM_TOMi_TGC0_INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.5.2 Container: TomChannel

This container contains the TOM Channel configuration. Each entry in this corresponds to the corresponding channel number. Multiplicity depends on TOM Channels 0 to 15.

4.2.5.2.1 Container: TomChannelUsage

This specifies the module which uses the current TOM Channel.

4.2.5.2.1.1 TomChannelUsage

Table 68 TomChannelUsage

Name:	TomChannelUsage		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: USED_BY_OTHER_DRIVERS_OR_UNUSED: Specifies that this Channel is either unused or configured by other modules(GPT/PWM) USED_BY_GTM_DRIVER: Specifies that this channel will be configured as a complex driver Default: USED_BY_OTHER_DRIVERS_OR_UNUSED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides as to if this Channel will be used as a Complex driver or for GPT or PWM. USED_BY_OTHER_DRIVERS_OR_UNUSED: Specifies that this Channel is either unused or configured by other modules(GPT/PWM). USED_BY_GTM_DRIVER: Specifies that this channel will be configured as a complex driver.		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	None		

4.2.5.2.2 Container: TomChannelEnable

This container contains the TOM Channel Enable configuration.

4.2.5.2.2.1 TomChannelEnable

Table 69 TomChannelEnable

Name:	TomChannelEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: AT_START: Specifies that this Channel is enabled at Start Itself i.e. Channel Enable will be set in ENDIS_STAT bits for the channel. ON_GLOBAL_TRIGGER: Specifies that this Channel is enabled only on a Global Trigger i.e. Channel Enable will be set in the ENDIS_CTRL bits of the channel. Hence On a Global Trigger, this channel will be enabled Default: AT_START		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This configuration decides when to enable the channel. There are two instances at which a channel can be enabled autonomously. One instance is enabling the channel at Start and the other is enabling the channel on a TGC Global Trigger. AT_START: Specifies that this Channel is enabled at Start Itself i.e. Channel Enable will be set in ENDIS_STAT bits for the channel. ON_GLOBAL_TRIGGER: Specifies that this Channel is enabled only on a Global Trigger i.e. Channel Enable will be set in the ENDIS_CTRL bits of the channel. Hence On a Global Trigger, this channel will be enabled.
SFR Modified	GTM_TOMi_TGCK_ENDIS_STAT
Scope:	Driver
Dependency:	TomChannelUsage should be USED_BY_GTM_DRIVER

4.2.5.2.2 TomChDisableOnTgcTrigger

Table 70 TomChDisableOnTgcTrigger

Name:	TomChDisableOnTgcTrigger		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Disable channel on TGC Trigger FALSE – Do not disable channel on TGC Trigger Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if an Enabled channel has to be Disabled on TGC Trigger. Channel Disable will be set in ENDIS_CTRL bits of the channel.		
SFR Modified	GTM_TOMi_TGCK_ENDIS_STAT		
Scope:	Driver		
Dependency:	TomChannelUsage should be USED_BY_GTM_DRIVER and TomChannelEnable set as AT_START.		

4.2.5.2.3 Container: TomChannelOutput

This container contains the TOM Channel Output configuration

4.2.5.2.3.1 TomChannelOutputControl

Table 71 TomChannelOutputControl

Name:	TomChannelOutputControl
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: DONT_ENABLE_CHANNEL_OUTPUT: Specifies not to enable the Channel Output i.e. Channel output is disabled in OUTEN_STAT and OUTEN_CTRL bits of the channel. ENABLE_OUTPUT_AT_START: Specifies to enable the Channel Output at start itself i.e. Channel output is enabled in OUTEN_STAT bits of the channel. ENABLE_OUTPUT_ON_GLOBAL_TRIGGER: Specifies to enable the Channel Output only on a Global Trigger i.e. Channel output is enabled in OUTEN_CTRL bits of the channel. Default: DONT_ENABLE_CHANNEL_OUTPUT		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides when to enable the channel output. DONT_ENABLE_CHANNEL_OUTPUT: Specifies not to enable the Channel Output i.e. Channel output is disabled in OUTEN_STAT and OUTEN_CTRL bits of the channel. ENABLE_OUTPUT_AT_START: Specifies to enable the Channel Output at start itself i.e. Channel output is enabled in OUTEN_STAT bits of the channel. ENABLE_OUTPUT_ON_GLOBAL_TRIGGER: Specifies to enable the Channel Output only on a Global Trigger i.e. Channel output is enabled in OUTEN_CTRL bits of the channel.		
SFR Modified	GTM_TOMi_TGCK_OUTEN_STAT		
Scope:	Driver		
Dependency:	TomChannelUsage should be 'USED_BY_GTM_DRIVER'		

4.2.5.2.3.2 TomChOutputDisableOnTgcTrig

Table 72 TomChOutputDisableOnTgcTrig

Name:	TomChOutputDisableOnTgcTrig		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Disable Output on TGC Trigger FALSE - Enable Output on TGC Trigger Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if the channel output has to be disabled on TGC Trigger. Will set "Disable Channel" in OUTEN_CTRL bits of the channel.		
SFR Modified	GTM_TOMi_TGCK_OUTEN_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage should be 'USED_BY_GTM_DRIVER' and TomChannelOutputControl set as 'ENABLE_OUTPUT_AT_START'		

4.2.5.2.3.3 TomChannelOutputSignalLevel

Table 73 TomChannelOutputSignalLevel

Name:	TomChannelOutputSignalLevel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: LOW: Specifies that the Initial State of the Channel Output will be Low. This resets GTM_TOM0_CHx_CTRL[SL] (x=0-15) HIGH: Specifies that the Initial State of the Channel Output will be High. This sets GTM_TOM0_CHx_CTRL[SL] (x=0-15) Default: LOW		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB
Description:	This decides the initial state of channel output. LOW: Specifies that the Initial State of the Channel Output will be Low. This resets CHx_CTRL[SL] (x=0-15). HIGH: Specifies that the Initial State of the Channel Output will be High. This Sets CHx_CTRL[SL] (x=0-15).		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage should be 'USED_BY_GTM_DRIVER' and TomChannelOutputControl not set as 'DONT_ENABLE_CHANNEL_OUTPUT'		

4.2.5.2.3.4 TomChannelPortPinSelect

Table 74 TomChannelPortPinSelect

Name:	TomChannelPortPinSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the output port and pin for the TOM Channel. This parameter is not applicable in case of configuration using bmd. While using Bmd, GtmToutConfiguration should be configured for Port assignment.		
SFR Modified	TOUTSELn		
Scope:	Driver		
Dependency:	None		

4.2.5.2.4 Container: TomChannelTimerParameters

This contains the TOM Channel Timer parameters configuration.

4.2.5.2.4.1 TomChannelCounterValCn0

Table 75 TomChannelCounterValCn0

Name:	TomChannelCounterValCn0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 65535 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the CN0 Counter value for the TOM Channel		
SFR Modified	GTM_TOMi_CHj_CN0		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.2 TomChannelPeriodCompareValCm0

Table 76 TomChannelPeriodCompareValCm0

Name:	TomChannelPeriodCompareValCm0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 65535 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the CM0 Compare value for the TOM Channel		
SFR Modified	GTM_TOMi_CHj_CM0		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.3 TomChannelDutyCycleCompareValCm1

Table 77 TomChannelDutyCycleCompareValCm1

Name:	TomChannelDutyCycleCompareValCm1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 65535 Default: 0		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the CM1 Compare value for the TOM Channel		
SFR Modified	GTM_TOMi_CHj_CM1		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.4 TomChPeriodCompareShadowValSr0

Table 78 TomChPeriodCompareShadowValSr0

Name:	TomChPeriodCompareShadowValSr0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 65535 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the SR0 Compare value for the TOM Channel.		
SFR Modified	GTM_TOMi_CHj_SR0		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.5 TomChDCycCompareShadowValSr1

Table 79 TomChDCycCompareShadowValSr1

Name:	TomChDCycCompareShadowValSr1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 65535 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the SR1 Compare value for the TOM Channel		
SFR Modified	GTM_TOMi_CHj_SR1		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.6 TomChUpdateEnableOnCn0Reset

Table 80 TomChUpdateEnableOnCn0Reset

Name:	TomChUpdateEnableOnCn0Reset		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – FALSE Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This will enable updating of the Compare Values with the Shadow Register Compare Values on every CN0 Reset. This will set the UPEN (Update Enable) bits for that particular channel		
SFR Modified	GTM_TOMi_TGCK_GLB_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.7 TomChEnableForceUpdate

Table 81 TomChEnableForceUpdate

Name:	TomChEnableForceUpdate		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – FALSE Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This will enable updating of the Compare Values with the Shadow Register Compare Values on a Global Trigger. This sets the FUPD (Force Update) bits for that particular channel.		
SFR Modified	GTM_TOMi_TGCK_FUPD_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.8 TomChResetCn0OnForceUpdate

Table 82 TomChResetCn0OnForceUpdate

Name:	TomChResetCn0OnForceUpdate		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – FALSE Default: FALSE		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	This specifies that CN0 Counter value should be reset on a Force Update request. This sets the RSTCN0_CHx bits for that particular channel.		
SFR Modified			
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.9 TomChannelClockSelect

Table 83 TomChannelClockSelect

Name:	TomChannelClockSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_FIXED_CLOCK_0: Fixed Clock 0 will be supplied to the Channel GTM_FIXED_CLOCK_1: Fixed Clock 1 will be supplied to the Channel GTM_FIXED_CLOCK_2: Fixed Clock 2 will be supplied to the Channel GTM_FIXED_CLOCK_3: Fixed Clock 3 will be supplied to the Channel GTM_FIXED_CLOCK_4: Fixed Clock 4 will be supplied to the Channel Default: GTM_FIXED_CLOCK_0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Clock Source for TOM Channel. Please note, this clock is the shadow clock for the module and the effect of this will not reflect immediately. The channel requires a trigger for this to take effect. In case of GPT or PWM channels, this will immediately take effect.		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.5.2.4.10 TomChannelCounterCn0Reset

Table 84 TomChannelCounterCn0Reset

Name:	TomChannelCounterCn0Reset		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: ON_COMPARE_MATCH_ON_CHANNEL: CN0 Counter will Reset whenever a compare match occurs in CCU0 ON_TRIGGER_FROM_PREV_CHANNEL: CN0 Counter will Reset whenever a Trigger occurs from the Previous Channel Default: ON_COMPARE_MATCH_ON_CHANNEL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	This decides when the CN0 Counter will be reset. This modifies the RST_CCU0 bit for the channel. This decides if a Compare match from previous channel resets CN0 or Trigger from Previous channels. ON_COMPARE_MATCH_ON_CHANNEL: CN0 Counter will Reset whenever a compare match occurs in CCU0 ON_TRIGGER_FROM_PREV_CHANNEL: CN0 Counter will Reset whenever a Trigger occurs from the Previous Channel		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.11 TomChannelExtCounterCn0Reset

Table 85 TomChannelExtCounterCn0Reset

Name:	TomChannelExtCounterCn0Reset		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: ON_COMPARE_MATCH_ON_CHANNEL: TRIG or previous channel is selected as trigger for CN0 TRIG_TIM_EXT_CAPTURE: TRIG from external capture signal of TIM Default: TRIG_ON_PREV_COMPARE_MATCH		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides which signal will be selected as trigger to reset cn0. ON_COMPARE_MATCH_ON_CHANNEL: TRIG or previous channel is selected as trigger for CN0 TRIG_TIM_EXT_CAPTURE: TRIG from external capture signal of TIM		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.12 TomChannelTriggerOutput

Table 86 TomChannelTriggerOutput

Name:	TomChannelTriggerOutput		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRIG_FROM_PREVIOUS_CHANNEL: Trigger output of this channel will be the same as the Trigger output of the previous Channel. CCU0_TRIG_OF_CHANNEL: Trigger output from this channel will be affected by the CCU0 Match result of the current channel. Default: TRIG_FROM_PREVIOUS_CHANNEL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the content of the Trigger Output from this Channel. The Trigger can hold information of Trigger from this Channel (CCU0 Comparison match) or Trigger information from the Previous Channel. This modifies TRIGOUT bit of the Channel Configuration.		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.13 TomChannelExtTriggerOutput

Table 87 TomChannelExtTriggerOutput

Name:	TomChannelExtTriggerOutput		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: EXT_TRIG_FROM_PREVIOUS_CHANNEL: previous channel trigger will be used as trigger output. EXT_TRIG_TIM: external trigger from TIM will be used as output. Default: EXT_TRIG_FROM_PREVIOUS_CHANNEL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides which signals will be used for triggering potential output signal TRIG_X.		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.14 TomChannelOneShotMode

Table 88 TomChannelOneShotMode

Name:	TomChannelOneShotMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE - Enable One Shot Mode for the Channel FALSE - Disable One Shot Mode for the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables One Shot Mode for the Channel. This sets OSM bit for the Channel		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.15 TomChannelTrigOneShotMode

Table 89 TomChannelTrigOneShotMode

Name:	TomChannelTrigOneShotMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable One Shot Mode trigger for the Channel FALSE - Disable One Shot Mode trigger for the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables One Shot Mode trigger for the Channel. This sets OSM_TRIG bit for the Channel.		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.5.2.4.16 TomChannelSpeMode

Table 90 TomChannelSpeMode

Name:	TomChannelSpeMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables Sensor Pattern Evaluation Mode for the Channel FALSE - Disables Sensor Pattern Evaluation Mode for the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	Enables/Disables Sensor Pattern Evaluation Mode for the Channel. This sets SPE bit for the channel. This is available only for channels 0 to 7		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.5.2.4.17 TomChannelGatedCounterMode

Table 91 TomChannelGatedCounterMode

Name:	TomChannelGatedCounterMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables Gated Counter Mode for the Channel FALSE - Disables Gated Counter Mode for the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Gated Counter Mode for the Channel. This sets GCM bit of the channel control. This is available only for channel 0 to 7		
SFR Modified	GTM_TOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.5.2.4.18 TomChannelBitReversalMode

Table 92 TomChannelBitReversalMode

Name:	TomChannelBitReversalMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables Bit Reversal Mode for the Channel FALSE - Disables Bit Reversal Mode for the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Bit Reversal Mode for the Channel. Sets BITREV bit of the channel. This is available only for Channel 15		
SFR Modified	GTM_TOMi_CH15_CTRL		
Scope:	Driver		

Dependency:	None
--------------------	------

4.2.5.2.5 Container: TomChannelInterrupt

This contains the TOM Channel Interrupt configuration.

4.2.5.2.5.1 TomIntEnableOnPeriodMatchCcu0

Table 93 TomIntEnableOnPeriodMatchCcu0

Name:	TomIntEnableOnPeriodMatchCcu0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables Interrupt on Period match or match in CCU0 FALSE - Disables Interrupt on Period match or match in CCU0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on Period match or match in CCU0 for the Channel.		
SFR Modified	GTM_TOMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.5.2.5.2 TomIntEnableOnDutyCycleMatchCcu1

Table 94 TomIntEnableOnDutyCycleMatchCcu1

Name:	TomIntEnableOnDutyCycleMatchCcu1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables Interrupt on DutyCycle match or match in CCU1 FALSE - Disables Interrupt on DutyCycle match or match in CCU1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on DutyCycle match or match in CCU1 for the Channel.		
SFR Modified	GTM_TOMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.5.2.5.3 TomInterruptMode

Table 95 TomInterruptMode

Name:	TomInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE – Level Mode chosen GTM_INTERRUPT_PULSE_MODE - Pulse Mode chosen GTM_INTERRUPT_PULSE_NOTIFY_MODE – Pulse-Notify Mode chosen GTM_INTERRUPT_SINGLE_PULSE_MODE – Single Pulse Mode chosen Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use.		
SFR Modified	GTM_TOMi_CHj_IRQ_MODE		
Scope:	Driver		
Dependency:	None		

4.2.5.2.5.4 TomNotification

Table 96 TomNotification

Name:	TomNotification		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: String Default: none		
Configuration Class:	Pre-compile	--	--
	Link Time	x	Variant LT
	Post build	--	--

Description:	Notification function name should be mentioned here which has to be called on an interrupt event of the current. The prototype of the notification function should be, <pre>void NotifFn(uint8 ModType, uint8 ModuleNo, uint8 ChanNo, uint8 IntSrc)</pre> ModType – 0 – TIM 1 – TOM 2 - ATOM ModuleNo – TIM/TOM/ATOM ModuleNo ChanNo - TIM/TOM/ATOM ChanNo IntSrc – InterruptType of TIM/TOM/ATOM In case of Post Build Loadable variant, address of the Notification function need to be provided.
SFR Modified	N/A
Scope:	Driver
Dependency:	None

4.2.6 Container: Atom

This container contains the ATOM Configuration for GTM. Maximum Multiplicity depends on the number of ATOM modules.

4.2.6.1 Container: AtomTriggerForAgc

This contains the AGC Trigger configuration

4.2.6.1.1 Container: AtomAgcTriggerByTimebase

This contains the AGC Timebase trigger configuration for GTM

4.2.6.1.1.1 AtomAgcTimeBaseTriggerEnable

Table 97 AtomAgcTimeBaseTriggerEnable

Name:	AtomAgcTimeBaseTriggerEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Timebase Trigger for the AGC FALSE - Disables the Timebase Trigger for the AGC Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Timebase Trigger for the AGC. This sets ACT_TB[TB_TRIG].		
SFR Modified	ACT_TB		

Scope:	Driver
Dependency:	None

4.2.6.1.1.2 AtomAgcTbuTimebaseSelect

Table 98 AtomAgcTbuTimebaseSelect

Name:	AtomAgcTbuTimebaseSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TBU_TS0_TIMEBASE: TBU_TS0 will be used to trigger the AGC TBU_TS1_TIMEBASE: TBU_TS1 will be used to trigger the AGC TBU_TS2_TIMEBASE: TBU_TS2 will be used to trigger the AGC (If available) Default: TBU_TS0_TIMEBASE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter selects the timebase which generates the Trigger for AGC. This modifies ACT_TB[TBU_SEL]. TBU_TS0_TIMEBASE: TBU_TS0 will be used to trigger the AGC. TBU_TS1_TIMEBASE: TBU_TS1 will be used to trigger the AGC. TBU_TS2_TIMEBASE: TBU_TS2 will be used to trigger the AGC. (If available)		
SFR Modified	ACT_TB		
Scope:	Driver		
Dependency:	None		

4.2.6.1.1.3 AtomAgcTimebaseMatchValue

Table 99 AtomAgcTimebaseMatchValue

Name:	AtomAgcTimebaseMatchValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the match value for the Timebase Trigger. This modifies ACT_TB[TB_VAL]. Range 0 to 0xFFFFFFFF.		
SFR Modified	ACT_TB		
Scope:	Driver		
Dependency:	None		

4.2.6.1.2 Container: AtomAgcIntTriggerSelect

This container contains the AGC Timebase trigger configuration for GTM

4.2.6.1.2.1 AtomAgcIntTriggerFromChannel0

Table 100 AtomAgcIntTriggerFromChannel0

Name:	AtomAgcIntTriggerFromChannel0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from ATOM Channel 0 FALSE - Disables the Internal Trigger from ATOM Channel 0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Internal Trigger from ATOM Channel 0. This sets INT_TRIG[INT_TRIG0] .		
SFR Modified	INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.6.1.2.2 AtomAgcIntTriggerFromChannel1

Table 101 AtomAgcIntTriggerFromChannel1

Name:	AtomAgcIntTriggerFromChannel1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from ATOM Channel 1 FALSE - Disables the Internal Trigger from ATOM Channel 1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB
Description:	Enables/Disables the Internal Trigger from ATOM Channel 1. This sets INT_TRIG[INT_TRIG1] .		
SFR Modified	INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.6.1.2.3 AtomAgcIntTriggerFromChannel2

Table 102 AtomAgcIntTriggerFromChannel2

Name:	AtomAgcIntTriggerFromChannel2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from ATOM Channel 2 FALSE - Disables the Internal Trigger from ATOM Channel 2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Internal Trigger from ATOM Channel 2. This sets INT_TRIG[INT_TRIG2] .		
SFR Modified	INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.6.1.2.4 AtomAgcIntTriggerFromChannel3

Table 103 AtomAgcIntTriggerFromChannel3

Name:	AtomAgcIntTriggerFromChannel3		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from ATOM Channel 3 FALSE - Disables the Internal Trigger from ATOM Channel 3 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Internal Trigger from ATOM Channel 3. This sets INT_TRIG[INT_TRIG3] .		
SFR Modified	INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.6.1.2.5 AtomAgcIntTriggerFromChannel4

Table 104 AtomAgcIntTriggerFromChannel4

Name:	AtomAgcIntTriggerFromChannel4		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables the Internal Trigger from ATOM Channel 4 FALSE - Disables the Internal Trigger from ATOM Channel 4 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB
Description:	Enables/Disables the Internal Trigger from ATOM Channel 4. This sets INT_TRIG[INT_TRIG4] .		
SFR Modified	INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.6.1.2.6 AtomAgcIntTriggerFromChannel5

Table 105 AtomAgcIntTriggerFromChannel5

Name:	AtomAgcIntTriggerFromChannel5		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from ATOM Channel 5 FALSE - Disables the Internal Trigger from ATOM Channel 5 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Internal Trigger from ATOM Channel 5. This sets INT_TRIG[INT_TRIG5] .		
SFR Modified	INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.6.1.2.7 AtomAgcIntTriggerFromChannel6

Table 106 AtomAgcIntTriggerFromChannel6

Name:	AtomAgcIntTriggerFromChannel6		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from ATOM Channel 6 FALSE - Disables the Internal Trigger from ATOM Channel 6 Default: FALSE		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Internal Trigger from ATOM Channel 6. This sets INT_TRIG[INT_TRIG6] .		
SFR Modified	INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.6.1.2.8 AtomAgcIntTriggerFromChannel7

Table 107 AtomAgcIntTriggerFromChannel7

Name:	AtomAgcIntTriggerFromChannel7		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Internal Trigger from ATOM Channel 7 FALSE - Disables the Internal Trigger from ATOM Channel 7 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Internal Trigger from ATOM Channel 7. This sets INT_TRIG[INT_TRIG7].		
SFR Modified	INT_TRIG		
Scope:	Driver		
Dependency:	None		

4.2.6.2 Container: AtomChannel

This container contains the ATOM AGC Channel configuration. Multiplicity depends on the number of ATOM channels present in each module.

4.2.6.2.1 Container: AtomChannelUsage

This specifies the module that uses the ATOM Channel.

4.2.6.2.1.1 AtomChannelUsage

Table 108 AtomChannelUsage

Name:	AtomChannelUsage
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: USED_BY_OTHER_DRIVERS_OR_UNUSED: Specifies that this Channel is either unused or configured by other modules(GPT/PWM) USED_BY_GTM_DRIVER: Specifies that this channel will be configured as a complex driver Default: USED_BY_OTHER_DRIVERS_OR_UNUSED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if this Channel will be used as a Complex driver or for GPT or PWM. USED_BY_OTHER_DRIVERS_OR_UNUSED: Specifies that this Channel is either unused or configured by other modules(GPT/PWM). USED_BY_GTM_DRIVER: Specifies that this channel will be configured as a complex driver.		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	None		

4.2.6.2.2 Container: AtomChannelEnable

This container contains the ATOM Channel Enable configuration

4.2.6.2.2.1 AtomChannelEnable

Table 109 AtomChannelEnable

Name:	AtomChannelEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: AT_START: Specifies that this Channel is enabled at Start Itself i.e. Channel Enable will be set in ENDIS_STAT bits for the channel. ON_GLOBAL_TRIGGER: Specifies that this Channel is enabled only on a Global Trigger i.e. Channel Enable will be set in the ENDIS_CTRL bits of the channel. Hence On a Global Trigger, this channel will be enabled. Default: AT_START		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This configuration decides when to enable the channel. There are two instances at which a channel can be enabled autonomously. One instance is enabling the channel at Start and the other is enabling the channel on an AGC Global Trigger. AT_START: Specifies that this Channel is enabled at Start Itself i.e. Channel Enable will be set in ENDIS_STAT bits for the channel. ON_GLOBAL_TRIGGER: Specifies that this Channel is enabled only on a Global Trigger i.e. Channel Enable will be set in the ENDIS_CTRL bits of the channel. Hence On a Global Trigger, this channel will be enabled.		
SFR Modified	GTM_TOMi_TGCK_ENDIS_STAT		
Scope:	Driver		

Dependency:	AtomChannelUsage should be USED_BY_GTM_DRIVER
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4.2.6.2.2.2 AtomChDisableOnAgcTrigger

Table 110 AtomChDisableOnAgcTrigger

Name:	AtomChDisableOnAgcTrigger		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Disable channel on AGC trigger FALSE – Channel not disabled on AGC Trigger Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if an enabled channel has to be disabled on AGC Trigger. Channel Disable will be set in ENDIS_CTRL bits of the channel.		
SFR Modified	GTM_ATOMi_AGCK_ENDIS_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage should be USED_BY_GTM_DRIVER and AtomChannelEnable set as AT_START		

4.2.6.2.3 Container: AtomChannelOutput

This container contains the ATOM Channel Output configuration

4.2.6.2.3.1 AtomChannelOutputControl

Table 111 AtomChannelOutputControl

Name:	AtomChannelOutputControl		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: DONT_ENABLE_CHANNEL_OUTPUT: Specifies not to enable the Channel Output i.e. Channel output is disabled in OUTEN_STAT and OUTEN_CTRL bits of the channel. ENABLE_OUTPUT_AT_START: Specifies to enable the Channel Output at start itself i.e. Channel output is enabled in OUTEN_STAT bits of the channel . ENABLE_OUTPUT_ON_GLOBAL_TRIGGER: Specifies to enable the Channel Output only on a Global Trigger i.e. Channel output is enabled in OUTEN_CTRL bits of the channel. Default: DONT_ENABLE_CHANNEL_OUTPUT		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This decides when to enable the channel output. DONT_ENABLE_CHANNEL_OUTPUT: Specifies not to enable the Channel Output i.e. Channel output is disabled in OUTEN_STAT and OUTEN_CTRL bits of the channel. ENABLE_OUTPUT_AT_START: Specifies to enable the Channel Output at start itself i.e. Channel output is enabled in OUTEN_STAT bits of the channel . ENABLE_OUTPUT_ON_GLOBAL_TRIGGER: Specifies to enable the Channel Output only on a Global Trigger i.e. Channel output is enabled in OUTEN_CTRL bits of the channel.
SFR Modified	GTM_ATOMi_AGCK_OUTEN_STAT
Scope:	Driver
Dependency:	AtomChannelUsage should be 'USED_BY_GTM_DRIVER'

4.2.6.2.3.2 AtomChOutputDisableOnAgcTrig

Table 112 AtomChOutputDisableOnAgcTrig

Name:	AtomChOutputDisableOnAgcTrig		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Channel Output Disable on AGC trigger FALSE - Channel Output not disabled on AGC trigger Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if the channel output has to be disabled on AGC Trigger. Will set "Disable Channel" in OUTEN_CTRL bits of the channel		
SFR Modified	GTM_ATOMi_AGCK_OUTEN_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage should be 'USED_BY_GTM_DRIVER' and AtomChannelOutputControl set as 'ENABLE_OUTPUT_AT_START'		

4.2.6.2.3.3 AtomChannelOutputSignalLevel

Table 113 AtomChannelOutputSignalLevel

Name:	AtomChannelOutputSignalLevel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: LOW: Specifies that the Initial State of the Channel Output will be Low. HIGH: Specifies that the Initial State of the Channel Output will be High. Default: LOW		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	X	Variant PB

Description:	This decides the initial state of channel output. LOW: Specifies that the Initial State of the Channel Output will be Low. This resets CHx_CTRL[SL] (x=0-7). HIGH: Specifies that the Initial State of the Channel Output will be High. This Sets CHx_CTRL[SL] (x=0-7).
SFR Modified	GTM_ATOMi_CHj_CTRL
Scope:	Driver
Dependency:	AtomChannelUsage should be 'USED_BY_GTM_DRIVER' and AtomChannelOutputControl not set as 'DONT_ENABLE_CHANNEL_OUTPUT'

4.2.6.2.3.4 AtomChannelPortPinSelect

Table 114 AtomChannelPortPinSelect

Name:	AtomChannelPortPinSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	x	Variant PC
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the output port and pin for the ATOM Channel. This parameter is not applicable in case of configuration using BMD. In such a case, GtmToutConfiguration should be used instead.		
SFR Modified	GTM_TOUTSELn		
Scope:	Driver		
Dependency:	None		

4.2.6.2.4 Container: AtomChannelTimerParameters

This contains the ATOM Channel Timer parameters configuration.

4.2.6.2.4.1 AtomChannelModeSelect

Table 115 AtomChannelModeSelect

Name:	AtomChannelModeSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: ATOM_SIG_OUTPUT_MODE_IMMEDIATE – SOMI Mode chosen ATOM_SIG_OUTPUT_MODE_COMPARE – SOMC Mode chosen ATOM_SIG_OUTPUT_MODE_PWM – SOMP Mode chosen ATOM_SIG_OUTPUT_MODE_SERIAL – SOMS Mode chosen Default: ATOM_SIG_OUTPUT_MODE_IMMEDIATE		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the operating mode for the ATOM Channel.		
SFR Modified	GTM_ATOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.2 AtomChannelCounterValCn0

Table 116 AtomChannelCounterValCn0

Name:	AtomChannelCounterValCn0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the CN0 Counter value for the ATOM Channel		
SFR Modified	GTM_ATOMi_CHj_CN0		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.3 AtomChannelCompareValueCm0

Table 117 AtomChannelCompareValueCm0

Name:	AtomChannelCompareValueCm0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
SFR Modified	GTM_ATOMi_CHj_CM0		
Description:	This parameter specifies the CM0 Compare value for the ATOM Channel		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.4 AtomChannelCompareValueCm1

Table 118 AtomChannelCompareValueCm1

Name:	AtomChannelCompareValueCm1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the CM1 Compare value for the ATOM Channel		
SFR Modified	GTM_ATOMi_CHj_CM1		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.5 AtomChShadowCompareValSr0

Table 119 AtomChShadowCompareValSr0

Name:	AtomChShadowCompareValSr0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
SFR Modified	GTM_ATOMi_CHj_SR0		
Description:	This parameter specifies the SR0 Compare value for the ATOM Channel		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.6 AtomChShadowCompareValSr1

Table 120 AtomChShadowCompareValSr1

Name:	AtomChShadowCompareValSr1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the SR1 Compare value for the ATOM Channel		
SFR Modified	GTM_ATOMi_CHj_SR1		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.7 AtomChUpdateEnableOnCn0Reset

Table 121 AtomChUpdateEnableOnCn0Reset

Name:	AtomChUpdateEnableOnCn0Reset		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Activate Update Enable for the Channel FALSE - Deactivate Update Enable for the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This will enable updation of the Compare Values with the Shadow Register Compare Values on every CN0 Reset. This will set the UPEN (Update Enable) bits for that particular channel		
SFR Modified	GTM_ATOMi_AGC_GLB_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.8 AtomChEnableForceUpdate

Table 122 AtomChEnableForceUpdate

Name:	AtomChEnableForceUpdate		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Force Update FALSE – Disable Force Update Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This will enable updation of the Compare Values with the Shadow Register Compare Values on a Global Trigger. This sets the FUPD (Force Update) bits for that particular channel.		
SFR Modified	GTM_ATOMi_AGC_FUPD_CTRL		
Scope:	Driver		

Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'
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4.2.6.2.4.9 AtomChResetCn0OnForceUpdate

Table 123 AtomChResetCn0OnForceUpdate

Name:	AtomChResetCn0OnForceUpdate		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Reset CN0 on Force Update FALSE – Do not reset CN0 on Force Update Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This specifies that CN0 Counter value should be reset on a Force Update request. This sets the RSTCN0_CHx bits for that particular channel		
SFR Modified	GTM_ATOMi_AGC_FUPD_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.10 AtomChannelClockSelect

Table 124 AtomChannelClockSelect

Name:	AtomChannelClockSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_CONFIGURABLE_CLOCK_0: Configurable Clock0 is chosen GTM_CONFIGURABLE_CLOCK_1: Configurable Clock1 is chosen GTM_CONFIGURABLE_CLOCK_2: Configurable Clock2 is chosen GTM_CONFIGURABLE_CLOCK_3: Configurable Clock3 is chosen GTM_CONFIGURABLE_CLOCK_4: Configurable Clock4 is chosen GTM_CONFIGURABLE_CLOCK_5: Configurable Clock5 is chosen GTM_CONFIGURABLE_CLOCK_6: Configurable Clock6 is chosen GTM_CONFIGURABLE_CLOCK_7: Configurable Clock7 is chosen Default: GTM_CONFIGURABLE_CLOCK_0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Clock Source for ATOM Channel. Please note, this clock is the shadow clock for the module and the effect of this will not reflect immediately. The channel requires a trigger for this to take effect. In case of GPT or PWM channels, this will immediately take effect.		
SFR Modified	GTM_ATOMi_CHj_CTRL		

Scope:	Driver
Dependency:	None

4.2.6.2.4.11 AtomChannelCounterCn0Reset

Table 125 AtomChannelCounterCn0Reset

Name:	AtomChannelCounterCn0Reset		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: ON_COMPARE_MATCH_ON_CHANNEL: CN0 Counter will Reset whenever a compare match occurs in CCU0 ON_TRIGGER_FROM_PREV_CHANNEL: CN0 Counter will Reset whenever a Trigger occurs from the Previous Channel Default: ON_COMPARE_MATCH_ON_CHANNEL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides when the CN0 Counter will be reset. This modifies the RST_CCU0 bit for the channel. This decides if a Compare match from previous channel resets CN0 or Trigger from Previous channels ON_COMPARE_MATCH_ON_CHANNEL: CN0 Counter will Reset whenever a compare match occurs in CCU0 ON_TRIGGER_FROM_PREV_CHANNEL: CN0 Counter will Reset whenever a Trigger occurs from the Previous Channel		
SFR Modified	RST_CCU0		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.4.12 AtomChannelTriggerOutput

Table 126 AtomChannelTriggerOutput

Name:	AtomChannelTriggerOutput		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRIG_FROM_PREVIOUS_CHANNEL: Trigger output of this channel will be the same as the Trigger output of the previous Channel CCU0_TRIG_OF_CHANNEL: Trigger output from this channel will be affected by the CCU0 Match result of the current channel Default: TRIG_FROM_PREVIOUS_CHANNEL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This decides the content of the Trigger Output from this Channel. The Trigger can hold information of Trigger from this Channel (CCU0 Comparison match) or Trigger information from the Previous Channel. This modifies TRIGOUT bit of the Channel Configuration.
SFR Modified	GTM_ATOMi_CHj_CTRL
Scope:	Driver
Dependency:	None

4.2.6.2.4.13 AtomChSomcTimebaseForComparison

Table 127 AtomChSomcTimebaseForComparison

Name:	AtomChSomcTimebaseForComparison		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TBU_TS1: TBU_TS1 is used for comparison TBU_TS2: TBU_TS2 is used for comparison Default: TBU_TS1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the timebase value to be used for Comparison when the channel is in SOMC mode. This modifies CHx_CTRL[TB12_SEL] (x=0-7).		
SFR Modified	GTM_ATOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage is 'USED_BY_GTM_DRIVER' and AtomChannelModeSelect is 'ATOM_SIG_OUTPUT_MODE_COMPARE'		

4.2.6.2.4.14 AtomChSomcCompareStrategy

Table 128 AtomChSomcCompareStrategy

Name:	AtomChSomcCompareStrategy		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TIMEBASE_GREATER_EQUAL_TO_CMx: Greater/equal compare against TBU time base values TIMEBASE_LESSER_EQUAL_TO_CMx: Less/equal compare against TBU time base values. Default: TIMEBASE_GREATER_EQUAL_TO_CMx		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This decides the comparison strategy to use in SOMC mode. TIMEBASE_GREATER_EQUAL_TO_CMx: Greater/equal compare against TBU time base values TIMEBASE_LESSER_EQUAL_TO_CMx: Less/equal compare against TBU time base values
SFR Modified	GTM_ATOMi_CHj_CTRL
Scope:	Driver
Dependency:	AtomChannelUsage is 'USED_BY_GTM_DRIVER' and AtomChannelModeSelect is 'ATOM_SIG_OUTPUT_MODE_COMPARE'

4.2.6.2.4.15 AtomModeControlBits

Table 129 AtomModeControlBits

Name:	AtomModeControlBits		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 31 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the ATOM Mode Control bits. This is not applicable for SOMP mode		
SFR Modified	GTM_ATOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomChannelModeSelect not equal to 'ATOM_SIG_OUTPUT_MODE_PWM'		

4.2.6.2.4.16 AtomChannelOneShotMode

Table 130 AtomChannelOneShotMode

Name:	AtomChannelOneShotMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable One Shot mode FALSE – Disable One Shot mode Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables One Shot Mode for the Channel. This sets OSM bit for the Channel.		
SFR Modified	GTM_ATOMi_CHj_CTRL		
Scope:	Driver		

Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'
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4.2.6.2.4.17 AtomServeLastAruCommStrategy

Table 131 AtomServeLastAruCommStrategy

Name:	AtomServeLastAruCommStrategy		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: FALSE – Capture SRx time stamps after CCU0 match event not provided to ARU TRUE – Capture SRx time stamps after CCU0 match event provided to ARU Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Serve last ARU communication strategy. A TRUE would set GTM_ATOMx_CHy_CTRL[SLA] Range: FALSE: Capture SRx time stamps after CCU0 match event not provided to ARU TRUE: Capture SRx time stamps after CCU0 match event provided to ARU This bit is only applicable in SOMC mode. Please note, that setting of this bit has only effect,when ACBI(4:2) is configured for serve last compare strategy ("100", "101", or "110"). FALSE: The captured time stamps in the shadow registers SRx are only provided after the CCU1 match occurred. The ACBO(4:3) bits always return "10" in that case. TRUE: The ATOM channel also provides the captured time stamps after the CCU0 match event to the ARU. The ACBO(4:3) bits are set to "01" in that case. After the CCU1 match event, the time stamps are captured again in the SRx registers and provided to the ARU. The ACBO(4:3) bits are set to "10". When the data in the shadow registers after the CCU0 match was not consumed by an ARU destination and the CCU1 match occurs, the data in the shadow registers is overwritten by the new captured time stamps. The ATOM channel does not request new data from the ARU when the CCU0 match values are read from an ARU destination.		
SFR Modified	GTM_ATOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomChannelModeSelect = 'ATOM_SIG_OUTPUT_MODE_COMPARE'		

4.2.6.2.5 Container: AtomChannelAruConfiguration

This container contains the ARU configuration for the ATOM Channel.

4.2.6.2.5.1 AtomChannelAruEnable

Table 132 AtomChannelAruEnable

Name:	AtomChannelAruEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable ARU for the Channel FALSE – Disable ARU for the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables ARU for the Channel. This sets ARU_EN bit for the channel		
SFR Modified	GTM_ATOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.5.2 AtomChannelCpuWriteRequest

Table 133 AtomChannelCpuWriteRequest

Name:	AtomChannelCpuWriteRequest		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Provides CPU Write request bit for late compare register update FALSE – Do not provides CPU Write request bit for late compare register update Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Provides CPU Write request bit for late compare register update. This is available only in SOMC mode		
SFR Modified	GTM_ATOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomChannelModeSelect = 'ATOM_SIG_OUTPUT_MODE_COMPARE'		

4.2.6.2.5.3 AtomChAruBlockingMode

Table 134 AtomChAruBlockingMode

Name:	AtomChAruBlockingMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: GTM_ABM_DISABLED: ARU blocking mode disabled: ATOM reads continuously from ARU and updates CM0, CM1 independent of pending compare match event GTM_ABM_ENABLED: ARU blocking mode enabled: after updating CM0,CM1 via ARU, no new data is read from ARU until compare match event is occurred Default: GTM_ABM_DISABLED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides ARU Blocking mode enabling for the channel. This updates the ABM bit in the Control Register for the Channel. This is applicable only in case of SOMC mode. ABM_DISABLED: ARU blocking mode disabled: ATOM reads continuously from ARU and updates CM0, CM1 independent of pending compare match event ABM_ENABLED: ARU blocking mode enabled: after updating CM0,CM1 via ARU, no new data is read from ARU until compare match event is occurred.		
SFR Modified	GTM_ATOMi_CHj_CTRL		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomChannelModeSelect = 'ATOM_SIG_OUTPUT_MODE_COMPARE'		

4.2.6.2.5.4 AtomChAruReadAddress0

Table 135 AtomChAruReadAddress0

Name:	AtomChAruReadAddress0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 511 Default: 510		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the ARU Read Address0. This read address is used by the ATOM channel to receive data from ARU immediately after the channel and ARU access is enabled		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.5.5 AtomChAruReadAddress1

Table 136 AtomChAruReadAddress1

Name:	AtomChAruReadAddress1		
Source:	This parameter is microcontroller specific.		

File:	Gtm_PBCfg.c		
Range:	Values: 0 to 511 Default: 510		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the ARU Read Address1. The ATOM channel switches to this read address, when requested in the ARU control bits 52 to 48 with the pattern "111--b". The channel switches back to the RDADDR0 after one ARU data package was received on RDADDR1. This is applicable only in SOMC mode		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomChannelModeSelect = 'ATOM_SIG_OUTPUT_MODE_COMPARE'		

4.2.6.2.6 Container: AtomChannelInterrupt

This container contains the Interrupt configuration for the ATOM Channel.

4.2.6.2.6.1 AtomGlobalCcuSel

Table 137 AtomGlobalCcuSel

Name:	AtomGlobalCcuSel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE –Seperate notifications to be configured for ATOM ccu0 and ccu1. FALSE – Single notification will be configured for ATOM ccu0 and ccu1. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables separate notifications configuration for ATOM CCU's. Note: This parameter will be marked true only if seperate notifications will be configured for CCU0 and CCU1		
SFR Modified	GTM_ATOMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.6.2.6.2 AtomIntEnableOnCcu0Trigger

Table 138 AtomIntEnableOnCcu0Trigger

Name:	AtomIntEnableOnCcu0Trigger
Source:	This parameter is microcontroller specific.

File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Interrupt on match in CCU0 FALSE - Disable Interrupt on match in CCU0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Interrupt on match in CCU0 for the Channel		
SFR Modified	GTM_ATOMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomGlobalCcuSel = false		

4.2.6.2.6.3 AtomIntEnableOnCcu1Trigger

Table 139 AtomIntEnableOnCcu1Trigger

Name:	AtomIntEnableOnCcu1Trigger		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Interrupt on match in CCU1 FALSE - Disable Interrupt on match in CCU1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Interrupt on match in CCU1 for the Channel.		
SFR Modified	GTM_ATOMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomGlobalCcuSel = false		

4.2.6.2.6.4 AtomInterruptMode

Table 140 AtomInterruptMode

Name:	AtomInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE: level mode is chosen GTM_INTERRUPT_PULSE_MODE: pulse mode is chosen GTM_INTERRUPT_PULSE_NOTIFY_MODE: pulse notify mode is chosen GTM_INTERRUPT_SINGLE_PULSE_MODE: single pulse mode is chosen Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use.		
SFR Modified	GTM_ATOMi_CHj_IRQ_MODE		
Scope:	Driver		
Dependency:	None		

4.2.6.2.6.5 AtomNotification

Table 141 AtomNotification

Name:	AtomNotification		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: String Default: none		
Configuration Class:	Pre-compile	--	--
	Link Time	x	Variant LT
	Post build	--	--
Description:	Notification function name should be mentioned here which has to be called on an interrupt event of the current. The prototype of the notification function should be, void NotifFn(uint8 ModType, uint8 ModuleNo, uint8 ChanNo, uint16 NotifRegVal) ModType – 0 – TIM 1 – TOM 2 - ATOM ModuleNo – TIM/TOM/ATOM ModuleNo ChanNo - TIM/TOM/ATOM ChanNo NotifRegVal– Notification Register value for TIM/TOM/ATOM register value.		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomGlobalCcuSel = false		

4.2.6.2.6.6 AtomCcu0Notification

Table 142 AtomCcu0Notification

Name:	AtomCcu0Notification		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: String Default: none		
Configuration Class:	Pre-compile	--	--

	Link Time	x	Variant LT
	Post build	--	--
Description:	Notification function name should be mentioned here which has to be called on an interrupt event of CCU0. The prototype of the notification function should be, void NotifFn (uint8 ModuleType, uint8 ModuleNo,uint8 ChannelNo,uint16 NotifRegVal) ModType – ATOM (Note: Modetype will always be ATOM when configured for this parameter. This parameter is retained for consistency purposes.) ModuleNo –ATOM ModuleNo ChanNo - ATOM ChanNo NotifRegVal – Notification Register value (It will be 0xFF (i.e. Invalid) when channel is configured for this param. This parameter is retained for consistency purposes.)		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomGlobalCcuSel = true		

4.2.6.2.6.7 AtomCcu1Notification

Table 143 AtomCcu0Notification

Name:	AtomCcu1Notification		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: String Default: none		
Configuration Class:	Pre-compile	--	--
	Link Time	x	Variant LT
	Post build	--	--
Description:	Notification function name should be mentioned here which has to be called on an interrupt event of CCU1. The prototype of the notification function should be, void NotifFn (uint8 ModuleType, uint8 ModuleNo,uint8 ChannelNo,uint16 NotifRegVal) ModType – ATOM (Note: Modetype will always be ATOM when configured for this parameter. This parameter is retained for consistency purposes.) ModuleNo –ATOM ModuleNo ChanNo - ATOM ChanNo NotifRegVal– Notification Register value (It will be 0xFF (i.e. Invalid) when channel is configured for this param. This parameter is retained for consistency purposes.)		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	AtomChannelUsage = 'USED_BY_GTM_DRIVER' and AtomGlobalCcuSel = true		

4.2.7 Container: Tim

This container contains the TIM Configuration for GTM. Multiplicity depends on the Number of TIM Channels.

4.2.7.1 Container: TimChannel

This container contains the TIM Channel Configuration. Each index corresponds to respective TIM Channel Number. Multiplicity depends on number of TIM Channels in each TIM module.

4.2.7.1.1 Container: TimChannelGeneral

This container contains the TIM Channel General Configuration

4.2.7.1.1.1 TimChannelUsage

Table 144 TimChannelUsage

Name:	TimChannelUsage		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: USED_BY_OTHER_DRIVERS_OR_UNUSED: Specifies that this Channel is either unused or configured by other modules (ICU). USED_BY_GTM_DRIVER: Specifies that this channel will be configured as a complex driver. Default: USED_BY_OTHER_DRIVERS_OR_UNUSED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides as to if this Channel will be used as a Complex driver or for ICU or will it not be used. USED_BY_OTHER_DRIVERS_OR_UNUSED: Specifies that this Channel is either unused or configured by other modules (ICU). USED_BY_GTM_DRIVER: Specifies that this channel will be configured as a complex driver.		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	None		

4.2.7.1.1.2 TimChannelEnable

Table 145 TimChannelEnable

Name:	TimChannelEnable
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: TRUE – Enable the TIM Channel FALSE – Disable the TIM Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the TIM Channel. This sets CHx_CTRL[TIM_EN] for the channel.		
SFR Modified	CHx_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.1.3 TimChannelInputSelect

Table 146 TimChannelInputSelect

Name:	TimChannelInputSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: INPUT_OF_CURRENT_TIM_CHANNEL: Input to the current TIM Channel will be the input assigned to this channel INPUT_OF_PREVIOUS_TIM_CHANNEL: Input to the current TIM channel will be the input assigned to the previous channel INPUT_OF_AUX_IN_TOM: Input to the current TIM channel will be the output of the corresponding TOM channel INPUT_OF_AUX_IN_ATOM: Input to the current TIM channel will be the output of the corresponding ATOM channel INPUT_OF_VAL: Input to the current TIM channel will be Fixed value selected by TimChannelValInitSignal Default: INPUT_OF_CURRENT_TIM_CHANNEL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Input for the channel INPUT_OF_CURRENT_TIM_CHANNEL: Input to the current TIM Channel will be the input assigned to this channel. INPUT_OF_PREVIOUS_TIM_CHANNEL: Input to the current TIM channel will be the input assigned to the previous channel. Please note, the Port and Pin number applicable in this case is the configuration in TimChannelPortPinSelect for the previous channel and not the Port and Pin configured for this channel. INPUT_OF_AUX_IN_TOM: Input to the current TIM channel will be the output of the corresponding TOM channel INPUT_OF_AUX_IN_ATOM: Input to the current TIM channel will be the output of the corresponding ATOM channel INPUT_OF_VAL: Input to the current TIM channel will be Fixed value selected by TimChannelValInitSignal.		
SFR Modified	GTM_TIMi_CHj_CTRL, GTM_TIMi_IN_SRC, GTM_TIMi_AUX_IN_SRC		

Scope:	Driver
Dependency:	None

4.2.7.1.1.4 TimChannelPortPinSelect

Table 147 TimChannelPortPinSelect

Name:	TimChannelPortPinSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details. Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the input port and pin of the TIM Channel. This value is applicable for this channel only if TimChannelInputSelect is configured as 'INPUT_OF_CURRENT_TIM_CHANNEL'		
SFR Modified	GTM_INOUTSEL_TIMi_INSEL		
Scope:	Driver		
Dependency:	TimChannelInputSelect		

4.2.7.1.1.5 TimChannelVallnitSignal

Table 148 TimChannelVallnitSignal

Name:	TimChannelVallnitSignal		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: VAL_LOW: Fixed value 0 will be selected VAL_HIGH: Fixed value 1 will be selected Default: VAL_LOW		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the signal level to detect for the TIM Channel VAL_LOW: Fixed value 0 will be selected VAL_HIGH: Fixed value 1 will be selected		
SFR Modified	GTM_TIMi_IN_SRC		
Scope:	Driver		
Dependency:	None		

4.2.7.1.1.6 TimChannelSignalEdgeSelect

Table 149 TimChannelSignalEdgeSelect

Name:	TimChannelSignalEdgeSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: FALLING_EDGE: Falling Edge in the input will be detected RISING_EDGE: Rising Edge in the input will be detected BOTH_EDGES: Both Edges in the input will be detected Default: FALLING_EDGE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the signal level to detect for the TIM Channel		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER' Both Edges is supported only in TIEM and TIPM modes in case of TIM0 and only in TIEM mode is case of other TIM modules.		

4.2.7.1.1.7 TimChannelClockSelect

Table 150 TimChannelClockSelect

Name:	TimChannelClockSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_CONFIGURABLE_CLOCK_0: Configurable Clock 0 will be supplied to this channel GTM_CONFIGURABLE_CLOCK_1: Configurable Clock 1 will be supplied to this channel GTM_CONFIGURABLE_CLOCK_2: Configurable Clock 2 will be supplied to this channel GTM_CONFIGURABLE_CLOCK_3: Configurable Clock 3 will be supplied to this channel GTM_CONFIGURABLE_CLOCK_4: Configurable Clock 4 will be supplied to this channel GTM_CONFIGURABLE_CLOCK_5: Configurable Clock 5 will be supplied to this channel GTM_CONFIGURABLE_CLOCK_6: Configurable Clock 6 will be supplied to this channel GTM_CONFIGURABLE_CLOCK_7: Configurable Clock 7 will be supplied to this channel Default: GTM_CONFIGURABLE_CLOCK_0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	X	Variant PB
Description:	This decides the Clock Source for TIM Channel		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.7.1.1.8 TimChannelGpr0InputSelect

Table 151 TimChannelGpr0InputSelect

Name:	TimChannelGpr0InputSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TIMEBASE_TBU_TS0: TBU_TS0 will be captured in GPR0 TIMEBASE_TBU_TS1: TBU_TS1 will be captured in GPR0 TIMEBASE_TBU_TS2: TBU_TS2 will be captured in GPR0 CNT_VALUE: CNT value will be captured in GPR0 ECNT_VALUE: ECNT value will be captured in GPR0 Default: TIMEBASE_TBU_TS0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the value captured in GPR0 register of TIM Channel		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	When TimChannelUsage = 'USED_BY_OTHER_DRIVERS_OR_UNUSED' i.e. if the channel is used for ICU purpose, CNT_VALUE cannot be configured in this.		

4.2.7.1.1.9 TimChannelGpr1InputSelect

Table 152 TimChannelGpr1InputSelect

Name:	TimChannelGpr1InputSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TIMEBASE_TBU_TS0: TBU_TS0 will be captured in GPR1 TIMEBASE_TBU_TS1: TBU_TS1 will be captured in GPR1 TIMEBASE_TBU_TS2: TBU_TS2 will be captured in GPR1 CNT_VALUE: CNT value will be captured in GPR1 ECNT_VALUE: ECNT value will be captured in GPR1 Default: TIMEBASE_TBU_TS0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
SFR Modified	GTM_TIMi_CHj_CTRL		

Description:	This decides the value captured in GPR1 register of TIM Channel
Scope:	Driver
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.7.1.1.10 TimChannelCntsInputSelect

Table 153 TimChannelCntsInputSelect

Name:	TimChannelCntsInputSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: CNT_VALUE: CNT Value will be captured in CNTS TIMEBASE_TBU_TS0: TBU_TS0 will be captured in CNTS Default: CNT_VALUE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the value captured in CNTS register of TIM Channel. This is not applicable in TIPM and TBCM mode		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER' and TimChannelModeSelect is not TIPM or TBCM mode.		

4.2.7.1.1.11 TimChannelCntsValue

Table 154 TimChannelCntsValue

Name:	TimChannelCntsValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the CNTS Value. This is available only in case of TIPM and TBCM modes		
SFR Modified	GTM_TIMi_CHj_CNTS		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER' and TimChannelModeSelect is not TIPM or TBCM mode.		

4.2.7.1.1.12 TimChannelTbu0Ts0BitsSelect

Table 155 TimChannelTbu0Ts0BitsSelect

Name:	TimChannelTbu0Ts0BitsSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: BITS_0_TO_23: Lower 24 bits will be captured in GPR0/GPR1 BITS_3_TO_26: upper 24 bits will be captured in GPR0/GPR1 Default: BITS_0_TO_23		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the resolution of TBU_TS0 that will be captured in GPR0 and GPR1 register of TIM Channel.		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	Applicable only for TIM0. Also, when any of the following condition is possible: a) TimChannelGpr0InputSelect = 'TIMEBASE_TBU_TS0' b) TimChannelGpr1InputSelect = 'TIMEBASE_TBU_TS0' when TimChannelUsage = 'USED_BY_GTM_DRIVER' and		

4.2.7.1.1.13 TimChannelModeSelect

Table 156 TimChannelModeSelect

Name:	TimChannelModeSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: PWM_MEASUREMENT_MODE_TPWM: TPWM mode is chosen PULSE_INTEGRATION_MODE_TPIM: TPIM mode is chosen INPUT_EVENT_MODE_TIEM: TIEM mode is chosen INPUT_PRESCALER_MODE_TIPM: TIPM mode is chosen BIT_COMPRESSION_MODE_TBCM: TBCM mode is chosen Default: PWM_MEASUREMENT_MODE_TPWM		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Operating Mode for the TIM Channel		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'. TBCM mode is applicable only for TIM Channel 0.		

4.2.7.1.1.14 TimChannelOneShotMode

Table 157 TimChannelOneShotMode

Name:	TimChannelOneShotMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable One Shot Mode FALSE – Disable One Shot Mode Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables One Shot Mode for the Channel.		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.1.15 TimChannelAruEnable

Table 158 TimChannelAruEnable

Name:	TimChannelAruEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable ARU for the channel FALSE - Disable ARU for the channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables ARU for the Channel. This sets bit ARU_EN for the channel		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.1.16 TimChannelExtCaptureModeEnable

Table 159 TimChannelExtCaptureModeEnable

Name:	TimChannelExtCaptureModeEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables the External Capture Mode for the Channel FALSE - Disable the External Capture Mode for the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables External Capture Mode for the Channel.This bit sets EXT_CAP_EN in TIM control register.		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.1.17 TimChannelExtCaptureSrc

Table 160 TimChannelExtCaptureSrc

Name:	TimChannelExtCaptureSrc		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SEL0_NEW_VAL_IRQ: NEW_VAL_IRQ of previous channel selected SEL1_AUX_IN: AUX_IN selected SEL2_CNTOFL_IRQ: CNTOFL_IRQ of previous channel selected SEL3_CICTRL_VALUE: CICTRL=1: use signal TIM_IN(x) as input for channel x and CICTRL=0: use signal TIM_IN(x-1) as input for channel x SEL4_ECNTOTL_IRQ: ECNTOTL_IRQ of previous channel selected SEL5_TODET_IRQ: TODET_IRQ of previous channel selected SEL6_GLITCHDET_IRQ: GLITCHDET_IRQ of previous channel selected SEL7_GPRzOF_IRQof: GPRzOF_IRQof previous channel selected Default: SEL0_NEW_VAL_IRQ		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Operating Mode for the TIM Channel		
SFR Modified	GTM_TIMi_CHj_ECTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'. TimChannelExtCaptureSrc is applicable only if TimChannelExtCaptureModeEnable is true.		

4.2.7.1.1.18 TimChannelEcntResetSetting

Table 161 TimChannelEcntResetSetting

Name:	TimChannelEcntResetSetting		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SEL0_ECNT_IN_OPERATING_MODE: ECNT counter operating in wrap around mode SEL1_ECNT_PERIODIC_SAMPLING_MODE: ECNT counter is reset with periodic sampling Default: SEL0_ECNT_IN_OPERATING_MODE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This Enables resetting the ECNT counter in periodic sampling mode		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'.		

4.2.7.1.1.19 TimChannelEcntOvflwSetting

Table 162 TimChannelEcntOvflwSetting

Name:	TimChannelEcntResetSetting		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SEL0_OVRFLOW_SIG_ON_ECNT_BITWIDTH_8: Overflow will be signaled on ECNT bit width = 8 SEL1_OVRFLOW_SIG_ON_EECNT_BITWIDTH: Overflow will be signaled on EECNT bit width (full range) Default: SEL0_OVRFLOW_SIG_ON_ECNT_BITWIDTH_8		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This Enables resetting the ECNT counter in periodic sampling mode		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'.		

4.2.7.1.2 Container: TimChannelFilterConfig

This holds the filter specific configurations for the channels.

4.2.7.1.2.1 TimChannelFilterEnable

Table 163 TimChannelFilterEnable

Name:	TimChannelFilterEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Filter for the channel FALSE – Disable Filter for the channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Filter for the Channel. This sets FLT_EN for the channel		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.7.1.2.2 TimChFilterCounterFreqSelect

Table 164 TimChFilterCounterFreqSelect

Name:	TimChFilterCounterFreqSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_CONFIGURABLE_CLOCK_0: Configurable Clock 0 clocks the Filter counter GTM_CONFIGURABLE_CLOCK_1: Configurable Clock 1 clocks the Filter counter GTM_CONFIGURABLE_CLOCK_6: Configurable Clock 6 clocks the Filter counter GTM_CONFIGURABLE_CLOCK_7: Configurable Clock 7 clocks the Filter counter Default: Z		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Filter Counter frequency for the TIM Channel. This modifies the FLT_CNT_FRQ for the channel		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.7.1.2.3 TimChFilterModeForRisingEdge

Table 165 TimChFilterModeForRisingEdge

Name:	TimChFilterModeForRisingEdge		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: IMMEDIATE_EDGE_PROPAGATION_MODE: Immediate Edge Propagation Mode is chosen DEGLITCH_WITH_UPDOWN_COUNTER: Deglitch Mode is chosen with Up/Down Counter DEGLITCH_WITH_HOLD_COUNTER: Deglitch mode is chosen with Hold counter Default: IMMEDIATE_EDGE_PROPAGATION_MODE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Filter Mode for Rising Edge of the TIM Channel Input		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.7.1.2.4 TimChFilterModeForFallingEdge

Table 166 TimChFilterModeForFallingEdge

Name:	TimChFilterModeForFallingEdge		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: IMMEDIATE_EDGE_PROPAGATION_MODE: Immediate Edge Propagation Mode is chosen DEGLITCH_WITH_UPDOWN_COUNTER: Deglitch Mode is chosen with Up/Down Counter DEGLITCH_WITH_HOLD_COUNTER: Deglitch mode is chosen with Hold counter Default: Z		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Filter Mode for Falling Edge of the TIM Channel Input		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.7.1.2.5 TimChFilterTimeForRisingEdge

Table 167 TimChFilterTimeForRisingEdge

Name:	TimChFilterTimeForRisingEdge		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Filter Time for Rising Edge of the TIM Channel Input		
SFR Modified	GTM_TIMi_CHj_FLT_RE		
Scope:	Driver		
Dependency:	None		

4.2.7.1.2.6 TimChFilterTimeForFallingEdge

Table 168 TimChFilterTimeForFallingEdge

Name:	TimChFilterTimeForFallingEdge		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Filter Time for Falling Edge of the TIM Channel Input		
SFR Modified	GTM_TIMi_CHj_FLT_FE		
Scope:	Driver		
Dependency:	None		

4.2.7.1.3 Container: TimChannelTimeoutDetection

This container holds the configuration for Timeout detection (TDU) for TIM channel

4.2.7.1.3.1 TimChTimeoutDetectionEnable

Table 169 TimChTimeoutDetectionEnable

Name:	TimChTimeoutDetectionEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SEL0_TIMEOUT_DISABLED: Timeout feature disabled SEL1_TIMEOUT_ENABLED_ON_BOTH_EDGES: Timeout feature enabled for both edges SEL2_TIMEOUT_ENABLED_ON_RISING_EDGES: Timeout feature enabled for rising edge only SEL3_TIMEOUT_ENABLED_ON_FALLING_EDGES: Timeout feature enabled for falling edge only Default: SEL0_TIMEOUT_DISABLED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Timeout detection for the Channel. This sets GTM_TIMi_CHx_TDU[TO_EN] (x=0-7) (i=0-3)		
SFR Modified	GTM_TIMi_CHj_TDU		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.3.2 TimChTimeoutClockSelect

Table 170 TimChTimeoutClockSelect

Name:	TimChTimeoutClockSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_CONFIGURABLE_CLOCK_0: Configurable Clock0 is supplied GTM_CONFIGURABLE_CLOCK_1: Configurable Clock1 is supplied GTM_CONFIGURABLE_CLOCK_2: Configurable Clock2 is supplied GTM_CONFIGURABLE_CLOCK_3: Configurable Clock3 is supplied GTM_CONFIGURABLE_CLOCK_4: Configurable Clock4 is supplied GTM_CONFIGURABLE_CLOCK_5: Configurable Clock5 is supplied GTM_CONFIGURABLE_CLOCK_6: Configurable Clock6 is supplied GTM_CONFIGURABLE_CLOCK_7: Configurable Clock7 is supplied Default: GTM_CONFIGURABLE_CLOCK_0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Timeout Clock for the TIM Channel		
SFR Modified	GTM_TIMi_CHj_CTRL		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.3.3 TimChTimeoutDuration

Table 171 TimChTimeoutDuration

Name:	TimChTimeoutDuration		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 255 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Timeout duration for the TIM Channel		

SFR Modified	GTM_TIMi_CHj_TDUV
Scope:	Driver
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.7.1.4 Container: TimChannelInterrupt

This container contains the Interrupt configuration for the TIM Channel

4.2.7.1.4.1 TimInterruptEnableOnNewVal

Table 172 TimInterruptEnableOnNewVal

Name:	TimInterruptEnableOnNewVal		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Interrupt on Detection of New Measurement Value FALSE – Disable Interrupt on Detection of New Measurement Value Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on Detection of New Measurement Value in the Channel.		
SFR Modified	GTM_TIMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.4.2 TimInterruptEnableOnEcntOfI

Table 173 TimInterruptEnableOnEcntOfI

Name:	TimInterruptEnableOnEcntOfI		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on overflow of Edge Counter Value FALSE - Disables Interrupt on overflow of Edge Counter Value Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on overflow of Edge Counter Value for the Channel		
SFR Modified	GTM_TIMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.4.3 TimInterruptEnableOnCntOfI

Table 174 TimInterruptEnableOnCntOfI

Name:	TimInterruptEnableOnCntOfI		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on overflow of Counter Value FALSE - Disables Interrupt on overflow of Counter Value Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on overflow of Counter Value for the Channel		
SFR Modified	GTM_TIMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.4.4 TimInterruptEnableOnGprxOfI

Table 175 TimInterruptEnableOnGprxOfI

Name:	TimInterruptEnableOnGprxOfI		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on overflow of GPRx Value FALSE - Disables Interrupt on overflow of GPRx Value Default: FALSE		
Configuration Class:	Pre-compile	x	Variant PC
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on overflow of GPRx Value for the Channel		
SFR Modified	GTM_TIMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.4.5 TimInterruptEnableOnToDet

Table 176 TimInterruptEnableOnToDet

Name:	TimInterruptEnableOnToDet		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables Interrupt on Timeout Detection FALSE - Disables Interrupt on Timeout Detection Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on Timeout Detection for the Channel		
SFR Modified	GTM_TIMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.4.6 TimInterruptEnableOnGlitchDet

Table 177 TimInterruptEnableOnGlitchDet

Name:	TimInterruptEnableOnGlitchDet		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Glitch Detection FALSE - Disables Interrupt on Glitch Detection Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on Glitch Detection in the Channel		
SFR Modified	GTM_TIMi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.4.7 TimInterruptMode

Table 178 TimInterruptMode

Name:	TimInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE: LEVEL MODE is used GTM_INTERRUPT_PULSE_MODE: PULSE MODE is used GTM_INTERRUPT_PULSE_NOTIFY_MODE: PULSE NOTIFY MODE is used GTM_INTERRUPT_SINGLE_PULSE_MODE: SINGLE PULSE MODE is used Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use		
SFR Modified	GTM_TIMi_CHj_IRQ_MODE		
Scope:	Driver		
Dependency:	None		

4.2.7.1.4.8 TimNotification

Table 179 TimNotification

Name:	TimNotification		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: String Default: none		
Configuration Class:	Pre-compile	--	--
	Link Time	x	Variant LT
	Post build	--	--
Description:	Notification function name should be mentioned here which has to be called on an interrupt event of the current. The prototype of the notification function should be, void NotifFn(uint8 ModType, uint8 ModuleNo, uint8 ChanNo, uint8 IntSrc) ModType – 0 – TIM 1 – TOM 2 - ATOM ModuleNo – TIM/TOM/ATOM ModuleNo ChanNo - TIM/TOM/ATOM ChanNo IntSrc – InterruptType of TIM/TOM/ATOM In case of Post Build Loadable variant, address of the Notification function need to be provided.		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	None		

4.2.7.1.5 Container: TimChannelErrorInterrupt

This container contains the Error Interrupt configuration for the TIM Channel

4.2.7.1.5.1 TimInterruptEnableOnNewValErr

Table 180 TimInterruptEnableOnNewValErr

Name:	TimInterruptEnableOnNewValErr
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Detection of New Measurement Value in the Channel Error FALSE - Disables Interrupt on Detection of New Measurement Value in the Channel Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on Detection of New Measurement Value in the Channel Error		
SFR Modified	GTM_TIMi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.5.2 TimInterruptEnableOnEcntOfErr

Table 181 TimInterruptEnableOnEcntOfErr

Name:	TimInterruptEnableOnEcntOfErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on overflow of Edge Counter Value for the Channel Error FALSE – Disables Interrupt on overflow of Edge Counter Value for the Channel Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on overflow of Edge Counter Value for the Channel Error		
SFR Modified	GTM_TIMi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.5.3 TimInterruptEnableOnCntOfErr

Table 182 TimInterruptEnableOnCntOfErr

Name:	TimInterruptEnableOnCntOfErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables Interrupt on overflow of Counter Value for the Channel Error FALSE – Disables Interrupt on overflow of Counter Value for the Channel Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on overflow of Counter Value for the Channel Error.		
SFR Modified	GTM_TIMi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.5.4 TimInterruptEnableOnGprxOfErr

Table 183 TimInterruptEnableOnGprxOfErr

Name:	TimInterruptEnableOnGprxOfErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on overflow of GPRx Value for the Channel Error. FALSE – Disables Interrupt on overflow of GPRx Value for the Channel Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on overflow of GPRx Value for the Channel Error.		
SFR Modified	GTM_TIMi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.7.1.5.5 TimInterruptEnableOnToDetErr

Table 184 TimInterruptEnableOnToDetErr

Name:	TimInterruptEnableOnToDetErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Timeout Detection for the Channel Error FALSE – Disables Interrupt on Timeout Detection for the Channel Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables/ Disables Interrupt on Timeout Detection for the Channel Error
SFR Modified	GTM_TIMi_CHj_EIRQ_EN
Scope:	Driver
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.7.1.5.6 TimInterruptEnableOnGlitchDetErr

Table 185 TimInterruptEnableOnGlitchDetErr

Name:	TimInterruptEnableOnGlitchDetErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Glitch Detection in the Channel Error. FALSE – Disables Interrupt on Glitch Detection in the Channel Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables Interrupt on Glitch Detection in the Channel Error		
SFR Modified	GTM_TIMi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	TimChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.8 Container: Aru

This container includes the configuration for ARU. Maximum multiplicity is 1. If this container is not included, then the registers values of ARU are retained as is.

4.2.8.1 Container: AruInterrupt

This container contains the Interrupt configuration for the ARU

4.2.8.1.1 AruNewData0InterruptEnable

Table 186 AruNewData0InterruptEnable

Name:	AruNewData0InterruptEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Interrupt when data transferred using DBG_ACCESS0 FALSE – Disable Interrupt when data transferred using DBG_ACCESS0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enable/disable Interrupt when data transferred using DBG_ACCESS0
SFR Modified	GTM_ARU_IRQ_EN
Scope:	Driver
Dependency:	None

4.2.8.1.2 AruNewData1InterruptEnable

Table 187 AruNewData1InterruptEnable

Name:	AruNewData1InterruptEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Interrupt when data transferred using DBG_ACCESS1 FALSE - Disable Interrupt when data transferred using DBG_ACCESS1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Interrupt when data transferred using DBG_ACCESS1		
SFR Modified	GTM_ARU_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.8.1.3 AruAccAckInterruptEnable

Table 188 AruAccAckInterruptEnable

Name:	AruAccAckInterruptEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Interrupt on AEI to ARU Access completion FALSE - Disable Interrupt on AEI to ARU Access completion Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Interrupt on AEI to ARU Access completion		
SFR Modified	GTM_ARU_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.8.1.4 AruInterruptMode

Table 189 AruInterruptMode

Name:	ArulInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE: LEVEL MODE is chosen GTM_INTERRUPT_PULSE_MODE: PULSE MODE is chosen GTM_INTERRUPT_PULSE_NOTIFY_MODE: PULSE NOTIFY MODE is chosen GTM_INTERRUPT_SINGLE_PULSE_MODE: SINGLE PULSE MODE is chosen Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use		
Scope:	Driver		
SFR Modified	GTM_ARU_IRQ_MODE		
Dependency:	None		

4.2.9 Container: Brc

This container includes the configuration for BRC. Maximum multiplicity is 1. If this container is not included, then the registers values of BRC are retained as is.

4.2.9.1 Container: BrcChannel

This container contains the BRC Channel Configuration. Multiplicity depends on the number of BRC sources available.

4.2.9.1.1 Container: BrcChannelGeneral

4.2.9.1.1.1 BrcChannelUsage

Table 190 BrcChannelUsage

Name:	BrcChannelUsage		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init Default: UNUSED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This decides if the channel will be used by GTM Driver or will be initialized elsewhere. UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init
SFR Modified	N/A
Scope:	Driver
Dependency:	None

4.2.9.1.1.2 BrcSourceAruAddress

Table 191 BrcSourceAruAddress

Name:	BrcSourceAruAddress		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 511 Default: 510		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the ARU read address to be used as data source for the input channel		
SFR Modified	GTM_BRC_SRCj_ADDR		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.1.3 BrcModeSelect

Table 192 BrcModeSelect

Name:	BrcModeSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: CONSISTENCY_MODE_DCM: Consistency Mode(DCM) is chosen MAXIMUM_THROUGHPUT_MODE_MTM: Maximum Throughput Mode (MTM) is chosen Default: CONSISTENCY_MODE_DCM		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Selection of the BRC Mode		
SFR Modified	GTM_BRC_SRCj_ADDR		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2 Container: BrcChannelDestinationSelect

This contains the BRC Channel Destination Selection.

4.2.9.1.2.1 BrcChDestinationAddr0Enable

Table 193 BrcChDestinationAddr0Enable

Name:	BrcChDestinationAddr0Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 0 for this BRC Channel FALSE – Disable Destination Address 0 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 0 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST0] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.2 BrcChDestinationAddr1Enable

Table 194 BrcChDestinationAddr1Enable

Name:	BrcChDestinationAddr1Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 1 for this BRC Channel FALSE – Disable Destination Address 1 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 1 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST1] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.3 BrcChDestinationAddr2Enable

Table 195 BrcChDestinationAddr2Enable

Name:	BrcChDestinationAddr2Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 2 for this BRC Channel FALSE – Disable Destination Address 2 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 2 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST2] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.4 BrcChDestinationAddr3Enable

Table 196 BrcChDestinationAddr3Enable

Name:	BrcChDestinationAddr3Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 3 for this BRC Channel FALSE – Disable Destination Address 3 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 3 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST3] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.5 BrcChDestinationAddr4Enable

Table 197 BrcChDestinationAddr4Enable

Name:	BrcChDestinationAddr4Enable		
Source:	This parameter is microcontroller specific.		

File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 4 for this BRC Channel FALSE – Disable Destination Address 4 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 4 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST4] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.6 BrcChDestinationAddr5Enable

Table 198 BrcChDestinationAddr5Enable

Name:	BrcChDestinationAddr5Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 5 for this BRC Channel FALSE – Disable Destination Address 5 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 5 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST5] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.7 BrcChDestinationAddr6Enable

Table 199 BrcChDestinationAddr6Enable

Name:	BrcChDestinationAddr6Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 6 for this BRC Channel FALSE – Disable Destination Address 6 for this BRC Channel Default: FALSE		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 6 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST6] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.8 BrcChDestinationAddr7Enable

Table 200 BrcChDestinationAddr7Enable

Name:	BrcChDestinationAddr7Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 7 for this BRC Channel FALSE – Disable Destination Address 7 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 7 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST7] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.9 BrcChDestinationAddr8Enable

Table 201 BrcChDestinationAddr8Enable

Name:	BrcChDestinationAddr8Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 8 for this BRC Channel FALSE – Disable Destination Address 8 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 8 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST8] (x=0-11)		

SFR Modified	GTM_BRC_SRCj_DEST
Scope:	Driver
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.9.1.2.10 BrcChDestinationAddr9Enable

Table 202 BrcChDestinationAddr9Enable

Name:	BrcChDestinationAddr9Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 9 for this BRC Channel FALSE – Disable Destination Address 9 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 9 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST9] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.11 BrcChDestinationAddr10Enable

Table 203 BrcChDestinationAddr10Enable

Name:	BrcChDestinationAddr10Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 10 for this BRC Channel FALSE – Disable Destination Address 10 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 10 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST10] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.12 BrcChDestinationAddr11Enable

Table 204 BrcChDestinationAddr11Enable

Name:	BrcChDestinationAddr11Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 11 for this BRC Channel FALSE – Disable Destination Address 11 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 11 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST11] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.13 BrcChDestinationAddr12Enable

Table 205 BrcChDestinationAddr12Enable

Name:	BrcChDestinationAddr12Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 12 for this BRC Channel FALSE – Disable Destination Address 12 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 12 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST12] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.14 BrcChDestinationAddr13Enable

Table 206 BrcChDestinationAddr13Enable

Name:	BrcChDestinationAddr13Enable
Source:	This parameter is microcontroller specific.

File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 13 for this BRC Channel FALSE – Disable Destination Address 13 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 13 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST13] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.15 BrcChDestinationAddr14Enable

Table 207 BrcChDestinationAddr14Enable

Name:	BrcChDestinationAddr14Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 14 for this BRC Channel FALSE – Disable Destination Address 14 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 14 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST14] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.16 BrcChDestinationAddr15Enable

Table 208 BrcChDestinationAddr15Enable

Name:	BrcChDestinationAddr15Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 15 for this BRC Channel FALSE – Disable Destination Address 15 for this BRC Channel Default: FALSE		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 15 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST15] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.17 BrcChDestinationAddr16Enable

Table 209 BrcChDestinationAddr16Enable

Name:	BrcChDestinationAddr16Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 16 for this BRC Channel FALSE – Disable Destination Address 16 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 16 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST16] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.18 BrcChDestinationAddr17Enable

Table 210 BrcChDestinationAddr17Enable

Name:	BrcChDestinationAddr17Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 17 for this BRC Channel FALSE – Disable Destination Address 17 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 17 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST17] (x=0-11)		

SFR Modified	GTM_BRC_SRCj_DEST
Scope:	Driver
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.9.1.2.19 BrcChDestinationAddr18Enable

Table 211 BrcChDestinationAddr18Enable

Name:	BrcChDestinationAddr18Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 18 for this BRC Channel FALSE – Disable Destination Address 18 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 18 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST18] (x=0-11)		
SFR Modified			
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.20 BrcChDestinationAddr19Enable

Table 212 BrcChDestinationAddr19Enable

Name:	BrcChDestinationAddr19Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 19 for this BRC Channel FALSE – Disable Destination Address 19 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 19 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST19] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.21 BrcChDestinationAddr20Enable

Table 213 BrcChDestinationAddr20Enable

Name:	BrcChDestinationAddr20Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 20 for this BRC Channel FALSE – Disable Destination Address 20 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 20 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST20] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.1.2.22 BrcChDestinationAddr21Enable

Table 214 BrcChDestinationAddr21Enable

Name:	BrcChDestinationAddr21Enable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Destination Address 21 for this BRC Channel FALSE – Disable Destination Address 21 for this BRC Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable Destination Address 21 for this BRC channel. By enabling this, Destination address 0 will be mapped to the BRC Channel Source. This sets GTM_BRC_SRCx_DEST[EN_DEST21] (x=0-11)		
SFR Modified	GTM_BRC_SRCj_DEST		
Scope:	Driver		
Dependency:	BrcChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.9.2 Container: BrcInterrupt

This contains the Interrupt configuration for the BRC

4.2.9.2.1 BrcEnableConfigurationErrorInt

Table 215 BrcEnableConfigurationErrorInt

Name:	BrcEnableConfigurationErrorInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Interrupt on Wrong Register Configuration FALSE - Disable Interrupt on Wrong Register Configuration Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Wrong Register Configuration		
SFR Modified	GTM_BRC_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.2 BrcEnSrc0DataInconsistencyErrInt

Table 216 BrcEnSrc0DataInconsistencyErrInt

Name:	BrcEnSrc0DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 0 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 0		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.3 BrcEnSrc1DataInconsistencyErrInt

Table 217 BrcEnSrc1DataInconsistencyErrInt

Name:	BrcEnSrc1DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 1 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 1		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.4 BrcEnSrc2DataInconsistencyErrInt

Table 218 BrcEnSrc2DataInconsistencyErrInt

Name:	BrcEnSrc2ataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 2 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 2		
SFR Modified			
Scope:	Driver		
Dependency:	None		

4.2.9.2.5 BrcEnSrc3DataInconsistencyErrInt

Table 219 BrcEnSrc3DataInconsistencyErrInt

Name:	BrcEnSrc3DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 3 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 3 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 3		

SFR Modified	GTM_BRC_IRQ_EN
Scope:	Driver
Dependency:	None

4.2.9.2.6 BrcEnSrc4DataInconsistencyErrInt

Table 220 BrcEnSrc4DataInconsistencyErrInt

Name:	BrcEnSrc4DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 4 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 4 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC 4		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.7 BrcEnSrc5DataInconsistencyErrInt

Table 221 BrcEnSrc5DataInconsistencyErrInt

Name:	BrcEnSrc5DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 5 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 5 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 5		
SFR Modified	GTM_BRC_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.8 BrcEnSrc6DataInconsistencyErrInt

Table 222 BrcEnSrc6DataInconsistencyErrInt

Name:	BrcEnSrc6DataInconsistencyErrInt
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 6 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 6 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 6		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.9 BrcEnSrc7DataInconsistencyErrInt

Table 223 BrcEnSrc7DataInconsistencyErrInt

Name:	BrcEnSrc7DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 7 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 7 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 7		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.10 BrcEnSrc8DataInconsistencyErrInt

Table 224 BrcEnSrc8DataInconsistencyErrInt

Name:	BrcEnSrc8DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 8 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 8 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 8		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.11 BrcEnSrc9DataInconsistencyErrInt

Table 225 BrcEnSrc9DataInconsistencyErrInt

Name:	BrcEnSrc9DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 9 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 9 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 9		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.12 BrcEnSrc10DataInconsistencyErrInt

Table 226 BrcEnSrc10DataInconsistencyErrInt

Name:	BrcEnSrc10DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 10 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 10 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 10		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.13 BrcEnSrc11DataInconsistencyErrInt

Table 227 BrcEnSrc11DataInconsistencyErrInt

Name:	BrcEnSrc11DataInconsistencyErrInt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enable Interrupt on Data Inconsistency on BRC Source 11 FALSE - Disable Interrupt on Data Inconsistency on BRC Source 11 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/disable Interrupt on Data Inconsistency on BRC Source 11		
SFR Modified	GTM_BRC_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.9.2.14 BrcInterruptMode

Table 228 BrcInterruptMode

Name:	BrcInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE: Level mode is chosen GTM_INTERRUPT_PULSE_MODE: Pulse mode is chosen GTM_INTERRUPT_PULSE_NOTIFY_MODE: Pulse Notify mode is chosen GTM_INTERRUPT_SINGLE_PULSE_MODE: Single Pulse mode is chosen Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use		
SFR Modified	GTM_BRC_IRQ_MODE		
Scope:	Driver		
Dependency:	None		

4.2.9.3 Container: BrcErrorInterrupt

This contains the Error Interrupt configuration for the BRC

4.2.9.3.1 BrcEnableDestErrorIntrpt

Table 229 BrcEnableDestErrorIntrpt

Name:	BrcEnableDestErrorIntrpt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DEST_ERR_EIRQ FALSE - Disables the Destination Error interrupt for BRC_DEST_ERR_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the Destination Error interrupt for BRC_DEST_ERR_EIRQ		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.2 BrcEnableDaInconsistencyDetd0

Table 230 BrcEnableDaInconsistencyDetd0

Name:	BrcEnableDaInconsistencyDetd0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID0 FALSE - Disables the Destination Error interrupt for BRC_DID0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID0		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.3 BrcEnableDaInconsistencyDetd1

Table 231 BrcEnableDaInconsistencyDetd1

Name:	BrcEnableDaInconsistencyDetd1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID1 FALSE - Disables the Destination Error interrupt for BRC_DID1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID1		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.4 BrcEnableDalInconsistencyDetd2

Table 232 BrcEnableDalInconsistencyDetd2

Name:	BrcEnableDalInconsistencyDetd2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID2 FALSE - Disables the Destination Error interrupt for BRC_DID2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID2		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.5 BrcEnableDalInconsistencyDetd3

Table 233 BrcEnableDalInconsistencyDetd3

Name:	BrcEnableDalInconsistencyDetd3		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID3 FALSE - Disables the Destination Error interrupt for BRC_DID3 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables/Disables the Destination Error interrupt for BRC_DID3
SFR Modified	BRC_DEST_ERR_EIRQ
Scope:	Driver
Dependency:	None

4.2.9.3.6 BrcEnableDalInconsistencyDetd4

Table 234 BrcEnableDalInconsistencyDetd4

Name:	BrcEnableDalInconsistencyDetd4		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID4 FALSE - Disables the Destination Error interrupt for BRC_DID4 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID2		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.7 BrcEnableDalInconsistencyDetd5

Table 235 BrcEnableDalInconsistencyDetd5

Name:	BrcEnableDalInconsistencyDetd5		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID5 FALSE - Disables the Destination Error interrupt for BRC_DID5 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID5		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.8 BrcEnableDaInconsistencyDetd6

Table 236 BrcEnableDaInconsistencyDetd2

Name:	BrcEnableDaInconsistencyDetd6		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID6 FALSE - Disables the Destination Error interrupt for BRC_DID6 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID6		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.9 BrcEnableDaInconsistencyDetd7

Table 237 BrcEnableDaInconsistencyDetd7

Name:	BrcEnableDaInconsistencyDetd7		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID7 FALSE - Disables the Destination Error interrupt for BRC_DID7 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID2		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.10 BrcEnableDaInconsistencyDetd8

Table 238 BrcEnableDaInconsistencyDetd8

Name:	BrcEnableDaInconsistencyDetd8		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID8 FALSE - Disables the Destination Error interrupt for BRC_DID8 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID8		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.11 BrcEnableDalInconsistencyDetd9

Table 239 BrcEnableDalInconsistencyDetd9

Name:	BrcEnableDalInconsistencyDetd9		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID9 FALSE - Disables the Destination Error interrupt for BRC_DID9 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID9		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.9.3.12 BrcEnableDalInconsistencyDetd10

Table 240 BrcEnableDalInconsistencyDetd10

Name:	BrcEnableDalInconsistencyDetd10		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID10 FALSE - Disables the Destination Error interrupt for BRC_DID10 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables/Disables the Destination Error interrupt for BRC_DID10
SFR Modified	BRC_DEST_ERR_EIRQ
Scope:	Driver
Dependency:	None

4.2.9.3.13 BrcEnableDalInconsistencyDetd11

Table 241 BrcEnableDalInconsistencyDetd11

Name:	BrcEnableDalInconsistencyDetd11		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE - Enables the Destination Error interrupt for BRC_DID11 FALSE - Disables the Destination Error interrupt for BRC_DID11 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Destination Error interrupt for BRC_DID11		
SFR Modified	BRC_DEST_ERR_EIRQ		
Scope:	Driver		
Dependency:	None		

4.2.10 Container: Fifo

This contains the FIFO configuration. Maximum multiplicity is 2 for TC29x and maximum multiplicity is 1 for all other derivatives. If this container is not included, then the registers values of FIFO are retained as it is.

4.2.10.1 Container: FifoChannel

This provides the FIFO Channel configuration. Multiplicity depends on the number of FIFO channels. Each index corresponds to respective FIFO Channel Number.

4.2.10.1.1 Container: FifoGeneral

This contains the FIFO General configuration

4.2.10.1.1.1 FifoChannelUsage

Table 242 FifoChannelUsage

Name:	FifoChannelUsage
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init Default: UNUSED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if the channel will be used by GTM Driver or will be initialized elsewhere. UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	None		

4.2.10.1.1.2 FifoStartAddress

Table 243 FifoStartAddress

Name:	FifoStartAddress		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 1023 Default: $128 * x$ where x is the FIFO channel Number		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Start Address of FIFO. Default value depends on the FIFO Channel		
SFR Modified	GTM_FIFOi_CHj_START_ADDR		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.1.3 FifoEndAddress

Table 244 FifoEndAddress

Name:	FifoEndAddress		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 1023 Default: $128 * (x+1) - 1$ where x is the FIFO channel Number		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the End Address of FIFO. Default value depends on the FIFO Channel		
SFR Modified	GTM_FIFOi_CHj_END_ADDR		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.1.4 FifoChannelModeSelect

Table 245 FifoChannelModeSelect

Name:	FifoChannelModeSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: FIFO_RING_BUFFER_MODE: Ring Buffer Mode is chosen FIFO_NORMAL_FIFO_OPERATION: Normal FIFO Operation is chosen Default: FIFO_RING_BUFFER_MODE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Mode for the FIFO Channel		
SFR Modified	GTM_FIFO0_CH1_CTRL		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.1.5 FifoUpperWatermarkAddress

Table 246 FifoUpperWatermarkAddress

Name:	FifoUpperWatermarkAddress		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 1023 Default: 96		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Upper Watermark Address for the FIFO Channel. This is applicable only in Normal Operation Mode		
SFR Modified	GTM_FIFOi_CHj_UPPER_WM		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER' and FifoChannelModeSelect = 'FIFO_NORMAL_FIFO_OPERATION'		

4.2.10.1.1.6 FifoLowerWatermarkAddress

Table 247 FifoLowerWatermarkAddress

Name:	FifoLowerWatermarkAddress		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 1023 Default: 32		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Lower Watermark Address for the FIFO Channel. This is applicable only in Normal Operation Mode		
SFR Modified	GTM_FIFOi_CHj_LOWER_WM		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER' and FifoChannelModeSelect = 'FIFO_NORMAL_FIFO_OPERATION'		

4.2.10.1.1.7 FifoGatePointerFromSlw2Lwu

Table 248 FifoGatePointerFromSlw2Lwu

Name:	FifoGatePointerFromSlw2Lwu		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 1023 Default: 96		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Gate Pointer from SLW to LWU. This is applicable only in Ring Buffer Mode		
SFR Modified	GTM_FIFOi_CHj_UPPER_WM		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER' and FifoChannelModeSelect = 'FIFO_NORMAL_FIFO_OPERATION'		

4.2.10.1.1.8 FifoGatePointerFromLwu2Uwe

Table 249 FifoGatePointerFromLwu2Uwe

Name:	FifoGatePointerFromLwu2Uwe		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: 0 to 1023 Default: 32		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Gate Pointer from LWU to UWE. This is applicable only in Ring Buffer Mode		
SFR Modified	GTM_FIFOi_CHj_LOWER_WM		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER' and FifoChannelModeSelect = 'FIFO_NORMAL_FIFO_OPERATION'		

4.2.10.1.1.9 FifoEnDirectRamWriteAccess

Table 250 FifoEnDirectRamWriteAccess

Name:	FifoEnDirectRamWriteAccess		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Direct RAM Write Access to FIFO Regions FALSE - Disable Direct RAM Write Access to FIFO Regions Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Direct RAM Write Access to FIFO Regions		
SFR Modified	GTM_FIFOi_CHj_CTRL		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.1.10 FifoRamAccessHighestPriority

Table 251 FifoRamAccessHighestPriority

Name:	FifoRamAccessHighestPriority		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: FIFO_PORT_ACCESS: FIFO Port Access has higher priority than AEI/IF Access AEI_IF_ACCESS: AEI-IF has higher priority than FIFO port access Default: FIFO_PORT_ACCESS		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This decides the RAM Access priority for the FIFO Channel. This decides if FIFO ports have higher priority or AEI-IF has higher priority in accessing the FIFO FIFO_PORT_ACCESS: FIFO Port Access has higher priority than AEI/IF Access. AEI_IF_ACCESS: AEI-IF has higher priority than FIFO port access
SFR Modified	GTM_FIFOi_CHj_CTRL
Scope:	Driver
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.10.1.2 Container: FifoChannelInterrupt

This container contains the Interrupt configuration for the FIFO Channel

4.2.10.1.2.1 FifoIntEnableOnFifoEmpty

Table 252 FifoIntEnableOnFifoEmpty

Name:	FifoIntEnableOnFifoEmpty		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Fifo Empty Condition FALSE - Disable Interrupt on Fifo Empty Condition Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on Fifo Empty Condition for the Channel		
SFR Modified	GTM_FIFOi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.2.2 FifoIntEnableOnFifoFull

Table 253 FifoIntEnableOnFifoFull

Name:	FifoIntEnableOnFifoFull		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Fifo Full Condition FALSE - Disables Interrupt on Fifo Full Condition Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on Fifo Full Condition for the Channel		

SFR Modified	GTM_FIFOi_CHj_IRQ_EN
Scope:	Driver
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.10.1.2.3 FifoIntEnOnFifoUndrrunLwrWtrmark

Table 254 FifoIntEnOnFifoUndrrunLwrWtrmark

Name:	FifoIntEnOnFifoUndrrunLwrWtrmark		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on FIFO underrun on Lower Watermark Address of the Channel FALSE - Disables Interrupt on FIFO underrun on Lower Watermark Address of the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on FIFO underrun on Lower Watermark Address of the Channel		
SFR Modified	GTM_FIFOi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.2.4 FifoIntEnOnFifoOvrrunUprrWtrmark

Table 255 FifoIntEnOnFifoOvrrunUprrWtrmark

Name:	FifoIntEnOnFifoOvrrunUprrWtrmark		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on FIFO overrun on Upper Watermark Address of the Channel FALSE - Disables Interrupt on FIFO overrun on Upper Watermark Address of the Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on FIFO overrun on Upper Watermark Address of the Channel		
SFR Modified	GTM_FIFOi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.2.5 FifoInterruptMode

Table 256 FifoInterruptMode

Name:	FifoInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE: Level Mode is used GTM_INTERRUPT_PULSE_MODE: Pulse mode is used GTM_INTERRUPT_PULSE_NOTIFY_MODE: Pulse notify mode is used GTM_INTERRUPT_SINGLE_PULSE_MODE: Single Pulse mode is used Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use		
Scope:	Driver		
SFR Modified	GTM_FIFOi_CHj_IRQ_MODE		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.2.6 FifoDmaHysterisisModeEnable

Table 257 FifoDmaHysterisisModeEnable

Name:	FifoDmaHysterisisModeEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables DMA Hysterisis Mode FALSE - Disables DMA Hysterisis Mode Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables DMA Hysterisis Mode for the Channel		
SFR Modified	GTM_FIFOi_CHj_IRQ_MODE		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.2.7 FifoHysterisModeDmaDirection

Table 258 FifoHysterisModeDmaDirection

Name:	FifoHysterisModeDmaDirection
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: DMA_DIRECTION_READ: In the case of DMA reading data from FIFO the DMA requests must be generated by the upper watermark. If the DMA hysteresis is enabled, the FIFO does not generate a new DMA request until the lower watermark is reached. DMA_DIRECTION_WRITE: In the case of DMA writing data to a FIFO the DMA requests must be generated by the lower watermark. If the DMA hysteresis is enabled, the FIFO does not generate a new DMA request until the upper watermark is reached Default: DMA_DIRECTION_READ		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if FIFO Read or Write is requested from DMA access. DMA_DIRECTION_WRITE: In the case of DMA writing data to a FIFO the DMA requests must be generated by the lower watermark. If the DMA hysteresis is enabled, the FIFO does not generate a new DMA request until the upper watermark is reached. DMA_DIRECTION_READ: In the case of DMA reading data from FIFO the DMA requests must be generated by the upper watermark. If the DMA hysteresis is enabled, the FIFO does not generate a new DMA request until the lower watermark is reached		
SFR Modified	GTM_FIFOi_CHj_IRQ_MODE		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.3 Container: FifoChannelErrorInterrupt

This container contains the Error Interrupt configuration for the FIFO Channel

4.2.10.1.3.1 FifoIntEnableOnFifoFullErr

Table 259 FifoIntEnableOnFifoFullErr

Name:	FifoIntEnableOnFifoFullErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Fifo Full Condition for the Channel Error FALSE - Disable Interrupt on Fifo Full Condition for the Channel Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables/Disable Interrupt on Fifo Full Condition for the Channel Error
SFR Modified	GTM_FIFOi_CHj_EIRQ_EN
Scope:	Driver
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.10.1.3.2 FifoIntEnOnFifoUndrrunLwrWtrmarkErr

Table 260 FifoIntEnOnFifoUndrrunLwrWtrmarkErr

Name:	FifoIntEnOnFifoUndrrunLwrWtrmarkErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on FIFO underrun on Lower Watermark Address of the Channel Error. FALSE - Disable Interrupt on FIFO underrun on Lower Watermark Address of the Channel Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable Interrupt on FIFO underrun on Lower Watermark Address of the Channel Error.		
SFR Modified	GTM_FIFOi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.10.1.3.3 FifoIntEnOnFifoUndrrunUprWtrmarkErr

Table 261 FifoIntEnOnFifoUndrrunUprWtrmarkErr

Name:	FifoIntEnOnFifoUndrrunUprWtrmarkErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on FIFO underrun on Upper Watermark Address of the Channel Error. FALSE - Disable Interrupt on FIFO underrun on Upper Watermark Address of the Channel Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable Interrupt on FIFO underrun on Lower Watermark Address of the Channel Error.		

SFR Modified	GTM_FIFOi_CHj_EIRQ_EN
Scope:	Driver
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'

4.2.10.1.3.4 FifoIntEnableOnFifoEmptyErr

Table 262 FifoIntEnableOnFifoEmptyErr

Name:	FifoIntEnableOnFifoEmptyErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Fifo Empty Condition for the Channel Error FALSE - Disable Interrupt on Fifo Empty Condition for the Channel Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable Interrupt on Fifo Empty Condition for the Channel Error		
SFR Modified	GTM_FIFOi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	FifoChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.11 Container: F2a

This container contains the Fifo to AEI Interface Configuration. Maximum multiplicity is 2 for TC29x and maximum multiplicity is 1 for all other derivatives. If this container is not included, then the registers values of F2A are retained as is.

4.2.11.1 Container: F2aChannel

This container contains the Fifo to AEI Interface Channel Configuration. Multiplicity depends on number of F2A channel. Each index corresponds to respective F2A Channel Number.

4.2.11.1.1 F2aChannelUsage

Table 263 F2aChannelUsage

Name:	F2aChannelUsage
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c
Range:	Values: UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init Default: UNUSED

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if the channel will be used by GTM Driver or will be initialized elsewhere. UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	None		

4.2.11.1.2 F2aChannelEnable

Table 264 F2aChannelEnable

Name:	F2aChannelEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the F2A Channel FALSE - Disables the F2A Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables the F2A Channel		
SFR Modified	GTM_F2A0_ENABLE		
Scope:	Driver		
Dependency:	F2aChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.11.1.3 F2aAruReadAddress

Table 265 F2aAruReadAddress

Name:	F2aAruReadAddress		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 511 Default: 510		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the ARU Read Address for the F2A Channel		
SFR Modified	GTM_F2A0_RD_CH0_ARU_RD_FIFO		
Scope:	Driver		

Dependency:	None
-------------	------

4.2.11.1.4 F2aAruDataTransferMode

Table 266 F2aAruDataTransferMode

Name:	F2aAruDataTransferMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: F2A_TRANSFER_LOWER_ARU_WORD: Transfer low word (ARU bits 23:0) from/to FIFO F2A_TRANSFER_UPPER_ARU_WORD: Transfer high word (ARU bits 47:24) from/to FIFO F2A_TRANSFER_BOTH_ARU_WORDS: Transfer both words from/to FIFO Default: F2A_TRANSFER_LOWER_ARU_WORD		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Transfer mode for 53 bit ARU data from/to FIFO. F2A_TRANSFER_LOWER_ARU_WORD: Transfer low word (ARU bits 23:0) from/to FIFO F2A_TRANSFER_UPPER_ARU_WORD: Transfer high word (ARU bits 47:24) from/to FIFO F2A_TRANSFER_BOTH_ARU_WORDS: Transfer both words from/to FIFO		
SFR Modified	GTM_F2A0_STR_CH0_STR_CFG		
Scope:	Driver		
Dependency:	F2aChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.11.1.5 F2aAruDataTransferDirection

Table 267 F2aAruDataTransferDirection

Name:	F2aAruDataTransferDirection		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: F2A_TRANSFER_ARU_TO_FIFO: Transport from ARU to FIFO F2A_TRANSFER_FIFO_TO_ARU: Transport from FIFO to ARU Default: F2A_TRANSFER_ARU_TO_FIFO		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the ARU Transfer direction needed for this F2A Channel i.e. ARU to FIFO or FIFO to ARU.		
SFR Modified	GTM_F2A0_STR_CH0_STR_CFG		
Scope:	Driver		
Dependency:	F2aChannelUsage = 'USED_BY_GTM_DRIVER'		

4.2.12 Container: Mcs

This container provides the Multi Channel Sequencer configuration. Multiplicity depends on the number of MCS modules. Each index corresponds to respective MCS module Number.

4.2.12.1 Container: McsGeneral

This container provides the General configuration for MCS

4.2.12.1.1 McsUsage

Table 268 McsUsage

Name:	McsUsage		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will used by Gtm Driver and will be initialized in Gtm_Init Default: USED_BY_OTHER_DRIVERS_OR_UNUSED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if the channel will be used by GTM Driver or will be initialized elsewhere. UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will used by Gtm Driver and will be initialized in Gtm_Init		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	None		

4.2.12.1.2 McsSchedulingScheme

Table 269 McsSchedulingScheme

Name:	McsSchedulingScheme		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: ACCELERATED_SCHEDULING_SCHEME: Accelerated Scheduling Scheme is used ROUND_ROBIN_SCHEDULING_SCHEME: Round Robin Scheduling Scheme Default: ACCELERATED_SCHEDULING_SCHEME		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This decides the Scheduling scheme to be used for the MCS Module
SFR Modified	GTM_MCSi_CTRL
Scope:	Driver
Dependency:	McsUsage = 'USED_BY_GTM_DRIVER'

4.2.12.1.3 McsHaltOnStackPointerOverflow

Table 270 McsHaltOnStackPointerOverflow

Name:	McsHaltOnStackPointerOverflow		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – MCS-channel will be disabled if a stack pointer counter over/underflow occurs FALSE - MCS-channel will not be disabled if a stack pointer counter over/underflow occurs Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	If enabled, MCS-channel will be disabled if a stack pointer counter over/underflow occurs		
SFR Modified	GTM_MCSi_CTRL		
Scope:	Driver		
Dependency:	McsUsage = 'USED_BY_GTM_DRIVER'		

4.2.12.2 Container: McsChannel

This container provides the Channel configuration for MCS. Multiplicity depends on the number of MCS channels. Each index corresponds to respective MCS Channel Number.

4.2.12.2.1 McsChannelEnable

Table 271 McsChannelEnable

Name:	McsChannelEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable the MCS Channel FALSE - Disable the MCS Channel Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables the MCS Channel		
SFR Modified	GTM_MCSi_CTRL		

Scope:	Driver
Dependency:	McsUsage = 'USED_BY_GTM_DRIVER'

4.2.12.2.2 McsChannelInterruptEnable

Table 272 McsChannelInterruptEnable

Name:	McsChannelInterruptEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable the MCS Channel Interrupt FALSE - Disable the MCS Channel Interrupt Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables the MCS Channel Interrupt		
SFR Modified	GTM_MCSi_CHj_IRQ_EN		
Scope:	Driver		
Dependency:	McsUsage = 'USED_BY_GTM_DRIVER'		

4.2.12.2.3 McsChStackErrorInterruptEnable

Table 273 McsChStackErrorInterruptEnable

Name:	McsChStackErrorInterruptEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the MCS Channel Stack Overflow/Underflow Interrupt FALSE - Disables the MCS Channel Stack Overflow/Underflow Interrupt Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ Disables the MCS Channel Stack Overflow/Underflow Interrupt		
SFR Modified	GTM_MCSi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	McsUsage = 'USED_BY_GTM_DRIVER'		

4.2.12.2.4 McsChMemoryErrorInterruptEnable

Table 274 McsChMemoryErrorInterruptEnable

Name:	McsChMemoryErrorInterruptEnable
Source:	This parameter is microcontroller specific.

File:	Gtm_PBCfg.c		
Range:	Values: TRUE – FALSE Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/ disables the MCS Channel Memory Access Out of Range Interrupt		
SFR Modified	GTM_MCSi_CHj_EIRQ_EN		
Scope:	Driver		
Dependency:	McsUsage = 'USED_BY_GTM_DRIVER'		

4.2.12.2.5 McsChannelInterruptMode

Table 275 McsChannelInterruptMode

Name:	McsChannelInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE: Level Mode is used GTM_INTERRUPT_PULSE_MODE: Pulse mode is used GTM_INTERRUPT_PULSE_NOTIFY_MODE: Pulse Notify mode is used GTM_INTERRUPT_SINGLE_PULSE_MODE: Single Pulse mode is used Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use		
SFR Modified	GTM_MCSi_CHj_EIRQ_MODE		
Scope:	Driver		
Dependency:	McsUsage = 'USED_BY_GTM_DRIVER'		

4.2.13 Container: Mcfg

This contains the Memory Configuration. Maximum multiplicity is 1. If this container is not included, then the registers values of MCFG are retained as is.

4.2.13.1 Container: McfgMemConfigForMcs

This contains the Memory Configuration for the MCS Channels. Multiplicity depends on the number of MCS modules. Each index corresponds to respective MCS module.

4.2.13.1.1 McfgMemoryLayoutConfiguration

Table 276 McfgMemoryLayoutConfiguration

Name:	McfgMemoryLayoutConfiguration		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: DEFAULT: For DEFAULT settings SWAP: To SWAP BORROW: To BORROW Default: DEFAULT		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the MCS0 Memory Configuration.		
SFR Modified	GTM_MCFG_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14 Container: Map

This contains the MAP Configuration. Maximum multiplicity is 1. If this container is not included, then the registers values of MAP are retained as is.

4.2.14.1 Container: MapTriggerSignal

This contains the Trigger Signal Configuration

4.2.14.1.1 MapTriggerSignalSelect

Table 277 MapTriggerSignalSelect

Name:	MapTriggerSignalSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_TIM0_CHANNEL0: Input processed by TIM0 Channel 0 will be used as the Trigger Signal TSPP0_SUBUNIT_OUTPUT: TSPP0 processed output will be used as the Trigger Signal Default: GTM_TIM0_CHANNEL0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	This decides the Channel output to be used as the Trigger Signal TIM0_CHANNEL_0: Input processed by TIM Channel 0 will be used as the Trigger Signal TSPP0_SUBUNIT_OUTPUT: TSPP0 processed output will be used as the Trigger Signal
SFR Modified	GTM_MAP_CTRL
Scope:	Driver
Dependency:	None

4.2.14.1.2 MapEnableTspp0SubUnit

Table 278 MapEnableTspp0SubUnit

Name:	MapEnableTspp0SubUnit		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable TSPP0 Sub-Unit FALSE – Disable TSPP0 Sub-Unit Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the TSPP0 Sub-Unit		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.1.3 MapDisableTim0Ch0InputToTspp0

Table 279 MapDisableTim0Ch0InputToTspp0

Name:	MapDisableTim0Ch0InputToTspp0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Disable TIM0 Channel 0 Input line to TSPP0 FALSE – Enable TIM0 Channel 0 Input line to TSPP0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the TIM0 Channel 0 Input line to TSPP0		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.1.4 MapDisableTim0Ch1InputToTspp0

Table 280 MapDisableTim0Ch1InputToTspp0

Name:	MapDisableTim0Ch1InputToTspp0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Disable the TIM0 Channel 1 Input line to TSPP0 FALSE - Enable the TIM0 Channel 1 Input line to TSPP0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the TIM0 Channel 1 Input line to TSPP0		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.1.5 MapDisableTim0Ch2InputToTspp0

Table 281 MapDisableTim0Ch2InputToTspp0

Name:	MapDisableTim0Ch2InputToTspp0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Disable the TIM0 Channel 2 Input line to TSPP0 FALSE - Enable the TIM0 Channel 2 Input line to TSPP0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/disable the TIM0 Channel 2 Input line to TSPP0		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.1.6 MapTim0Ch6InputLevelSel

Table 282 MapTim0Ch6InputLevelSel

Name:	MapTim0Ch6InputLevelSel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TIM0_IN6_LEVEL0: STIM0_IN6 input level '0' encodes TRIGGER in forward direction TIM0_IN6_LEVEL1: STIM0_IN6 input level '1' encodes TRIGGER in forward direction Default: TIM0_IN6_LEVEL0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Decides the Level encoding Selection for forward direction in TIM0_IN6.		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		

4.2.14.1.7 MapTspp0DirSignalLevelDefinition

Table 283 MapTspp0DirSignalLevelDefinition

Name:	MapTspp0DirSignalLevelDefinition		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SPE0_DIR: Signal is routed through as is INVERT_OF_SPE0_DIR: SPEx_DIR signal is inverted and used Default: SPE0_DIR		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Level Definition for the Direction Signal. SPE0_DIR: SPE0_DIR Signal is routed through as is. INVERT_OF_SPE0_DIR: SPE0_DIR signal is inverted and used		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.2 Container: MapStateSignal

This contains the State Signal Configuration

4.2.14.2.1 MapStateSignalSelect

Table 284 MapStateSignalSelect

Name:	MapStateSignalSelect
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: GTM_TIM0_CHANNEL1: Input processed by TIM0 Channel 1 will be used as the State Signal GTM_TIM0_CHANNEL2: Input processed by TIM0 Channel 2 will be used as the State Signal GTM_TIM0_CHANNEL3: Input processed by TIM0 Channel 3 will be used as the State Signal GTM_TIM0_CHANNEL4: Input processed by TIM0 Channel 4 will be used as the State Signal GTM_TIM0_CHANNEL5: Input processed by TIM0 Channel 5 will be used as the State Signal TSPP1_SUBUNIT_OUTPUT: TSPP1 processed output will be used as the Trigger Signal Default: GTM_TIM0_CHANNEL1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Channel output to be used as the State Signal		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.2.2 MapEnableTspp1SubUnit

Table 285 MapEnableTspp1SubUnit

Name:	MapEnableTspp1SubUnit		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable TSPP1 Sub-Unit FALSE – Disable TSPP1 Sub-Unit Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the TSPP1 Sub-Unit		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.2.3 MapDisableTim0Ch3InputToTspp1

Table 286 MapDisableTim0Ch3InputToTspp1

Name:	MapDisableTim0Ch3InputToTspp1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Disable the TIM0 Channel 3 Input line to TSPP1 FALSE - Enable the TIM0 Channel 3 Input line to TSPP1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the TIM0 Channel 3 Input line to TSPP1		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.2.4 MapDisableTim0Ch4InputToTspp1

Table 287 MapDisableTim0Ch4InputToTspp1

Name:	MapDisableTim0Ch4InputToTspp1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Disable the TIM0 Channel 4 Input line to TSPP1 FALSE - Enable the TIM0 Channel 4 Input line to TSPP1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the TIM0 Channel 4 Input line to TSPP1		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.14.2.5 MapDisableTim0Ch5InputToTspp1

Table 288 MapDisableTim0Ch5InputToTspp1

Name:	MapDisableTim0Ch5InputToTspp1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Disable the TIM0 Channel 5 Input line to TSPP1 FALSE - Enable the TIM0 Channel 5 Input line to TSPP1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enable/Disable the TIM0 Channel 5 Input line to TSPP1		

SFR Modified	GTM_MAP_CTRL
Scope:	Driver
Dependency:	None

4.2.14.2.6 MapTspp1DirSignalLevelDefinition

Table 289 MapTspp1DirSignalLevelDefinition

Name:	MapTspp1DirSignalLevelDefinition		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SPE1_DIR: SPE1_DIR Signal is routed through as is INVERT_OF_SPE1_DIR: SPE1_DIR signal is inverted and used. Default: SPE1_DIR		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Level Definition for the Direction Signal SPE1_DIR: SPE1_DIR Signal is routed through as is. INVERT_OF_SPE1_DIR: SPE1_DIR signal is inverted and used		
SFR Modified	GTM_MAP_CTRL		
Scope:	Driver		
Dependency:	None		

4.2.15 Container: Dpll

Configuration for the DPLL Sub-Module. Maximum multiplicity is 1. If this container is not included, then the registers values of DPLL are retained as is.

4.2.15.1 Container: DpllGeneral

General Configuration for DPLL Sub-Module.

4.2.15.1.1 DpllEnable

Table 290 DpllEnable

Name:	DpllEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the DPLL Sub-Module FALSE - Disables the DPLL Sub-Module Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	Enables the DPLL Sub-Module		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.1.2 DpllReferenceMode

Table 291 DpllReferenceMode

Name:	DpllReferenceMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: NORMAL_MODE – Dpll work in Normal Mode EMERGENCY_MODE – Dpll work in Emergency Mode. Default: NORMAL_MODE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the input for Sub_Inc1 Generator. This updates GTM_DPLL_CTRL_0[RMO] NORMAL_MODE: the signal TRIGGER is used to generate the SUB_INC1 signals EMERGENCY_MODE: For SMC=0; signal STATE is used to generate the SUB_INC1 signals; For SMC=1: signal TRIGGER is used to generate the SUB_INC1 signals and STATE is used to generate the SUB_INC2 signals		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.1.3 DpllModeAndCorrectionStrategySel

Table 292 DpllModeAndCorrectionStrategySel

Name:	DpllModeAndCorrectionStrategySel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: AUTOENDMODE_NEW_PULSES_DELAY AUTOENDMODE_WITH_IMMED_PULSES CONTINUOUS_MODE Default: AUTOENDMODE_NEW_PULSES_DELAY		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Mode Selection and Correction Strategy for DPLL AUTOENDMODE_NEW_PULSES_DELAY : This makes DPLL_CTRL_1[DMO] = 0 AND DPLL_CTRL_1[COA] = 0 Automatic End Mode chosen with the following Correction Strategy. The maximum pulse frequency of the system clock will be used to make up for missing pulses from last increment; the output of the calculated new pulses will be delayed accordingly and the FFs in the pulse generation unit will be reset before sending new pulses AUTOENDMODE_WITH_IMMED_PULSES: This makes DPLL_CTRL_1[DMO] = 0 AND DPLL_CTRL_1[COA] = 1 Automatic End Mode chosen with the following Correction Strategy. Missing pulses of the last increment are distributed evenly to the next increment, calculations are done when the next valid input event appears. The number of missing sub-pulses will be determined by the pulse counter difference between the last two valid TRIGGER/STATE events respectively; the FFs in the pulse generation unit are not reset before sending new pulses. CONTINUOUS_MODE: Continuous Mode Chosen. This makes DPLL_CTRL_1[DMO] = 1
SFR Modified	GTM_DPLL_CTRL_1
Scope:	Driver
Dependency:	None

4.2.15.1.4 DpllInputFilterPositionContent

Table 293 DpllInputFilterPositionContent

Name:	DpllInputFilterPositionContent		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TIME_RELATED_VALUES POSITION_RELATED_VALUES Default: TIME_RELATED_VALUES		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Decides if TRIGGER_FT and STATE_FT contain Time Related or Position Related information. TIME_RELATED_VALUES: This makes DPLL_CTRL_0 [IFP] = 0. This would mean TRIGGER_FT and STATE_FT stores the number of time stamp clocks. POSITION_RELATED_VALUES: This makes DPLL_CTRL_0[IFP] = 1. This would mean TRIGGER_FT and STATE_FT stores the number of SUB_INC1 (or SUB_INC2 in the case SMC=1) pulses respectively		

SFR Modified	DPLL_CTRL_0
Scope:	Driver
Dependency:	None

4.2.15.1.5 DpllSynchronousMotorControl

Table 294 DpllSynchronousMotorControl

Name:	DpllSynchronousMotorControl		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRIGGER_INPUT_NOT_USED_FOR_SMC TRIGGER_IS_COMBINED_SENSOR_SIG. Default: TRIGGER_INPUT_NOT_USED_FOR_SMC		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	TRIGGER_INPUT_NOT_USED_FOR_SMC: This makes DPLL_CTRL_0[SMC] = 0. This would mean the TRIGGER input is not used for SMC. TRIGGER_IS_COMBINED_SENSOR_SIG: This makes DPLL_CTRL_0[SMC] = 1. This would mean the TRIGGER input reflects a combined sensor signal for SMC		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.1.6 DpllInputDirDetStrategy

Table 295 DpllInputDirDetStrategy

Name:	DpllInputDirDetStrategy		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – The input direction is detected using TDIR input signal also in the case SMC=0 FALSE - The input direction is detected comparing the THMI value with the duration between valid and invalid slope of TRIGGER Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
SFR Modified	GTM_DPLL_CTRL_1		
Description:	This parameter specifies the input detection method.		
Scope:	Driver		

Dependency:	DpllSynchronousMotorControl == TRIGGER_INPUT_NOT_USED_FOR_SMC
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4.2.15.1.7 DpllLockingCondition

Table 296 DpllLockingCondition

Name:	DpllLockingCondition		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – locking condition definition is one times missing TRIGGERS as expected by the profile in HALF_SCALE (one gap) FALSE - locking condition definition is n-1 times missing TRIGGERS as expected by the profile in HALF_SCALE (one additional tooth) Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the locking condition method.		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.2 Container: DpllTriggerSignal

Configuration for DPLL Sub-Module Trigger Signal.

4.2.15.2.1 Container: DpllTriggerGeneral

General Configuration for DPLL Sub-Module Trigger Signal

4.2.15.2.1.1 DpllTriggerSignalEnable

Table 297 DpllTriggerSignalEnable

Name:	DpllTriggerSignalEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enable Trigger Signal for DPLL FALSE - Disable Trigger Signal for DPLL Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Trigger Signal for DPLL. This affects DPLL_CTRL_0[TEN]		
SFR Modified	GTM_DPLL_CTRL_0		

Scope:	Driver
Dependency:	None

4.2.15.2.1.2 DpllTriggerSignalActiveEdge

Table 298 DpllTriggerSignalActiveEdge

Name:	DpllTriggerSignalActiveEdge		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: NONE : None of the Edge Information will be used. This makes DPLL_CTRL_1[TSL] = 0 RISING_EDGE: Rising Edge Information will be used. This makes DPLL_CTRL_1[TSL] = 1 FALLING_EDGE: Falling Edge Information will be used. This makes DPLL_CTRL_1[TSL] = 2 BOTH_EDGES: Both the Edge Information will be used. This makes DPLL_CTRL_1[TSL] = 3 Default: FALLING_EDGE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Decides the Active Edge for the Trigger Signal. NONE : None of the Edge Information will be used. This makes DPLL_CTRL_1[TSL] = 0 RISING_EDGE: Rising Edge Information will be used. This makes DPLL_CTRL_1[TSL] = 1 FALLING_EDGE: Falling Edge Information will be used. This makes DPLL_CTRL_1[TSL] = 2 BOTH_EDGES: Both the Edge Information will be used. This makes DPLL_CTRL_1[TSL] = 3		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.3 DpllTriggerEventsPerHalfScale

Table 299 DpllTriggerEventsPerHalfScale

Name:	DpllTriggerEventsPerHalfScale		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 1 to 511 Default: 60		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	This parameter specifies the number of Trigger Events per Half Scale. This updates the Register content as follows $DPLL_CTRL_0[TNU] = (DPLLTriggerEventsPerHalfScale - 1)$		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.4 DpllVirtualIncrRange

Table 300 DpllVirtualIncrRange

Name:	DpllVirtualIncrRange		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: VIRT_INCR_FOR_HALF_SCALE : Virtual Increment count in DPLLVirtualIncrementCount is applicable for Half Scale VIRT_INCR_FOR_FULL_SCALE : Virtual Increment count in DPLLVirtualIncrementCount is applicable for Full Scale Default: VIRT_INCR_FOR_HALF_SCALE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	his value decides if DPLLVirtualIncrementCount means the sum of all systematic missing STATE events in HALF_SCALE or FULL_SCALE VIRT_INCR_FOR_HALF_SCALE : Virtual Increment count in DPLLVirtualIncrementCount is applicable for Half Scale VIRT_INCR_FOR_FULL_SCALE : Virtual Increment count in DPLLVirtualIncrementCount is applicable for Full Scale This updates GTM_DPLL_CTRL_1[SYSF]		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.5 DpllVirtualIncrementCount

Table 301 DpllVirtualIncrementCount

Name:	DpllVirtualIncrementCount		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0– 31 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	This parameter specifies the number of virtual increments, sum of all systematic missing STATE events in HALF_SCALE (for DPLLVirtualIncrRange =VIRT_INCR_FOR_HALF_SCALE) or FULL_SCALE (for DPLLVirtualIncrRange =VIRT_INCR_FOR_FULL_SCALE). This updates GTM_DPLL_CTRL_1[SYN_NT]		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.6 DpllTrigSigTimeStampResolution

Table 302 DpllTrigSigTimeStampResolution

Name:	DpllTrigSigTimeStampResolution		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SAME_AS_TBU_TS0 : The TRIGGER input time stamp resolution is same as the resolution of the used DPLL input TBU_TS0 bits. This makes DPLL_CTRL_1[TS0_HRT] = 0. IS_8_TIMES_MORE_THAN_TBU_TS0: The TRIGGER input time stamps has an 8 times higher resolution as the TBU_TS0 input. Default: SAME_AS_TBU_TS0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Decides the TRIGGER input time stamp resolution with respect to the resolution of the used DPLL input TBU_TS0 bits. SAME_AS_TBU_TS0 : The TRIGGER input time stamp resolution is same as the resolution of the used DPLL input TBU_TS0 bits. This makes DPLL_CTRL_1[TS0_HRT] = 0. IS_8_TIMES_MORE_THAN_TBU_TS0: The TRIGGER input time stamps has an 8 times higher resolution as the TBU_TS0 input. This makes DPLL_CTRL_1[TS0_HRT] = 1.		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.7 DpllTriggerUseAdaptMode

Table 303 DpllTriggerUseAdaptMode

Name:	DpllTriggerUseAdaptMode
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: TRUE – Use Adapt Mode FALSE – Do not use Adapt Mode Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Use of adaptation information of TRIGGER. This makes DPLL_CTRL_0[AMT] = 1. By Enabling this, the values ADT_T[i] are considered to calculate the SUB_INC1 pulses in normal mode and for SMC=1		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.8 DpllTriggerUseDelayInfo

Table 304 DpllTriggerUseDelayInfo

Name:	DpllTriggerUseDelayInfo		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Use Delay Information FALSE – Do not use Delay Information Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Use of input delay information transmitted in FT part of the TRIGGER signal. This makes DPLL_CTRL_0[IDT] = 1. By Enabling this, Up to 24 bits of the FT part contain the delay value of the input signal, concerning the corresponding edge		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.9 DpllPlausibilityValueContent

Table 305 DpllPlausibilityValueContent

Name:	DpllPlausibilityValueContent		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: POSITION_RELATED_INFO: The plausibility value is position related (PVT contains the number of SUB_INC1 pulses). This makes DPLL_CTRL_1[PIT] = 0. TIME_RELATED_INFO: The plausibility value is time related (the PVT value is to be multiplied with the expected duration of the current increment CDT_T[i] and divided by 1024). This makes DPLL_CTRL_1[PIT] = 1 Default: POSITION_RELATED_INFO		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Decides the Info held in Plausibility value PVT to next valid TRIGGER. POSITION_RELATED_INFO: The plausibility value is position related (PVT contains the number of SUB_INC1 pulses). This makes DPLL_CTRL_1[PIT] = 0. TIME_RELATED_INFO: The plausibility value is time related (the PVT value is to be multiplied with the expected duration of the current increment CDT_T[i] and divided by 1024). This makes DPLL_CTRL_1[PIT] = 1		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.10 DpllTrigIsNuteEqualToFullScale

Table 306 DpllTrigIsNuteEqualToFullScale

Name:	DpllTrigIsNuteEqualToFullScale		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – NUTE is equal to FULL SCALE FALSE – NUTE is not equal to FULL SCALE Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if NUTE must be equal to FULL_SCALE. This sets GTM_DPLL_NUTC[FST]. TRUE: NUTE is Equal to FULL SCALE FALSE: NUTE is less than FULL SCALE		
SFR Modified	GTM_DPLL_NUTC		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.11 DpllTriggerSignalNuteValue

Table 307 DpllTriggerSignalNuteValue

Name:	DpllTriggerSignalNuteValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 1023 Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the number of last nominal increments to be considered for the calculations. This sets GTM_DPLL_NUTC[WNUT] and updates GTM_DPLL_NUTC[NUTE]		
SFR Modified	GTM_DPLL_NUTC		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.12 DpllTriggerSignalVtnValue

Table 308 DpllTriggerSignalVtnValue

Name:	DpllTriggerSignalVtnValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 63 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Number of virtual increments in the current NUTE. This sets GTM_DPLL_NUTC[WVTN] and updates GTM_DPLL_NUTC[VTN]		
SFR Modified	GTM_DPLL_NUTC		
Scope:	Driver		
Dependency:	None		

4.2.15.2.1.13 DpllTriggerSignalNtiCntValue

Table 309 DpllTriggerSignalNtiCntValue

Name:	DpllTriggerSignalNtiCntValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: 0 to 1023 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Number of active TRIGGER events after which DPLL_CDTI interrupt will be triggered. Updates GTM_DPLL_NTI_CNT[NTI_CNT]		
SFR Modified	GTM_DPLL_NTI_CNT		
Scope:	Driver		
Dependency:	None		

4.2.15.2.2 Container: DpllActualRamPointerAddress

General Configuration for DPLL Sub-Module Trigger Signal

4.2.15.2.2.1 DpllActRamPtrAddOffForDtAndRdt

Table 310 DpllActRamPtrAddOffForDtAndRdt

Name:	DpllActRamPtrAddOffForDtAndRdt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 1023 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Actual RAM pointer address value offset for DT_T[i] and RDT_T[i] in FULL_SCALE for 2*(TNU+1-SYN_NT) TRIGGER events. This pointer is used for the RAM region 2 subsections 2a and 2d. This sets GTM_DPLL_APT[WAPT] and updates GTM_DPLL_APT[APT]		
SFR Modified	GTM_DPLL_APT		
Scope:	Driver		
Dependency:	None		

4.2.15.2.2.2 DpllActRamPtrAddOffForTsf

Table 311 DpllActRamPtrAddOffForTsf

Name:	DpllActRamPtrAddOffForTsf		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 255 Default: 0		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Actual RAM pointer address value for TSF_T[i]. Actual RAM pointer address of TRIGGER events in FULL_SCALE for 2*(TNU+1) TRIGGER periods; this pointer is used for the RAM region 2b. This sets GTM_DPLL_APT[WAPT_2b] and updates GTM_DPLL_APT[APT_2b]		
SFR Modified	GTM_DPLL_APT		
Scope:	Driver		
Dependency:	None		

4.2.15.2.2.3 DpllActRamPtrAddOffForAdt

Table 312 DpllActRamPtrAddOffForAdt

Name:	DpllActRamPtrAddOffForAdt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 1023 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Actual RAM pointer address value for ADT_T[i]. Actual RAM pointer address value of TRIGGER adapt events in FULL_SCALE for 2*(TNU+1-SYN_NT) TRIGGER periods depending on the size of the used RAM 2; this pointer is used for the RAM region 2 for the subsection 2c only. This updates GTM_DPLL_APT_2c[APT_2c]		
SFR Modified	GTM_DPLL_APT_2c		
Scope:	Driver		
Dependency:	None		

4.2.15.3 Container: DpllStateSignal

Configuration for DPLL Sub-Module State Signal

4.2.15.3.1 Container: DpllStateGeneral

General Configuration for DPLL Sub-Module State Signal

4.2.15.3.1.1 DpllStateSignalEnable

Table 313 DpllStateSignalEnable

Name:	DpllStateSignalEnable
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: TRUE – Enable State Signal FALSE - Disable State Signal Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables State Signal for DPLL. This affects DPLL_CTRL_0[SEN]		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.2 DpllStateSignalActiveEdge

Table 314 DpllStateSignalActiveEdge

Name:	DpllStateSignalActiveEdge		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: NONE : None of the Edge Information will be used. This makes DPLL_CTRL_1[SSL] = 0 RISING_EGDE: Rising Edge Information will be used. This makes DPLL_CTRL_1[SSL] = 1 FALLING_EDGE: Falling Edge Information will be used. This makes DPLL_CTRL_1[SSL] = 2 BOTH_EDGES: Both the Edge Information will be used. This makes DPLL_CTRL_1[SSL] = 3 Default: BOTH_EDGES		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Decides the Active Edge for the State Signal. NONE : None of the Edge Information will be used. This makes DPLL_CTRL_1[SSL] = 0 RISING_EGDE: Rising Edge Information will be used. This makes DPLL_CTRL_1[SSL] = 1 FALLING_EDGE: Falling Edge Information will be used. This makes DPLL_CTRL_1[SSL] = 2 BOTH_EDGES: Both the Edge Information will be used. This makes DPLL_CTRL_1[SSL] = 3		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.3 DpllStateEventsPerHalfScale

Table 315 DpllStateEventsPerHalfScale

Name:	DpllStateEventsPerHalfScale		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 1 to 31 Default: 24		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the number of State Events per Half Scale. This updates the register content as follows $DPLL_CTRL_0[SNU] = (DPLLStateEventsPerHalfScale - 1)$		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.4 DpllVirtualIncPerHalfScale

Table 316 DpllVirtualIncPerHalfScale

Name:	DpllVirtualIncPerHalfScale		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 31 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the number of Virtual Increments or Missing Events per Half Scale. This updates the Register content $DPLL_CTRL_1[SYN_NS]$		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.5 DpllStateSigTimeStampResolution

Table 317 DpllStateSigTimeStampResolution

Name:	DpllStateSigTimeStampResolution		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: SAME_AS_TBU_TS0 : The STATE input time stamp resolution is same as the resolution of the used DPLL input TBU_TS0 bits. This makes DPLL_CTRL_1[TS0_HRS] = 0. IS_8_TIMES_MORE_THAN_TBU_TS0: The STATE input time stamps has a 8 times higher resolution as the TBU_TS0 input. This makes DPLL_CTRL_1[TS0_HRS] = 1 Default: SAME_AS_TBU_TS0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Decides the STATE input time stamp resolution with respect to the resolution of the used DPLL input TBU_TS0 bits. SAME_AS_TBU_TS0 : The STATE input time stamp resolution is same as the resolution of the used DPLL input TBU_TS0 bits. This makes DPLL_CTRL_1[TS0_HRS] = 0. IS_8_TIMES_MORE_THAN_TBU_TS0: The STATE input time stamps has a 8 times higher resolution as the TBU_TS0 input. This makes DPLL_CTRL_1[TS0_HRS] = 1		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.6 DpllStateUseAdaptMode

Table 318 DpllStateUseAdaptMode

Name:	DpllStateUseAdaptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Use Adaptation Information FALSE – Do not use Adaptation Information Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Use of adaptation information of STATE. This makes DPLL_CTRL_0[AMS] = 1. By Enabling this, the values ADT_S[i] are considered to calculate SUB_INC1 pulses in emergency mode (SMC=0) or SUB_INC2 pulses for SMC=1		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.7 DpllStateUseDelayInfo

Table 319 DpllStateUseDelayInfo

Name:	DpllStateUseDelayInfo		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Use Delay Information FALSE – Do not use Delay Information Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Use of input delay information transmitted in FT part of the STATE signal. This makes DPLL_CTRL_0[IDS] = 1. By Enabling this, Up to 24 bits of the FT part contain the delay value of the input signal, concerning the corresponding edge		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.8 DpllStateIsNuseEqualToFullScale

Table 320 DpllStateIsNuseEqualToFullScale

Name:	DpllStateIsNuseEqualToFullScale		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – NUSE is equal to Full Scale FALSE - NUSE is not equal to Full Scale Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if NUSE must be equal to FULL_SCALE. This sets GTM_DPLL_NUSC[FSS]. TRUE: NUSE is Equal to FULL SCALE FALSE: NUSE is less than FULL SCALE		
SFR Modified	GTM_DPLL_NUSC		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.9 DpllStateSignalNuseValue

Table 321 DpllStateSignalNuseValue

Name:	DpllStateSignalNuseValue
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 63 Default: 1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the number of last nominal increments to be considered for the calculations. This sets GTM_DPLL_NUSC[WNUS] and updates GTM_DPLL_NUSC[NUSE]		
SFR Modified	GTM_DPLL_NUSC		
Scope:	Driver		
Dependency:	None		

4.2.15.3.1.10 DpllStateSignalVsnValue

Table 322 DpllStateSignalVsnValue

Name:	DpllStateSignalVsnValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 63 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Number of virtual increments in the current NUSE. This sets GTM_DPLL_NUSC[WVSN] and updates GTM_DPLL_NUSC[VSN]		
SFR Modified	GTM_DPLL_NUSC		
Scope:	Driver		
Dependency:	None		

4.2.15.3.2 Container: DpllActualRamPointerAddress

General Configuration for DPLL Sub-Module State Signal

4.2.15.3.2.1 DpllActRamPtrAddOffForDtAndRdt

Table 323 DpllActRamPtrAddOffForDtAndRdt

Name:	DpllActRamPtrAddOffForDtAndRdt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: 0 to 63 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Actual RAM pointer address value for DT_S[i] and RDT_S[i]. Actual RAM pointer and synchronization position/value of STATE events in FULL_SCALE for up to 64 STATE events but limited to 2*(SNU+1-SYN_NS) in normal and emergency mode; this pointer is used for the RAM region 1c1 and 1c4. This sets GTM_DPLL_APS[WAPS] and updates GTM_DPLL_APS[APS]		
SFR Modified	GTM_DPLL_APS		
Scope:	Driver		
Dependency:	None		

4.2.15.3.2.2 DpllActRamPtrAddOffForTsf

Table 324 DpllActRamPtrAddOffForTsf

Name:	DpllActRamPtrAddOffForTsf		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 63 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Address pointer STATE for RAM region 1c2 Actual RAM pointer address value for TSF_S[i] Initial value: zero (0x00). Actual RAM pointer and synchronization position/value of STATE events in FULL_SCALE for up to 64 STATE events but limited to 2*(SNU+1) in normal and emergency mode; this pointer is used for the RAM region 1c2. This sets GTM_DPLL_APS[WAPS_1c2] and updates GTM_DPLL_APT[APS_1c2]		
SFR Modified	GTM_DPLL_APT		
Scope:	Driver		
Dependency:	None		

4.2.15.3.2.3 DpllActRamPtrAddOffForAdt

Table 325 DpllActRamPtrAddOffForAdt

Name:	DpllActRamPtrAddOffForAdt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: 0 to 63 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Address pointer STATE for RAM region 1c3. Actual RAM pointer address value for ADT_S[i]. Initial value: zero (0x00). Actual RAM pointer and synchronization position/value of STATE events in FULL_SCALE for up to 64 STATE events but limited to 2*(SNU+1-SYN_NS) in normal and emergency mode; this pointer is used for the RAM region 1c3. This updates GTM_DPLL_APS_1c3[APS_1c3]		
SFR Modified	GTM_DPLL_APS_1c3		
Scope:	Driver		
Dependency:	None		

4.2.15.4 Container: DpIISubInc1SignalConfig

This contains the configurations for Sub_Inc1 Signal

4.2.15.4.1 DpIIEnableSubInc1Generator

Table 326 DpIIEnableSubInc1Generator

Name:	DpIIEnableSubInc1Generator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Sub_Inc1 Generator FALSE - Disables the Sub_Inc1 Generator Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables the Sub_Inc1 Generator. This sets Updates DPLL_CTRL_1[SGE1]		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.4.2 DpIIPulseCntBtwnTrigEvntsMlt

Table 327 DpIIPulseCntBtwnTrigEvntsMlt

Name:	DpIIPulseCntBtwnTrigEvntsMlt		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: 0 to 1023 Default: 600		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the number of SUB_INC1 pulses between two TRIGGER events in normal mode (1..1024). This makes DPLL_CTRL_0[MLT] = DPLLPulseCntBtwnTrigEvntsMLT - 1. For emergency mode the number of SUB_INC1 pulses between two STATE events is calculated by the CPU using the formula $MLS1 = (MLT + 1) * (TNU + 1) / (SNU + 1)$ in order to get the same number of SUB_INC1 pulses for FULL_SCALE		
SFR Modified	GTM_DPLL_CTRL_0		
Scope:	Driver		
Dependency:	None		

4.2.15.4.3 DpllUseDirectLoadMode

Table 328 DpllUseDirectLoadMode

Name:	DpllUseDirectLoadMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Direct Load Mode for Sub_Inc1 generation FALSE - Disables Direct Load Mode for Sub_Inc1 generation Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Direct Load Mode for Sub_Inc1 generation. If enabled, the ADD_IN_LD value is used for the SUB_INC1 generation. This sets DPLL_CTRL_1[DLM1]		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.4.4 DpllDirectLoadValue

Table 329 DpllDirectLoadValue

Name:	DpllDirectLoadValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	This parameter specifies the Direct load Value used for Sub_inc1 Generation during Direct Load Mode. This updates GTM_ADD_IN_LD1[ADD_IN_LD_1]		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.4.5 DpllUsePulseCorrectionMode

Table 330 DpllUsePulseCorrectionMode

Name:	DpllUsePulseCorrectionMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Pulse Correction Mode for Sub_Inc1 generation FALSE - Disables Pulse Correction Mode for Sub_Inc1 generation Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Pulse Correction Mode for Sub_Inc1 generation. If enabled, the DPLL uses the correction value stored in MPVAL1 in normal and emergency mode. This sets DPLL_CTRL_1[PCM1]		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.5 Container: DpllSubInc2SignalConfig

This contains the configurations for Sub_Inc2 Signal

4.2.15.5.1 DpllEnableSubInc2Generator

Table 331 DpllEnableSubInc2Generator

Name:	DpllEnableSubInc2Generator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Sub_Inc2 Generator FALSE - Disables the Sub_Inc2 Generator Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables the Sub_Inc2 Generator. This sets Updates DPLL_CTRL_1[SGE2]. This will also makes DPLL_CTRL_1[SMC] = 1		

SFR Modified	GTM_DPLL_CTRL_1
Scope:	Driver
Dependency:	None

4.2.15.5.2 DpllUseDirectLoadMode

Table 332 DpllUseDirectLoadMode

Name:	DpllUseDirectLoadMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Direct Load Mode for Sub_Inc2 generation FALSE - Disables Direct Load Mode for Sub_Inc2 generation Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Direct Load Mode for Sub_Inc2 generation. If enabled, the ADD_IN_LD value is used for the SUB_INC1 generation. This sets Updates DPLL_CTRL_1[DLM2]		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.5.3 DpllDirectLoadValue

Table 333 DpllDirectLoadValue

Name:	DpllDirectLoadValue		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 0xFFFFFFFF Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Direct load Value used for Sub_inc2 Generation during Direct Load Mode. This updates GTM_ADD_IN_LD2[ADD_IN_LD_2]		
Scope:	Driver		
Dependency:	None		

4.2.15.5.4 DpllUsePulseCorrectionMode

Table 334 DpllUsePulseCorrectionMode

Name:	DpllUsePulseCorrectionMode
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Pulse Correction Mode for Sub_Inc1 generation FALSE - Disables Pulse Correction Mode for Sub_Inc1 generation Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Pulse Correction Mode for Sub_Inc1 generation. If enabled, the DPLL uses the correction value stored in MPVAL1 in normal and emergency mode. This sets DPLL_CTRL_1[PCM2]		
SFR Modified	GTM_DPLL_CTRL_1		
Scope:	Driver		
Dependency:	None		

4.2.15.6 Container: DpllEnableAction

This Container holds the Action Enabling configuration for DPLL. This has multiplicity of 24, each corresponding to DPLL Action depending on the index number.

4.2.15.6.1 DpllEnableAction

Table 335 DpllEnableAction

Name:	DpllEnableAction		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Corresponding DPLL Action FALSE - Disables the Corresponding DPLL Action Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables the Corresponding DPLL Action. This sets GTM_DPLL_CTRL_j[WADx] and then sets GTM_DPLL_CTRL_2[AENx] (j = 2- 4), (x = 0 - 23) where j and x depend on the Action Index		
SFR Modified	GTM_DPLL_CTRL_2		
Scope:	Driver		
Dependency:	None		

4.2.15.7 Container: DpllRAMRegion2Address

This Container holds the Action Enabling configuration for DPLL.

4.2.15.7.1 DpllOffsetSizeForRegion2

Table 336 DpllOffsetSizeForRegion2

Name:	DpllOffsetSizeForRegion2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: OFFSET_SIZE_128 : Offset size 128 of RAM region 2. OFFSET_SIZE_256 : Offset size 256 of RAM region 2. OFFSET_SIZE_512 : Offset size 512 of RAM region 2. OFFSET_SIZE_1024 : Offset size 1024 of RAM region 2. Default: OFFSET_SIZE_512		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This specifies the Offset size of RAM region 2. This updates the Register Bits GTM_DPLL_OSW[OSS]. OFFSET_SIZE_128 : Offset size 128 of RAM region 2. OFFSET_SIZE_256 : Offset size 256 of RAM region 2. OFFSET_SIZE_512 : Offset size 512 of RAM region 2. OFFSET_SIZE_1024 : Offset size 1024 of RAM region 2.		
SFR Modified	GTM_DPLL_OSW		
Scope:	Driver		
Dependency:	None		

4.2.15.8 Container: DpllAruldToInSigPmtr

This defines the ID information to the input signal PMTR_x from the ARU. This updates GTM_DPLL_ID_PMTR_x [ID_PMTR_x] (x= 0 - 23). This has multiplicity of 24, each corresponding to PMTR information depending on the index number. DpllAruldToInSigPmtr_x corresponds to PMTR_x where x = 0 - 23

4.2.15.8.1 DpllAruldInfoToInputSignalPmtr

Table 337 DpllAruldInfoToInputSignalPmtr

Name:	DpllAruldInfoToInputSignalPmtr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 511 Default: 510		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines the ID information to the input signal PMTR_x from the ARU . This updates GTM_DPLL_ID_PMTR_x [ID_PMTR_x] (x= 0 - 23)		

SFR Modified	GTM_DPLL_ID_PMTR_i
Scope:	Driver
Dependency:	None

4.2.15.9 Container: DpllInterrupt

This container holds the Interrupt Mode and Enables for DPLL Interrupt Sources.

4.2.15.9.1 DpllEnIntOnDpllDisable

Table 338 DpllEnIntOnDpllDisable

Name:	DpllEnIntOnDpllDisable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on DPLL Disabling Event FALSE - Disables interrupt on DPLL Disabling Event Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on DPLL Disabling Event. This announces the switch off of the DEN bit. Sets DPLL_IRQ_EN[PDI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.2 DpllEnIntOnDpllEnable

Table 339 DpllEnIntOnDpllEnable

Name:	DpllEnIntOnDpllEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on DPLL Enabling Event FALSE - Disables interrupt on DPLL Enabling Event Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on DPLL Enabling Event. This announces the switch on of the DEN bit. Sets DPLL_IRQ_EN[PEI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.3 DpllEnIntOnTrigMinHoldTimeErr

Table 340 DpllEnIntOnTrigMinHoldTimeErr

Name:	DpllEnIntOnTrigMinHoldTimeErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on violation of minimum hold time of TRIGGER FALSE - Disables interrupt on violation of minimum hold time of TRIGGER Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on violation of minimum hold time of TRIGGER. Sets DPLL_IRQ_EN[TINI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.4 DpllEnIntOnTrigMaxHoldTimeErr

Table 341 DpllEnIntOnTrigMaxHoldTimeErr

Name:	DpllEnIntOnTrigMaxHoldTimeErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on violation of Maximum hold time of TRIGGER FALSE - Disables interrupt on violation of Maximum hold time of TRIGGER Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on violation of Maximum hold time of TRIGGER. Sets DPLL_IRQ_EN[TAXI_IRQ_EN]		
SFR Modified	DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.5 DpllEnIntOnStateInactiveSlope

Table 342 DpllEnIntOnStateInactiveSlope

Name:	DpllEnIntOnStateInactiveSlope		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables interrupt when an inactive slope of STATE is detected FALSE - Disables interrupt when an inactive slope of STATE is detected Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt when an inactive slope of STATE is detected. Sets DPLL_IRQ_EN[SISI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.6 DpLlEnIntOnTriggerInactiveSlope

Table 343 DpLlEnIntOnTriggerInactiveSlope

Name:	DpLlEnIntOnTriggerInactiveSlope		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when an inactive slope of TRIGGER is detected FALSE - Disables interrupt when an inactive slope of TRIGGER is detected Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt when an inactive slope of TRIGGER is detected. Sets DPLL_IRQ_EN[TISI_IRQ_EN]		
SFR Modified	DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.7 DpLlEnIntOnMissingStateSignal

Table 344 DpLlEnIntOnMissingStateSignal

Name:	DpLlEnIntOnMissingStateSignal		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on missing STATE Signal FALSE - Disables interrupt on missing STATE Signal Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	Enables interrupt on missing STATE Signal. Sets DPLL_IRQ_EN[MSI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.8 DpllEnIntOnMissingTriggerSignal

Table 345 DpllEnIntOnMissingTriggerSignal

Name:	DpllEnIntOnMissingTriggerSignal		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on missing TRIGGER Signal FALSE - Disables interrupt on missing TRIGGER Signal Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on missing TRIGGER Signal. Sets DPLL_IRQ_EN[MTI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.9 DpllEnIntOnStateActiveSlope

Table 346 DpllEnIntOnStateActiveSlope

Name:	DpllEnIntOnStateActiveSlope		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when an active slope of TRIGGER is detected FALSE - Disables interrupt when an active slope of TRIGGER is detected Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt when an active slope of TRIGGER is detected. Sets DPLL_IRQ_EN[TASI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.10 DpllEnIntOnTriggerActiveSlope

Table 347 DpllEnIntOnTriggerActiveSlope

Name:	DpllEnIntOnTriggerActiveSlope		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when an active slope of TRIGGER is detected FALSE - Disables interrupt when an active slope of TRIGGER is detected Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt when an active slope of TRIGGER is detected. Sets DPLL_IRQ_EN[TASI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.11 DpllEnIntOnPlausWindowViolation

Table 348 DpllEnIntOnPlausWindowViolation

Name:	DpllEnIntOnPlausWindowViolation		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Plausibility Window(PVT) violation FALSE - Disables interrupt on Plausibility Window(PVT) violation Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on Plausibility Window(PVT) violation. Sets DPLL_IRQ_EN[PWI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.12 DpllEnIntOnRamWriteAccessToReg2

Table 349 DpllEnIntOnRamWriteAccessToReg2

Name:	DpllEnIntOnRamWriteAccessToReg2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables interrupt on RAM Write Access to Region 2 FALSE - Disables interrupt on RAM Write Access to Region 2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on RAM Write Access to Region 2. Sets DPLL_IRQ_EN[W2I_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.13 DpllEnIntOnRamWriteAccToReg1b1c

Table 350 DpllEnIntOnRamWriteAccToReg1b1c

Name:	DpllEnIntOnRamWriteAccToReg1b1c		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on RAM Write Access to Region 1b and 1c FALSE - Disables interrupt on RAM Write Access to Region 1b and 1c Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on RAM Write Access to Region 1b and 1c. Sets DPLL_IRQ_EN[W1I_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.14 DpllEnIntOnSubInc1LockGet

Table 351 DpllEnIntOnSubInc1LockGet

Name:	DpllEnIntOnSubInc1LockGet		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on getting lock on Sub Inc1 FALSE - Disables interrupt on getting lock on Sub Inc1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables interrupt on getting lock on Sub Inc1. Sets DPLL_IRQ_EN[GL1I_IRQ_EN]
SFR Modified	GTM_DPLL_IRQ_EN
Scope:	Driver
Dependency:	None

4.2.15.9.15 DpllEnIntOnSubInc1LockLost

Table 352 DpllEnIntOnSubInc1LockLost

Name:	DpllEnIntOnSubInc1LockLost		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on losing lock on Sub Inc1 FALSE - Disables interrupt on losing lock on Sub Inc1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on losing lock on Sub Inc1. Sets DPLL_IRQ_EN[LL1I_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.16 DpllEnIntOnError

Table 353 DpllEnIntOnError

Name:	DpllEnIntOnError		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on losing lock on Sub Inc1 FALSE - Disables interrupt on losing lock on Sub Inc1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Error interrupt. Sets DPLL_IRQ_EN[EI_IRQ_EN]. This represents Error during configuration or operation resulting in unexpected values		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.17 DpllEnIntOnSubInc2LockGet

Table 354 DpllEnIntOnSubInc2LockGet

Name:	DpllEnIntOnSubInc2LockGet		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on getting lock on Sub Inc2 FALSE - Disables interrupt on getting lock on Sub Inc2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on getting lock on Sub Inc2. Sets DPLL_IRQ_EN[GL2I_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.18 DpllEnIntOnSubInc2LockLost

Table 355 DpllEnIntOnSubInc2LockLost

Name:	DpllEnIntOnSubInc2LockLost		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on losing lock on Sub Inc2 FALSE - Disables interrupt on losing lock on Sub Inc2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on losing lock on Sub Inc2. Sets DPLL_IRQ_EN[LL2I_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.19 DpllEnIntOnTriggerEvent0

Table 356 DpllEnIntOnTriggerEvent0

Name:	DpllEnIntOnTriggerEvent0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables interrupt on Trigger Event 0 FALSE - Disables interrupt on Trigger Event 0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on Trigger Event 0. Sets DPLL_IRQ_EN[TE0I_IRQ_EN]		
SFR Modified	DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.20 DpllEnIntOnTriggerEvent1

Table 357 DpllEnIntOnTriggerEvent1

Name:	DpllEnIntOnTriggerEvent1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Trigger Event 1 FALSE - Disables interrupt on Trigger Event 1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on Trigger Event 1. Sets DPLL_IRQ_EN[TE1I_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.21 DpllEnIntOnTriggerEvent2

Table 358 DpllEnIntOnTriggerEvent2

Name:	DpllEnIntOnTriggerEvent2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Trigger Event 2 FALSE - Disables interrupt on Trigger Event 2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on Trigger Event 2. Sets DPLL_IRQ_EN[TE2I_IRQ_EN]		

SFR Modified	GTM_DPLL_IRQ_EN
Scope:	Driver
Dependency:	None

4.2.15.9.22 DpLlEnIntOnTriggerEvent3

Table 359 DpLlEnIntOnTriggerEvent3

Name:	DpLlEnIntOnTriggerEvent3		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Trigger Event 3 FALSE - Disables interrupt on Trigger Event 3 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on Trigger Event 3. Sets DPLL_IRQ_EN[TE3I_IRQ_EN]		
SFR Modified	DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.23 DpLlEnIntOnTriggerEvent4

Table 360 DpLlEnIntOnTriggerEvent4

Name:	DpLlEnIntOnTriggerEvent4		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Trigger Event 4 FALSE - Disables interrupt on Trigger Event 4 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on Trigger Event 4. Sets DPLL_IRQ_EN[TE4I_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.24 DpLlEnIntOnTriggerDurCalcDone

Table 361 DpLlEnIntOnTriggerDurCalcDone

Name:	DpLlEnIntOnTriggerDurCalcDone
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on calculated TRIGGER duration requested while NTI_CNT is zero FALSE - Disables interrupt on calculated TRIGGER duration requested while NTI_CNT is zero Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on calculated TRIGGER duration requested while NTI_CNT is zero. Sets DPLL_IRQ_EN[CDTI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.25 DpllEnIntOnStateDurCalcDone

Table 362 DpllEnIntOnStateDurCalcDone

Name:	DpllEnIntOnStateDurCalcDone		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on calculated STATE duration requested FALSE - Disables interrupt on calculated STATE duration requested Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt on calculated STATE duration requested. Sets DPLL_IRQ_EN[CDSI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.26 DpllEnIntOnTriggerOutOfRange

Table 363 DpllEnIntOnTriggerOutOfRange

Name:	DpllEnIntOnTriggerOutOfRange		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables interrupt when TRIGGER is out of range FALSE - Disables interrupt when TRIGGER is out of range Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt when TRIGGER is out of range. Sets DPLL_IRQ_EN[TORI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.27 DpllEnIntOnStateOutOfRange

Table 364 DpllEnIntOnStateOutOfRange

Name:	DpllEnIntOnStateOutOfRange		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when STATE is out of range FALSE - Disables interrupt when STATE is out of range Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables interrupt when STATE is out of range. Sets DPLL_IRQ_EN[SORI_IRQ_EN]		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.9.28 DpllEnIntOnTriggerDirChange

Table 365 DpllEnIntOnTriggerDirChange

Name:	DpllEnIntOnTriggerDirChange		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when a direction change of TRIGGER is detected FALSE - Disables interrupt when a direction change of TRIGGER is detected Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables interrupt when a direction change of TRIGGER is detected. Sets DPLL_IRQ_EN[DCGI_IRQ_EN]
SFR Modified	GTM_DPLL_IRQ_EN
Scope:	Driver
Dependency:	None

4.2.15.9.29 DpllInterruptMode

Table 366 DpllInterruptMode

Name:	DpllInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE: Level Mode is used GTM_INTERRUPT_PULSE_MODE: Pulse mode is used GTM_INTERRUPT_PULSE_NOTIFY_MODE: Pulse Notify mode is used GTM_INTERRUPT_SINGLE_PULSE_MODE: Single Pulse mode is used Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use		
SFR Modified	GTM_DPLL_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10 Container: DpllErrorInterrupt

This container holds the Error Interrupt Mode and Enables for DPLL Interrupt Sources.

4.2.15.10.1 DpllEnErrIntOnDpllDisable

Table 367 DpllEnErrIntOnDpllDisable

Name:	DpllEnErrIntOnDpllDisable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on DPLL Disabling Event Error FALSE - Disables interrupt on DPLL Disabling Event Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	Enables/Disables interrupt on DPLL Disabling Event Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.2 DpllEnErrIntOnDpllEnable

Table 368 DpllEnErrIntOnDpllEnable

Name:	DpllEnErrIntOnDpllEnable		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on DPLL Enabling Event Error. FALSE - Disables interrupt on DPLL Enabling Event Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on DPLL Enabling Event Error.		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.3 DpllEnErrIntOnTrigMinHoldTimeErr

Table 369 DpllEnErrIntOnTrigMinHoldTimeErr

Name:	DpllEnErrIntOnTrigMinHoldTimeErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on violation of minimum hold time of TRIGGER Error FALSE - Disables interrupt on violation of minimum hold time of TRIGGER Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on violation of minimum hold time of TRIGGER Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.4 DpllEnErrIntOnTrigMaxHoldTimeErr

Table 370 DpllEnErrIntOnTrigMaxHoldTimeErr

Name:	DpllEnErrIntOnTrigMaxHoldTimeErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on violation of maximum hold time of TRIGGER Error FALSE - Disables interrupt on violation of maximum hold time of TRIGGER Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on violation of maximum hold time of TRIGGER Error		
SFR Modified			
Scope:	Driver		
Dependency:	None		

4.2.15.10.5 DpllEnErrIntOnStateInactiveSlope

Table 371 DpllEnErrIntOnStateInactiveSlope

Name:	DpllEnErrIntOnStateInactiveSlope		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when an inactive slope of STATE is detected Error FALSE - Disables interrupt when an inactive slope of STATE is detected Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt when an inactive slope of STATE is detected Error.		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.6 DpllEnErrIntOnTriggerInactiveSlope

Table 372 DpllEnErrIntOnTriggerInactiveSlope

Name:	DpllEnErrIntOnTriggerInactiveSlope		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables interrupt when an inactive slope of TRIGGER is detected Error FALSE - Disables interrupt when an inactive slope of TRIGGER is detected Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt when an inactive slope of TRIGGER is detected Error.		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.7 DpllEnErrIntOnMissingStateSignal

Table 373 DpllEnErrIntOnMissingStateSignal

Name:	DpllEnErrIntOnMissingStateSignal		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on missing STATE Signal Error FALSE - Disables interrupt on missing STATE Signal Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on missing STATE Signal Error.		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.8 DpllEnErrIntOnMissingTriggerSignal

Table 374 DpllEnErrIntOnMissingTriggerSignal

Name:	DpllEnErrIntOnMissingTriggerSignal		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on missing TRIGGER Signal Error FALSE - Disables interrupt on missing TRIGGER Signal Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables/Disables interrupt on missing TRIGGER Signal Error.
SFR Modified	GTM_DPLL_EIRQ_EN
Scope:	Driver
Dependency:	None

4.2.15.10.9 DpIIEnErrIntOnStateActiveSlope

Table 375 DpIIEnErrIntOnStateActiveSlope

Name:	DpIIEnErrIntOnStateActiveSlope		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when an active slope of STATE is detected Error FALSE - Disables interrupt when an active slope of STATE is detected Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt when an active slope of STATE is detected Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.10 DpIIEnErrIntOnTriggerActiveSlope

Table 376 DpIIEnErrIntOnTriggerActiveSlope

Name:	DpIIEnErrIntOnTriggerActiveSlope		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when an active slope of TRIGGER is detected Error FALSE - Disables interrupt when an active slope of TRIGGER is detected Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt when an active slope of TRIGGER is detected Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.11 DpllEnErrIntOnPlausWindowViolation

Table 377 DpllEnErrIntOnPlausWindowViolation

Name:	DpllEnErrIntOnPlausWindowViolation		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Plausibility Window(PVT) violation Error FALSE - Disables interrupt on Plausibility Window(PVT) violation Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on Plausibility Window(PVT) violation Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.12 DpllEnErrIntOnRamWriteAccessToReg2

Table 378 DpllEnErrIntOnRamWriteAccessToReg2

Name:	DpllEnErrIntOnRamWriteAccessToReg2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on RAM Write Access to Region 2 Error FALSE - Disables interrupt on RAM Write Access to Region 2 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on RAM Write Access to Region 2 Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.13 DpllEnErrIntOnRamWriteAccToReg1b1c

Table 379 DpllEnErrIntOnRamWriteAccToReg1b1c

Name:	DpllEnErrIntOnRamWriteAccToReg1b1c		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables interrupt on RAM Write Access to Region 1b and 1c Error FALSE - Disables interrupt on RAM Write Access to Region 1b and 1c Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on RAM Write Access to Region 1b and 1c Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.14 DpllEnErrIntOnSubInc1LockGet

Table 380 DpllEnErrIntOnSubInc1LockGet

Name:	DpllEnErrIntOnSubInc1LockGet		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on getting lock on Sub Inc1 Error FALSE - Disable interrupt on getting lock on Sub Inc1 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt on getting lock on Sub Inc1 Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.15 DpllEnErrIntOnSubInc1LockLost

Table 381 DpllEnErrIntOnSubInc1LockLost

Name:	DpllEnErrIntOnSubInc1LockLost		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on lost lock on Sub Inc1 Error FALSE - Disable interrupt on lost lock on Sub Inc1 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables/Disable interrupt on lost lock on Sub Inc1 Error
SFR Modified	GTM_DPLL_EIRQ_EN
Scope:	Driver
Dependency:	None

4.2.15.10.16 DpllEnErrIntOnError

Table 382 DpllEnErrIntOnError

Name:	DpllEnErrIntOnError		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Error interrupt Error FALSE - Disables Error interrupt Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Error interrupt Error.		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.17 DpllEnErrIntOnSubInc2LockGet

Table 383 DpllEnErrIntOnSubInc2LockGet

Name:	DpllEnErrIntOnSubInc2LockGet		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on getting lock on Sub Inc2 Error FALSE - Disable interrupt on getting lock on Sub Inc2 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt on getting lock on Sub Inc2 Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.18 DpllEnErrIntOnSubInc2LockLost

Table 384 DpllEnErrIntOnSubInc2LockLost

Name:	DpllEnErrIntOnSubInc2LockLost		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on lost lock on Sub Inc2 Error FALSE - Disable interrupt on lost lock on Sub Inc2 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt on lost lock on Sub Inc2 Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.19 DpllEnErrIntOnTriggerEvent0

Table 385 DpllEnErrIntOnTriggerEvent0

Name:	DpllEnErrIntOnTriggerEvent0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Trigger Event 0 Error FALSE - Disable interrupt on Trigger Event 0 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt on Trigger Event 0 Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.20 DpllEnErrIntOnTriggerEvent1

Table 386 DpllEnErrIntOnTriggerEvent1

Name:	DpllEnErrIntOnTriggerEvent1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables interrupt on Trigger Event 1 Error FALSE - Disable interrupt on Trigger Event 1 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt on Trigger Event 1 Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.21 DpllEnErrIntOnTriggerEvent2

Table 387 DpllEnErrIntOnTriggerEvent2

Name:	DpllEnErrIntOnTriggerEvent2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Trigger Event 2 Error FALSE - Disable interrupt on Trigger Event 2 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt on Trigger Event 2 Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.22 DpllEnErrIntOnTriggerEvent3

Table 388 DpllEnErrIntOnTriggerEvent3

Name:	DpllEnErrIntOnTriggerEvent3		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Trigger Event 3 Error FALSE - Disable interrupt on Trigger Event 3 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Enables/Disable interrupt on Trigger Event 3 Error
SFR Modified	GTM_DPLL_EIRQ_EN
Scope:	Driver
Dependency:	None

4.2.15.10.23 DpllEnErrIntOnTriggerEvent4

Table 389 DpllEnErrIntOnTriggerEvent4

Name:	DpllEnErrIntOnTriggerEvent4		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on Trigger Event 4 Error FALSE - Disable interrupt on Trigger Event 4 Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt on Trigger Event 4 Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.24 DpllEnErrIntOnTriggerDurCalcDone

Table 390 DpllEnErrIntOnTriggerDurCalcDone

Name:	DpllEnErrIntOnTriggerDurCalcDone		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on calculated TRIGGER duration requested while NTI_CNT is zero Error FALSE - Disables interrupt on calculated TRIGGER duration requested while NTI_CNT is zero Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on calculated TRIGGER duration requested while NTI_CNT is zero Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.25 DpllEnErrIntOnStateDurCalcDone

Table 391 DpllEnErrIntOnStateDurCalcDone

Name:	DpllEnErrIntOnStateDurCalcDone		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt on calculated STATE duration requested while NTI_CNT is zero Error FALSE - Disables interrupt on calculated STATE duration requested while NTI_CNT is zero Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt on calculated STATE duration requested while NTI_CNT is zero Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.26 DpllEnErrIntOnTriggerOutOfRange

Table 392 DpllEnErrIntOnTriggerOutOfRange

Name:	DpllEnErrIntOnTriggerOutOfRange		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when TRIGGER is out of range Error FALSE - Disable interrupt when TRIGGER is out of range Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt when TRIGGER is out of range Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.27 DpllEnErrIntOnStateOutOfRange

Table 393 DpllEnErrIntOnStateOutOfRange

Name:	DpllEnErrIntOnStateOutOfRange		
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when STATE is out of range Error FALSE - Disable interrupt when STATE is out of range Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disable interrupt when STATE is out of range Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.15.10.28 DpllEnErrIntOnTriggerDirChange

Table 394 DpllEnErrIntOnTriggerDirChange

Name:	DpllEnErrIntOnTriggerDirChange		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables interrupt when a direction change of TRIGGER is detected Error FALSE - Disables interrupt when a direction change of TRIGGER is detected Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables interrupt when a direction change of TRIGGER is detected Error		
SFR Modified	GTM_DPLL_EIRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.16 Container: Spe

This container decides the configuration for SPE. Multiplicity depends on the number of SPE modules. Each index corresponds to respective SPE module Number.

4.2.16.1 Container: SpeGeneral

This contains the general configurations for SPE.

4.2.16.1.1 SpeUsage

Table 395 SpeUsage

Name:	SpeUsage		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init Default: UNUSED		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides if the channel will be used by GTM Driver or will be initialized elsewhere. UNUSED: Channel is not used. USED_BY_GTM_DRIVER: Channel will be used by Gtm Driver and will be initialized in Gtm_Init		
SFR Modified	N/A		
Scope:	Driver		
Dependency:	None		

4.2.16.1.2 SpeEnableModule

Table 396 SpeEnableModule

Name:	SpeEnableModule		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the SPE Module FALSE - Disables the SPE Module Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the SPE Module		
SFR Modified	GTM_SPEi_CTRL_STAT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.1.3 SpeTimChannelsSelect

Table 397 SpeTimChannelsSelect

Name:	SpeTimChannelsSelect
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: For SPE 0, SEL0_TIM0_CH0_TO_2 (Default) SEL1_TIM1_CH0_TO_2 For SPE1, SEL0_TIM0_CH3_TO_5 (Default) SEL1_TIM1_CH3_TO_5		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	TIM channels Select for the SPE module		
SFR Modified	GTM_SPEi_CTRL_STAT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.1.4 SpeEnableInput0

Table 398 SpeEnableInput0

Name:	SpeEnableInput0		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the SPE Modules Input 0 FALSE - Disables the SPE Modules Input 0 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables the SPE Modules Input 0. This updates GTM_SPE0_CTRL_STAT[SIE0]		
SFR Modified	GTM_SPEi_CTRL_STAT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.1.5 SpeEnableInput1

Table 399 SpeEnableInput1

Name:	SpeEnableInput1		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE – Enables the SPE Modules Input 1 FALSE - Disables the SPE Modules Input 1 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables the SPE Modules Input 1. This updates GTM_SPE0_CTRL_STAT[SIE0]		
SFR Modified	GTM_SPEi_CTRL_STAT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.1.6 SpeEnableInput2

Table 400 SpeEnableInput2

Name:	SpeEnableInput2		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the SPE Modules Input 2 FALSE - Disables the SPE Modules Input 2 Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables the SPE Modules Input 2. This updates CTRL_STAT[SIE0]		
SFR Modified	GTM_SPEi_CTRL_STAT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.1.7 SpeTriggerInputSelect

Table 401 SpeTriggerInputSelect

Name:	SpeTriggerInputSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SPE_NIPD: NIPD(New valid input pattern detected) Signal is used as Trigger TOM_CH0_TRIG_CCU0: Trigger from TOM Channel 0 CCU0 is used as Trigger TOM_CH0_TRIG_CCU1: Trigger from TOM Channel 0 CCU1 is used as Trigger TOM_CH2_TRIG_CCU1: Trigger from TOM Channel 2 CCU1 is used as Trigger Note: For SPE0, TOM0 is used. For SPE1, TOM1 is used Default: SPE_NIPD		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	SPE Trigger Input Select for the SPE module. This updates GTM_SPE0_CTRL_STAT[TRIG_SEL] SPE_NIPD: NIPD(New valid input pattern detected) Signal is used as Trigger TOM_CH0_TRIG_CCU0: Trigger from TOM Channel 0 CCU0 is used as Trigger TOM_CH0_TRIG_CCU1: Trigger from TOM Channel 0 CCU1 is used as Trigger TOM_CH2_TRIG_CCU1: Trigger from TOM Channel 2 CCU1 is used as Trigger For SPE0, TOM0 is used. For SPE1, TOM1 is used		
SFR Modified	GTM_SPEi_CTRL_STAT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.1.8 SpeEnableFastShutOff

Table 402 SpeEnableFastShutOff

Name:	SpeEnableFastShutOff		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Fast Shutoff Mode FALSE - Disables Fast Shutoff Mode Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Fast Shutoff Mode		
SFR Modified	GTM_SPEi_CTRL_STAT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.1.9 SpeInputSigRevCntr

Table 403 SpeInputSigRevCntr

Name:	SpeInputSigRevCntr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 16777216 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

	Link Time	--	--
	Post build	x	Variant PB
Description:	Specifies the value of SPE revolution counter.		
SFR Modified	GTM_SPEi_REV_CNT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.1.10 SpelInputSigRevCntrCmpVal

Table 404 SpelInputSigRevCntrCmpVal

Name:	SpelInputSigRevCntrCmpVal		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 16777216 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	The interrupt SPEi RCMP is raised when the SPEi_REV_CNT value equals the SPEi_REV_CMP register.		
SFR Modified	GTM_SPE0_REV_CMP		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.2 Container: SpelInputPattern

This contains the configuration for the Input Pattern. Multiplicity corresponds to 8 Input Patterns. SpelInputPattern_x where x corresponds to Input pattern number.

4.2.16.2.1 SpelsPatternValid

Table 405 SpelsPatternValid

Name:	SpelsPatternValid		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Specifies that the pattern given in SpelInputPattern is valid FALSE - Specifies that the pattern given in SpelInputPattern is not valid Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Determines the validity of the pattern		
SFR Modified	GTM_SPEi_PAT		

Scope:	Driver
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER

4.2.16.2.2 SpeInputPattern

Table 406 SpeInputPattern

Name:	SpeInputPattern		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: 0 to 7 Default: 0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This parameter specifies the Pattern. This is of the form Bit 0 = Tim Input 0 Bit 1 = Tim Input 1 Bit 2 = Tim Input 2. This Updates PAT[IPx_PAT] (x= 0-7)		
SFR Modified	GTM_SPEi_OUT_PATj		
Scope:	Driver		
Dependency:	SpelsPatternValid = 'true' and SpeUsage = USED_BY_GTM_DRIVER		

4.2.16.3 Container: SpeOutputConfig

This contains the configurations for the SPE Output. Multiplicity depends on the number of Outputs.

4.2.16.3.1 SpeOutputSelect

Table 407 SpeOutputSelect

Name:	SpeOutputSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TOM_CH0_SOUR: set SPE_OUT to TOM_CH0_SOUR TOM_CH1_SOUR: set SPE_OUT to TOM_CH1_SOUR LEVEL_0: set SPE_OUT to "0" LEVEL_1: set SPE_OUT to "1" Default: TOM_CH0_SOUR		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB

Description:	Selects the output for the SPE TOM_CH0_SOUR: set SPE_OUT to TOM_CH0_SOUR TOM_CH1_SOUR: set SPE_OUT to TOM_CH1_SOUR LEVEL_0: set SPE_OUT to "0" LEVEL_1: set SPE_OUT to "1"
SFR Modified	GTM_SPE0_OUT_CTRL
Scope:	Driver
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER

4.2.16.3.2 SpeFastShutoffLevel

Table 408 SpeFastShutoffLevel

Name:	SpeFastShutoffLevel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: LOW_LEVEL: Fast Shut off Value for the Channel is Logic "0" HIGH_LEVEL: Fast Shut off Value for the Channel is Logic "1" Default: LOW_LEVEL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Selects the Fast Shut-Off level at output for the SPE LOW_LEVEL: Fast Shut off Value for the Channel is Logic "0" HIGH_LEVEL: Fast Shut off Value for the Channel is Logic "1"		
SFR Modified			
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.4 Container: SpeInterrupt

This container contains the Interrupt configuration for the SPE Module.

4.2.16.4.1 SpeEnableIntOnNewPatternDetect

Table 409 SpeEnableIntOnNewPatternDetect

Name:	SpeEnableIntOnNewPatternDetect
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c
Range:	Values: TRUE – Enables Interrupt on detection of a New Pattern FALSE - Disables Interrupt on detection of a New Pattern Default: FALSE

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Interrupt on detection of a New Pattern		
SFR Modified	GTM_SPEi_CTRL_STAT		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.4.2 SpeEnableIntOnSpeDirBitChange

Table 410 SpeEnableIntOnSpeDirBitChange

Name:	SpeEnableIntOnSpeDirBitChange		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on SPE_DIR bit changed on behalf of new Input Pattern FALSE - Disables Interrupt on SPE_DIR bit changed on behalf of new Input Pattern Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Interrupt on SPE_DIR bit changed on behalf of new Input Pattern		
SFR Modified	GTM_SPEi_IRQ_EN		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.4.3 SpeEnableIntOnWrongPattern

Table 411 SpeEnableIntOnWrongPattern

Name:	SpeEnableIntOnWrongPattern		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Wrong or invalid pattern detected at input FALSE - Enables Interrupt on Wrong or invalid pattern detected at input Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Interrupt on Wrong or invalid pattern detected at input		
Scope:	Driver		

SFR Modified	GTM_SPEi_IRQ_EN
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER

4.2.16.4.4 SpeEnableIntOnBouncingInput

Table 412 SpeEnableIntOnBouncingInput

Name:	SpeEnableIntOnBouncingInput		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Detection of Bouncing Input Detection FALSE - Enables Interrupt on Detection of Bouncing Input Detection Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Interrupt on Detection of Bouncing Input Detection		
SFR Modified	GTM_SPEi_IRQ_EN		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.4.5 SpeEnableIntOnRegComp

Table 413 SpeEnableIntOnRegComp

Name:	SpeEnableIntOnRegComp		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on the SPEi_REV_CNT value equals the SPEi_REV_CMP register. FALSE - Disables Interrupt on the SPEi_REV_CNT value equals the SPEi_REV_CMP register. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on the SPEi_REV_CNT value equals the SPEi_REV_CMP register.		
SFR Modified	GTM_SPEi_REV_CMP		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.4.6 SpeInterruptMode

Table 414 SpeInterruptMode

Name:	SpeInterruptMode		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: GTM_INTERRUPT_LEVEL_MODE GTM_INTERRUPT_PULSE_MODE GTM_INTERRUPT_PULSE_NOTIFY_MODE GTM_INTERRUPT_SINGLE_PULSE_MODE Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use		
SFR Modified	GTM_SPEi_IRQ_MODE		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.5 Container: SpeErrorInterrupt

This container contains the Error Interrupt configuration for the SPE Module.

4.2.16.5.1 SpeEnableIntOnNewPatternDetectErr

Table 415 SpeEnableIntOnNewPatternDetectErr

Name:	SpeEnableIntOnNewPatternDetectErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on detection of a New Pattern Error. FALSE - Disables Interrupt on detection of a New Pattern Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on detection of a New Pattern Error		
SFR Modified	GTM_SPEi_EIRQ_EN		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.5.2 SpeEnableIntOnSpeDirBitChangeErr

Table 416 SpeEnableIntOnSpeDirBitChangeErr

Name:	SpeEnableIntOnSpeDirBitChangeErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on SPE_DIR bit changed on behalf of new Input Pattern Error. FALSE - Disables Interrupt on SPE_DIR bit changed on behalf of new Input Pattern Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on SPE_DIR bit changed on behalf of new Input Pattern Error.		
SFR Modified	GTM_SPEi_EIRQ_EN		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.5.3 SpeEnableIntOnWrongPatternErr

Table 417 SpeEnableIntOnWrongPatternErr

Name:	SpeEnableIntOnWrongPatternErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Wrong or invalid pattern detected at input Error. FALSE - Disables Interrupt on Wrong or invalid pattern detected at input Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on Wrong or invalid pattern detected at input Error.		
SFR Modified	GTM_SPEi_EIRQ_EN		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.5.4 SpeEnableIntOnBouncingInputErr

Table 418 SpeEnableIntOnBouncingInputErr

Name:	SpeEnableIntOnBouncingInputErr		
Source:	This parameter is microcontroller specific.		

File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Detection of Bouncing Input Detection Error. FALSE - Disables Interrupt on Detection of Bouncing Input Detection Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on Detection of Bouncing Input Detection Error.		
SFR Modified	GTM_SPEi_EIRQ_EN		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.16.5.5 SpeEnableIntOnRegCompErr

Table 419 SpeEnableIntOnRegCompErr

Name:	SpeEnableIntOnRegCompErr		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on the SPEi_REV_CNT value equals the SPEi_REV_CMP register Error. FALSE - Disables Interrupt on the SPEi_REV_CNT value equals the SPEi_REV_CMP register Error. Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables Interrupt on the SPEi_REV_CNT value equals the SPEi_REV_CMP register Error.		
SFR Modified	GTM_SPEi_EIRQ_EN		
Scope:	Driver		
Dependency:	SpeUsage must be USED_BY_GTM_DRIVER		

4.2.17 Container: Cmp

This contains the configuration for CMP module. Maximum multiplicity is 1. If this container is not included, then the registers values of CMP are retained as is.

4.2.17.1 Container: CmpTomChanComparator

This contains the configuration for CMP module for TOM Channel. Multiplicity depends on the number of TOM comparators.

4.2.17.1.1 CmpEnableComparator

Table 420 CmpEnableComparator

Name:	CmpEnableComparator		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Comparator FALSE - Disables the Comparator Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Comparator		
SFR Modified	GTM_CMP_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.1.2 CmpEnableInterruptOnError

Table 421 CmpEnableInterruptOnError

Name:	CmpEnableInterruptOnError		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables Interrupt on Error FALSE - Disables Interrupt on Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Interrupt on Error		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.2 Container: CmpAtomChanComparator

This contains the configuration for CMP module for ATOM Channel. Multiplicity depends on the number of ATOM comparators.

4.2.17.2.1 CmpEnableComparator

Table 422 CmpEnableComparator

Name:	CmpEnableComparator
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Enables the Comparator FALSE - Disables the Comparator Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the Comparator		
SFR Modified	GTM_CMP_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.2 CmpEnableInterruptOnError

Table 423 CmpEnableInterruptOnError

Name:	CmpEnableInterruptOnError		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables Interrupt on Error FALSE: Disables Interrupt on Error Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables Interrupt on Error		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.3 Container: CmpInterrupt

This container contains the Interrupt configuration for the CMP Channel

4.2.17.3.1 CmpInterruptMode

Table 424 CmpInterruptMode

Name:	CmpInterruptMode
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: GTM_INTERRUPT_LEVEL_MODE: Level Mode GTM_INTERRUPT_PULSE_MODE: Pulse Mode GTM_INTERRUPT_PULSE_NOTIFY_MODE: Pulse-Notify Mode GTM_INTERRUPT_SINGLE_PULSE_MODE: Single-Pulse Mode Default: GTM_INTERRUPT_LEVEL_MODE Note: It is recommended to use GTM_INTERRUPT_PULSE_NOTIFY_MODE.		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This decides the Interrupt mode to use		
SFR Modified	GTM_CMP_IRQ_MODE		
Scope:	Driver		
Dependency:	None		

4.2.17.4 Container: CmpErrorInterrupt

This container contains the Error Interrupt configuration for the CMP Channel

4.2.17.4.1 CmpEnableABWC0InterruptSource

Table 425 CmpEnableABWC0InterruptSource

Name:	CmpEnableABWC0InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC0 interrupt source for CMP_EIRQ FALSE: Disables the ABWC0 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC0 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.2 CmpEnableABWC1InterruptSource

Table 426 CmpEnableABWC1InterruptSource

Name:	CmpEnableABWC1InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC1 interrupt source for CMP_EIRQ FALSE: Disables the ABWC1 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC1 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.3 CmpEnableABWC2InterruptSource

Table 427 CmpEnableABWC2InterruptSource

Name:	CmpEnableABWC2InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC2 interrupt source for CMP_EIRQ FALSE: Disables the ABWC2 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC2 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.4 CmpEnableABWC3InterruptSource

Table 428 CmpEnableABWC3InterruptSource

Name:	CmpEnableABWC3InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE: Enables the ABWC3 interrupt source for CMP_EIRQ FALSE: Disables the ABWC3 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC3 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.5 CmpEnableABWC4InterruptSource

Table 429 CmpEnableABWC4InterruptSource

Name:	CmpEnableABWC4InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC4 interrupt source for CMP_EIRQ FALSE: Disables the ABWC4 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC4 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.6 CmpEnableABWC5InterruptSource

Table 430 CmpEnableABWC5InterruptSource

Name:	CmpEnableABWC5InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC5 interrupt source for CMP_EIRQ FALSE: Disables the ABWC5 interrupt source for CMP_EIRQ Default: FALSE		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC5 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.7 CmpEnableABWC6InterruptSource

Table 431 CmpEnableABWC6InterruptSource

Name:	CmpEnableABWC6InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC6 interrupt source for CMP_EIRQ FALSE: Disables the ABWC6 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC6 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.8 CmpEnableABWC7InterruptSource

Table 432 CmpEnableABWC7InterruptSource

Name:	CmpEnableABWC7InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC7 interrupt source for CMP_EIRQ FALSE: Disables the ABWC7 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
SFR Modified	GTM_CMP_IRQ_EN		
Description:	Enables/Disables the ABWC7 interrupt source for CMP_EIRQ		

Scope:	Driver
Dependency:	None

4.2.17.4.9 CmpEnableABWC8InterruptSource

Table 433 CmpEnableABWC8InterruptSource

Name:	CmpEnableABWC8InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC8 interrupt source for CMP_EIRQ FALSE: Disables the ABWC8 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC8 interrupt source for CMP_EIRQ		
SFR Modified			
Scope:	Driver		
Dependency:	None		

4.2.17.4.10 CmpEnableABWC9InterruptSource

Table 434 CmpEnableABWC9InterruptSource

Name:	CmpEnableABWC9InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC9 interrupt source for CMP_EIRQ FALSE: Disables the ABWC9 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC9 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.11 CmpEnableABWC10InterruptSource

Table 435 CmpEnableABWC10InterruptSource

Name:	CmpEnableABWC10InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC10 interrupt source for CMP_EIRQ FALSE: Disables the ABWC10 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC10 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.12 CmpEnableABWC11InterruptSource

Table 436 CmpEnableABWC11InterruptSource

Name:	CmpEnableABWC11InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the ABWC11 interrupt source for CMP_EIRQ FALSE: Disables the ABWC11 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the ABWC11 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.13 CmpEnableTBWC0InterruptSource

Table 437 CmpEnableTBWC0InterruptSource

Name:	CmpEnableTBWC0InterruptSource		
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC0 interrupt source for CMP_EIRQ FALSE: Disables the TBWC0 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC0 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.14 CmpEnableTBWC1InterruptSource

Table 438 CmpEnableTBWC1InterruptSource

Name:	CmpEnableTBWC1InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC1 interrupt source for CMP_EIRQ FALSE: Disables the TBWC1 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC1 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.15 CmpEnableTBWC2InterruptSource

Table 439 CmpEnableTBWC2InterruptSource

Name:	CmpEnableTBWC2InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		

Range:	Values: TRUE: Enables the TBWC2 interrupt source for CMP_EIRQ FALSE: Disables the TBWC2 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC2 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.16 CmpEnableTBWC3InterruptSource

Table 440 CmpEnableTBWC3InterruptSource

Name:	CmpEnableTBWC3InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC3 interrupt source for CMP_EIRQ FALSE: Disables the TBWC3 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC3 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.17 CmpEnableTBWC4InterruptSource

Table 441 CmpEnableTBWC4InterruptSource

Name:	CmpEnableTBWC4InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC4 interrupt source for CMP_EIRQ FALSE: Disables the TBWC4 interrupt source for CMP_EIRQ Default: FALSE		

Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC4 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.18 CmpEnableTBWC5InterruptSource

Table 442 CmpEnableTBWC5InterruptSource

Name:	CmpEnableTBWC5InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC5 interrupt source for CMP_EIRQ FALSE: Disables the TBWC5 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC5 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.19 CmpEnableTBWC6InterruptSource

Table 443 CmpEnableTBWC6InterruptSource

Name:	CmpEnableTBWC6InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC6 interrupt source for CMP_EIRQ FALSE: Disables the TBWC6 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC6 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		

Scope:	Driver
Dependency:	None

4.2.17.4.20 CmpEnableTBWC7InterruptSource

Table 444 CmpEnableTBWC7InterruptSource

Name:	CmpEnableTBWC7InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC7 interrupt source for CMP_EIRQ FALSE: Disables the TBWC7 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC7 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.21 CmpEnableTBWC8InterruptSource

Table 445 CmpEnableTBWC8InterruptSource

Name:	CmpEnableTBWC8InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC8 interrupt source for CMP_EIRQ FALSE: Disables the TBWC8 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC8 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.22 CmpEnableTBWC9InterruptSource

Table 446 CmpEnableTBWC9InterruptSource

Name:	CmpEnableTBWC9InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC0 interrupt source for CMP_EIRQ FALSE: Disables the TBWC0 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC0 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.23 CmpEnableTBWC10InterruptSource

Table 447 CmpEnableTBWC10InterruptSource

Name:	CmpEnableTBWC10InterruptSource		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC10 interrupt source for CMP_EIRQ FALSE: Disables the TBWC10 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC10 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.17.4.24 CmpEnableTBWC11InterruptSource

Table 448 CmpEnableTBWC11InterruptSource

Name:	CmpEnableTBWC11InterruptSource		
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Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE: Enables the TBWC11 interrupt source for CMP_EIRQ FALSE: Disables the TBWC11 interrupt source for CMP_EIRQ Default: FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	Enables/Disables the TBWC11 interrupt source for CMP_EIRQ		
SFR Modified	GTM_CMP_IRQ_EN		
Scope:	Driver		
Dependency:	None		

4.2.18 Container: GtmConnections

This container defines the connections between GTM and other Modules. Maximum Multiplicity is 1. If this container is not included, then the related registers values are retained as is.

4.2.18.1 Container: GtmToutConnections

This container defines the connections between GTM and Port. Maximum Multiplicity is the number of TOUT registers. If this container is not included, then the GTM to PORT interface registers values are retained as is. This container is applicable only in case of configuration using BMD. In case of Xdm configuration, TOM and ATOM modules provide a configuration for this purpose.

4.2.18.2 Container: GtmConnectionToMsc

This container defines the connections between GTM and MSC. Maximum Multiplicity is 1. If this container is not included, then the GTM to MSC interface registers values are retained as is.

4.2.18.2.1 Container: GtmSelectSet1Output

This container defines the output for Set 1. Multiplicity is 16, corresponding to the 16 outputs of Set1. GtmSelectSet1Output_x where x corresponds to the Set1 output.

4.2.18.2.1.1 GtmSelectSet1Output

Table 449 GtmSelectSet1Output

Name:	GtmSelectSet1Output		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--

	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines the Output for Set 1		
SFR Modified	GTM_MSCSET_1S0_CON1		
Scope:	Driver		
Dependency:			

4.2.18.2.2 Container: GtmSelectSet2Output

This container defines the output for Set 2. Multiplicity is 16, corresponding to the 16 outputs of Set2. GtmSelectSet2Output_x where x corresponds to the Set2 output.

4.2.18.2.2.1 GtmSelectSet2Output

Table 450 GtmSelectSet2Output

Name:	GtmSelectSet2Output		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This container defines the Output for Set 2		
SFR Modified	GTM_MSCSET_1S0_CON2		
Scope:	Driver		
Dependency:			

4.2.18.2.3 Container: GtmSelectSet3Output

This container defines the output for Set 3. Multiplicity is 16, corresponding to the 16 outputs of Set3. GtmSelectSet3Output_x where x corresponds to the Set3 output.

4.2.18.2.3.1 GtmSelectSet3Output

Table 451 GtmSelectSet3Output

Name:	GtmSelectSet3Output		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--

	Post build	x	Variant PB
Description:	This container defines the Output for Set 3		
SFR Modified	GTM_MSCSET_1S0_CON3		
Scope:	Driver		
Dependency:			

4.2.18.2.4 Container: GtmSelectSet4Output

This container defines the output for Set 4. Multiplicity is 16, corresponding to the 16 outputs of Set4. GtmSelectSet4Output_x where x corresponds to the Set4 output.

4.2.18.2.4.1 GtmSelectSet4Output

Table 452 GtmSelectSet4Output

Name:	GtmSelectSet4Output		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This container defines the Output for Set 4		
SFR Modified	GTM_MSCSET_1S0_CON4		
Scope:	Driver		
Dependency:			

4.2.18.2.5 Container: GtmConnectionToMsc

This container defines the input connection to MSC. Multiplicity depends on the number of MSC modules.

4.2.18.2.5.1 Container: GtmOutSelForMscInLow

This container defines the GTM output Set Connections to Input Low connections of MSC. GtmOutSelForMscInLow_x where x = 0 to 15, the lower 16 inputs to MSC.

4.2.18.2.5.1.1 GtmOutSelForMscInputLow

Table 453 GtmOutSelForMscInputLow

Name:	GtmOutSelForMscInputLow
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: USE_SET1_SIGNAL USE_SET2_SIGNAL USE_SET3_SIGNAL USE_SET4_SIGNAL Default: USE_SET1_SIGNAL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This container defines the GTM output Set Connections to Input Low (i.e. 0-15)connections of MSC		
SFR Modified	GTM_MSC1INLCON		
Scope:	Driver		
Dependency:	None		

4.2.18.2.5.2 Container: GtmOutSelForMscInHigh

This container defines the GTM output Set Connections to Input High connections of MSC. GtmOutSelForMscInHigh_x where x = 0 to 15, the higher 16 inputs to MSC

4.2.18.2.5.2.1 GtmOutSelForMscInHigh

Table 454 GtmOutSelForMscInHigh

Name:	GtmOutSelForMscInHigh		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: USE_SET1_SIGNAL USE_SET2_SIGNAL USE_SET3_SIGNAL USE_SET4_SIGNAL Default: USE_SET1_SIGNAL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This container defines the GTM output Set Connections to Input High (i.e. 0-15)connections of MSC		
SFR Modified	GTM_MSC1INHCON		
Scope:	Driver		
Dependency:	None		

4.2.18.3 Container: GtmConnectionToDsadc

This container defines the Connections between GTM and DSADC. Maximum multiplicity is 1. If this container is not included, then the GTM to DSADC interface registers values are retained as is.

4.2.18.3.1.1 GtmDsadcInputToTim0Channel

Table 455 GtmDsadcInputToTim0Channel

Name:	GtmDsadcInputToTim0Channel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SEL0_DSADC0_CONNECTED_TO_SRM0: DSADC0 will be connected to SRM0 SEL1_DSADC1_CONNECTED_TO_SRM0: DSADC1 will be connected to SRM0 SEL2_DSADC2_CONNECTED_TO_SRM0: DSADC2 will be connected to SRM0 SEL3_DSADC3_CONNECTED_TO_SRM0: DSADC3 will be connected to SRM0 SEL4_DSADC4_CONNECTED_TO_SRM0: DSADC4 will be connected to SRM0 SEL5_DSADC5_CONNECTED_TO_SRM0: DSADC5 will be connected to SRM0 SEL10_DSADCY_CONNECTED_TO_SAUL: DSADCY will be connected to SRM0 SEL11_DSADCY_CONNECTED_TO_SBLL: DSADCY will be connected to SRM0 SEL12_DSADCZ_CONNECTED_TO_SAUL: DSADCZ will be connected to SRM0 SEL13_DSADCZ_CONNECTED_TO_SBLL: DSADCZ will be connected to SRM0 SEL15_DRIVE_0: Drive Logic 0 to SRM0 Default: SEL0_DSADC0_CONNECTED_TO_SRM0 Note: (only for ChannelNumber = 0 to 5, for ChannelNumber = 6,7 SAUL and SBLL connections not possible): if ChannelNumber%2==0 Then Y=ChannelNumber and Z=Y+1 Else Z=ChannelNumber and Y=Z+1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines the Connections between DSADC and TIM Module 1 Channel x where x is the index number for this parameter and between DSADC and SAUL, SBLL		
SFR Modified	GTM_DSADCINSEL0		
Scope:	Driver		
Dependency:	None		

4.2.18.3.2 Container: GtmDsadcInputToTim1Ch

This defines the Connections between DSADC and TIM1 Channels. Multiplicity depends on the number of DSADC to TIM connections. GtmDsadcInputToTim1Ch_x, where x = 0- 7, Channel number for TIM Module 1.

4.2.18.3.2.1 GtmDsadcInputToTim1Channel

Table 456 GtmDsadcInputToTim1Channel

Name:	GtmDsadcInputToTim1Channel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SEL0_DSADC0_CONNECTED_TO_SRM0: DSADC0 will be connected to SRM0 SEL1_DSADC1_CONNECTED_TO_SRM0: DSADC1 will be connected to SRM0 SEL2_DSADC2_CONNECTED_TO_SRM0: DSADC2 will be connected to SRM0 SEL3_DSADC3_CONNECTED_TO_SRM0: DSADC3 will be connected to SRM0 SEL4_DSADC4_CONNECTED_TO_SRM0: DSADC4 will be connected to SRM0 SEL5_DSADC5_CONNECTED_TO_SRM0: DSADC5 will be connected to SRM0 SEL10_DSADCY_CONNECTED_TO_SAUL:DSADCY will be connected to SRM0 SEL11_DSADCY_CONNECTED_TO_SBLL: DSADCY will be connected to SRM0 SEL12_DSADCZ_CONNECTED_TO_SAUL: DSADCZ will be connected to SRM0 SEL13_DSADCZ_CONNECTED_TO_SBLL: DSADCZ will be connected to SRM0 SEL15_DRIVE_0: Drive Logic 0 to SRM0 Default: SEL0_DSADC0_CONNECTED_TO_ SRM0 Note: (only for ChannelNumber = 0 to 5, for ChannelNumber = 6,7 SAUL and SBLL connections not possible): if ChannelNumber%2==0 Then Y=ChannelNumber and Z=Y+1 Else Z=ChannelNumber and Y=Z+1		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines the Connections between DSADC and TIM Module 1 Channel x where x is the index number for this parameter and between DSADC and SAUL, SBLL		
SFR Modified	GTM_DSADCINSEL1		
Scope:	Driver		
Dependency:	None		

4.2.18.3.3 Container: GtmDsadcInputToTim2Ch

This defines the Connections between DSADC and TIM2 Channels. Multiplicity depends on the number of DSADC to TIM connections. GtmDsadcInputToTim2Ch_x, where x = 0- 7, Channel number for TIM Module 2.

4.2.18.3.3.1 GtmDsadcInputToTim2Channel

Table 457 GtmDsadcInputToTim2Channel

Name:	GtmDsadcInputToTim2Channel		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: SEL0_DSADC0_CONNECTED_TO_SRM0: DSADC0 will be connected to SRM0 SEL1_DSADC1_CONNECTED_TO_SRM0: DSADC1 will be connected to SRM0 SEL2_DSADC2_CONNECTED_TO_SRM0: DSADC2 will be connected to SRM0 SEL3_DSADC3_CONNECTED_TO_SRM0: DSADC3 will be connected to SRM0 SEL4_DSADC4_CONNECTED_TO_SRM0: DSADC4 will be connected to SRM0 SEL5_DSADC5_CONNECTED_TO_SRM0: DSADC5 will be connected to SRM0 SEL15_DRIVE_0: Drive Logic 0 to SRM0 Default: SEL0_DSADC0_CONNECTED_TO_SRM0		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines the Connections between DSADC and TIM Module 2 Channel x where x is the index number for this parameter		
SFR Modified	GTM_DSADCINSEL2		
Scope:	Driver		
Dependency:	None		

4.2.18.3.4 Container: GtmOutputSelfForDsadc

This defines which TOM / ATOM channel output is used as DSADCx trigger. Multiplicity depends on the number of DSADC modules. GtmOutputSelfForDsadc_x where x is the DSADC module number 0 – 5.

4.2.18.3.4.1 GtmDsadcTrigger0Select

Table 458 GtmDsadcTrigger0Select

Name:	GtmDsadcTrigger0Select
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: SEL0_TOM0_6_CONNECTED_TO_DSADC: TOM0 Channel 6 is used as DSADC Trigger 0 SEL1_TOM0_7_CONNECTED_TO_DSADC: TOM0 Channel 7 is used as DSADC Trigger 0 SEL2_TOM0_13_CONNECTED_TO_DSADC: TOM0 Channel 13 is used as DSADC Trigger 0 SEL3_TOM0_14_CONNECTED_TO_DSADC: TOM0 Channel 14 is used as DSADC Trigger 0 SEL4_ATOM0_4_CONNECTED_TO_DSADC: ATOM0 Channel 4 is used as DSADC Trigger 0 SEL5_ATOM0_5_CONNECTED_TO_DSADC: ATOM0 Channel 5 is used as DSADC Trigger 0 SEL6_ATOM0_6_CONNECTED_TO_DSADC: ATOM0 Channel 6 is used as DSADC Trigger 0 SEL7_ATOM0_7_CONNECTED_TO_DSADC: ATOM0 Channel 7 is used as DSADC Trigger 0 Default: SEL0_TOM0_6_CONNECTED_TO_DSADC		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines which TOM / ATOM channel output is used as DSADCx trigger0		
SFR Modified	GTM_DSADCOUTSEL00		
Scope:	Driver		
Dependency:	None		

4.2.18.3.4.2 GtmDsadcTrigger1Select

Table 459 GtmDsadcTrigger1Select

Name:	GtmDsadcTrigger1Select
Source:	This parameter is microcontroller specific.
File:	Gtm_PBCfg.c

Range:	Values: SEL0_TOM2_6_CONNECTED_TO_DSADC: TOM2 Channel 6 is used as DSADC Trigger 1 SEL1_TOM2_7_CONNECTED_TO_DSADC: TOM2 Channel 7 is used as DSADC Trigger 1 SEL2_TOM2_13_CONNECTED_TO_DSADC: TOM2 Channel 13 is used as DSADC Trigger 1 SEL3_TOM2_14_CONNECTED_TO_DSADC: TOM2 Channel 14 is used as DSADC Trigger 1 SEL4_ATOM2_4_CONNECTED_TO_DSADC: ATOM2 Channel 4 is used as DSADC Trigger 1 SEL5_ATOM2_5_CONNECTED_TO_DSADC: ATOM2 Channel 5 is used as DSADC Trigger 1 SEL6_ATOM2_6_CONNECTED_TO_DSADC: ATOM2 Channel 6 is used as DSADC Trigger 1 SEL7_ATOM2_7_CONNECTED_TO_DSADC: ATOM2 Channel 7 is used as DSADC Trigger 1 Default: SEL0_TOM2_6_CONNECTED_TO_DSADC		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines which TOM / ATOM channel output is used as DSADCx trigger1		
SFR Modified	GTM_DSADCOUTSEL10		
Scope:	Driver		
Dependency:	None		

4.2.18.4 Container: GtmTriggerForAdc

This Specifies the TOM/ATOM channels that can be used as ADC Triggers. Multiplicity depends on the number of ADC modules.

4.2.18.4.1 GtmTrigger0Select

Table 460 GtmTrigger0Select

Name:	GtmTrigger0Select		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This Specifies the TOM/ATOM channel that will be used as ADC Trigger0		
SFR Modified	GTM_ADCTRIG0OUT0		
Scope:	Driver		
Dependency:			

4.2.18.4.2 GtmTrigger1Select

Table 461 GtmTrigger1Select

Name:	GtmTrigger1Select		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This Specifies the TOM/ATOM channel that will be used as ADC Trigger1		
SFR Modified	GTM_ADCTRIG1OUT0		
Scope:	Driver		
Dependency:			

4.2.18.5 Container: GtmTtcanTrigger

This defines the GTM Triggers to TTCAN. Multiplicity depends on the number of TTCAN modules.

4.2.18.5.1 GtmTriggerSelect

Table 462 GtmTriggerSelect

Name:	GtmTriggerSelect		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This Specifies the TOM/ATOM channel that will be used as TTCAN Trigger		
SFR Modified	GTM_CANOUTSEL		
Scope:	Driver		
Dependency:			

4.2.18.6 Container: GtmPsi5Triggers

This defines the TOM /ATOM channel output that will be used as PSI5x and PSI5Sx triggers.

4.2.18.7 Container: GtmPsi5Trigger

This defines which TOM /ATOM channel output is used as PSI5x trigger. Multiplicity depends on the number of PSI5x modules.

4.2.18.7.1 GtmPsi5Trigger

Table 463 GtmPsi5Trigger

Name:	GtmPsi5Trigger		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines which TOM / ATOM channel output is used as PSI5x trigger		
SFR Modified	GTM_PSI5OUTSEL0		
Scope:	Driver		
Dependency:	none		

4.2.18.8 Container: GtmPsi5STrigger

This defines which TOM /ATOM channel output is used as PSI5Sx trigger. Multiplicity depends on the number of PSI5S modules.

4.2.18.8.1 GtmPsi5STrigger

Table 464 GtmPsi5STrigger

Name:	GtmPsi5STrigger		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: Refer [1] for details Default: Refer [1] for details		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines which TOM / ATOM channel output is used as PSI5Sx trigger		
SFR Modified	GTM_PSI5SOUTSEL0		
Scope:	Driver		
Dependency:			

4.2.19 Container: GtmAccessEnable

This defines the Access Control of Gtm Module from different Bus Master. Maximum multiplicity is 1. If this container is not included, then the registers value of Access Enable is retained as is.

4.2.19.1 Container: GtmEnableAccForTagId

This defines the Access Control of Gtm Module from different Bus Masters. Multiplicity is 32. GtmEnableAccForTagId_x where x corresponds to the bus master with that Tag id.

4.2.19.1.1 GtmEnableAccessForTagId

Table 465 GtmEnableAccessForTagId

Name:	GtmEnableAccessForTagId		
Source:	This parameter is microcontroller specific.		
File:	Gtm_PBCfg.c		
Range:	Values: TRUE – Access to GTM module allowed from the Bus Master with this Tag ID FALSE - Access to GTM module denied from the Bus Master with this Tag ID Default: TRUE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	Variant PB
Description:	This defines the Access Control of Gtm Module from different Bus Master		
SFR Modified	GTM_ACCEN0		
Scope:	Driver		
Dependency:			

5 API Documentation

The API documentation describes the syntax and semantics of the API functions as well as it gives some additional implementation specific behavior description not defined by the AUTOSAR ICU specification.

Implementation specific type definitions, which are required for configuration and convenience, are also documented.

5.1 API type definition

5.1.1 Gtm_ConfigType

Table 466 Gtm_ConfigType

Syntax:	Gtm_ConfigType
Type:	Struct
File:	Gtm.h
Range:	This type definition is used to configure over all GTM configuration.
Description:	Type to define the GTM Driver configuration that holds the pointer to actual instance.

5.2 API Functions

5.2.1 Gtm_Init

Table 467 Specification of API Gtm_Init

Service Name:	Gtm_Init	
Syntax:	Std_ReturnType Gtm_Init (const Gtm_ConfigType *ConfigPtr)	
Source:	This is a vendor defined API.	
Service ID:	--	
Sync / Async:	Synchronous	
Re-entrancy:	Non- Reentrant	
Parameters (in):	ConfigPtr - Pointer to GTM Configuration Structure	
Parameters (out):	None	
Return Value:	E_OK	If GTM hardware is successfully initialized.
	E_NOT_OK	If GTM hardware is not initialized successfully.
Description:	All configured resources of all the Sub-Modules of GTM are initialized and non-configured resources are left as is. If McuResetSfrAtInit is marked as true then GTM module kernel reset will be performed before clock to GTM is enabled. (Refer Mcu user manual for McuResetSfrAtInit configuration parameter). Gtm_Init can be called only from the MCU Driver with Mcu_Init. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then Gtm_Init Api will run in user1 or supervisory mode depending on McuUserModelInitApiEnable. McuUserModelInitApiEnable is 'true' then Gtm_Init API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModelInitApiEnable is 'true' then Gtm_Init API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuRunningInUser0Mode and McuUserModelInitApiEnable configuration parameters).	
Caveats:	It should be ensured that no MCS RAM Reset is ongoing during the initialization process. SMU ECC alarms for MCS, DPLL and FIFO RAM occurred during Gtm_Init can be ignored or SMU ECC alarms can be avoided by disabling these alarms before calling Gtm_Init and alarms can be re-enabled once Gtm_Init is done. Following services should be called after calling Gtm_Init and before accessing RAM of FIFO, MCS and DPLL. Gtm_FlushFifo Gtm_ClearMcsRam Gtm_InitDpllRam.	
Configuration:	--	
DET:	None	

Service Name:	Gtm_Init
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeInitApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeInitApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeInitApiEnable is 'False'.

5.2.2 Gtm_Delnit

Table 468 Specification of API Gtm_Delnit

Service Name:	Gtm_Delnit
Syntax:	void Gtm_Delnit (void)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Non- Reentrant
Parameters (in):	None
Parameters (out):	None
Return Value:	None
Description:	All resources of all the Sub-Modules of GTM and global variables are uninitialized by kernel reset. All the interrupt SCR registers configured by Gtm module are deinitialized. Gtm_Delnit can be called only from the MCU Driver with Mcu_Delnit. Availability of Gtm_Delnit api will be controlled by McuDelnitApi configuration parameter. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then Gtm_Delnit Api will run in user1 or supervisory mode depending on McuUserModeDelnitApiEnable. McuUserModeDelnitApiEnable is 'true' then Gtm_Delnit API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeDelnitApiEnable is 'true' then Gtm_Delnit API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuRunningInUser0Mode and McuUserModeDelnitApiEnable configuration parameters).

Service Name:	Gtm_DelInit
Caveats:	It should be ensured that no MCS RAM Reset is ongoing during the de-initialization process.
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeDelInitApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeDelInitApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeDelInitApiEnable is 'False'.

5.2.3 Gtm_InitCheck

Table 469 Specification of API Gtm_InitCheck

Service Name:	Gtm_InitCheck
Syntax:	void Gtm_InitCheck(const Gtm_ConfigType *ConfigPtr)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Non- Reentrant
Parameters (in):	ConfigPtr - Pointer to GTM Configuration Structure
Parameters (out):	None
Return Value:	None
Description:	All the configured resources are initialized to the SFR's by Gtm_Init(). In Gtm_InitCheck SFR's which are essential for proper functioning of safety modules such as GPT/ICU/PWM are verified to the same config pointers. Gtm_InitCheck can be called only from the MCU Driver with Mcu_InitCheck. Same Configuration Pointer will be used to Initialize Mcu_Init and to call Mcu_Initcheck. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then Gtm_InitCheck Api will run is user1 or supervisory mode depending on McuUserModeInitApiEnable. McuUserModeInitApiEnable is 'true' then Gtm_InitCheck API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeInitApiEnable is 'true' then Gtm_InitCheck API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true').

Service Name:	Gtm_InitCheck
	(Refer Mcu user manual for McuRunningInUser0Mode and McuUserModeInitApiEnable configuration parameters).
Caveats:	None
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeInitApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeInitApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeInitApiEnable is 'False'.

5.2.4 Gtm_SetTomHostTrigger

Table 470 Specification of API Gtm_SetTomHostTrigger

Service Name:	Gtm_SetTomHostTrigger
Syntax:	<pre>inline void Gtm_SetTomHostTrigger (uint8 ModuleNo, uint8 TgcNumber)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TOM Module Number TgcNumber – TOM TGC Number
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the Host Trigger Bit corresponding to the Tgc If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false').

Service Name:	Gtm_SetTomHostTrigger
	McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.5 Gtm_SetAtomHostTrigger

Table 471 Specification of API Gtm_SetAtomHostTrigger

Service Name:	Gtm_SetTomHostTrigger
Syntax:	<pre>inline void Gtm_SetAtomHostTrigger (uint8 ModuleNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – ATOM Module Number
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the Host Trigger Bit corresponding to the Module. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false').

Service Name:	Gtm_SetTomHostTrigger
	McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.6 Gtm_SetTomCounterCn0

Table 472 Specification of API Gtm_SetTomCounterCn0

Service Name:	Gtm_SetTomCounterCn0
Syntax:	<pre>inline void Gtm_SetTomCounterCn0 (uint8 ModuleNo, uint8 ChannelNo, uint16 Cn0Value)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TOM Module Number ChannelNo – TOM Channel Number Cn0Value – CN0 Value to be written
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of CN0 of the TOM channel with the value given by Cn0Value. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode

Service Name:	Gtm_SetTomCounterCn0
	is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.7 Gtm_SetAtomCounterCn0

Table 473 Specification of API Gtm_SetAtomCounterCn0

Service Name:	Gtm_SetAtomCounterCn0
Syntax:	<pre>inline void Gtm_SetAtomCounterCn0 (uint8 ModuleNo, uint8 ChannelNo, uint32 Cn0Value)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – ATOM Module Number ChannelNo – ATOM Channel Number Cn0Value – CN0 Value to be written

Service Name:	Gtm_SetAtomCounterCn0
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of CN0 of the ATOM channel with the value given by Cn0Value. Cn0Value must be of max 24 bits. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.8 Gtm_GetTomCounterCn0

Table 474 Specification of API Gtm_GetTomCounterCn0

Service Name:	Gtm_GetTomCounterCn0
Syntax:	<pre>inline uint16 Gtm_GetTomCounterCn0 (uint8 ModuleNo, uint8 ChannelNo,)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous

Service Name:	Gtm_GetTomCounterCn0
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None
Return Value:	Cn0 value
Description:	This inline function returns the value of CN0 of the TOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.9 Gtm_GetAtomCounterCn0

Table 475 Specification of API Gtm_GetAtomCounterCn0

Service Name:	Gtm_GetAtomCounterCn0
Syntax:	<pre>inline uint32 Gtm_GetAtomCounterCn0 (uint8 ModuleNo, uint8 ChannelNo,</pre>

Service Name:	Gtm_GetAtomCounterCn0
	)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – ATOM Module Number ChannelNo – ATOM Channel Number
Parameters (out):	None
Return Value:	Cn0 value
Description:	This inline function returns the value of CN0 of the ATOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.10 Gtm_SetTomCompareValCm0

Table 476 Specification of API Gtm_SetTomCompareValCm0

Service Name:	Gtm_SetTomCompareValCm0
Syntax:	inline void Gtm_SetTomCompareValCm0

Service Name:	Gtm_SetTomCompareValCm0
	<pre>(uint8 ModuleNo, uint8 ChannelNo, uint16 Cm0Value)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –TOM Module Number ChannelNo – TOM Channel Number Cm0Value – CM0 value to be written
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of CM0 of the TOM channel with the value given by Cm0Value. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.11 Gtm_SetAtomCompareValCm0

Table 477 Specification of API Gtm_SetAtomCompareValCm0

Service Name:	Gtm_SetAtomCompareValCm0
Syntax:	<pre>inline void Gtm_SetAtomCompareValCm0 (uint8 ModuleNo, uint8 ChannelNo, uint32 Cm0Value)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – ATOM Module Number ChannelNo – ATOM Channel Number Cm0Value – CM0 value to be written
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of CM0 of the ATOM channel with the value given by Cm0Value. Cm0Value must be of max 24 bits. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'.

Service Name:	Gtm_SetAtomCompareValCm0
	This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.12 Gtm_GetTomCompareValCm0

Table 478 Specification of API Gtm_GetTomCompareValCm0

Service Name:	Gtm_GetTomCompareValCm0
Syntax:	<pre>inline uint16 Gtm_GetTomCompareValCm0 (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None
Return Value:	CM0 Value
Description:	This inline function returns the value of CM0 of the TOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'.

Service Name:	Gtm_GetTomCompareValCm0
	This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.13 Gtm_GetAtomCompareValCm0

Table 479 Specification of API Gtm_GetAtomCompareValCm0

Service Name:	Gtm_GetAtomCompareValCm0
Syntax:	<pre>inline uint32 Gtm_GetAtomCompareValCm0 (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –ATOM Module Number ChannelNo – ATOM Channel Number
Parameters (out):	None
Return Value:	CM0 Value
Description:	This inline function returns the value of CM0 of the ATOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided -

Service Name:	Gtm_GetAtomCompareValCm0
	configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'.
	This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'.
	This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.14 Gtm_SetTomCompareValCm1

Table 480 Specification of API Gtm_SetTomCompareValCm1

Service Name:	Gtm_SetTomCompareValCm1
Syntax:	<pre>inline void Gtm_SetTomCompareValCm1 (uint8 ModuleNo, uint8 ChannelNo, uint16 Cm1Value)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –TOM Module Number ChannelNo – TOM Channel Number Cm1Value – CM1 value to be written
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of CM1 of the TOM channel with the value given by Cm1Value. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--

Service Name:	Gtm_SetTomCompareValCm1
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.15 Gtm_SetAtomCompareValCm1

Table 481 Specification of API Gtm_SetAtomCompareValCm1

Service Name:	Gtm_SetAtomCompareValCm1
Syntax:	<pre>inline void Gtm_SetAtomCompareValCm1 (uint8 ModuleNo, uint8 ChannelNo, uint32 Cm1Value)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –ATOM Module Number ChannelNo – ATOM Channel Number Cm1Value – CM1 value to be written
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of CM1 of the ATOM channel with the value given by Cm1Value. CM1Value must be limited to 24 bits. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode,

Service Name:	Gtm_SetAtomCompareValCm1
	else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.16 Gtm_GetTomCompareValCm1

Table 482 Specification of API Gtm_GetTomCompareValCm1

Service Name:	Gtm_GetTomCompareValCm1
Syntax:	<pre>inline uint16 Gtm_GetTomCompareValCm1 (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None
Return Value:	CM1 value
Description:	This inline function returns the value of CM1 of the TOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is

Service Name:	Gtm_GetTomCompareValCm1
	'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.17 Gtm_GetAtomCompareValCm1

Table 483 Specification of API Gtm_GetAtomCompareValCm1

Service Name:	Gtm_GetAtomCompareValCm1
Syntax:	<pre>inline uint32 Gtm_GetAtomCompareValCm1 (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –ATOM Module Number ChannelNo – ATOM Channel Number
Parameters (out):	None
Return Value:	CM1 value
Description:	This inline function returns the value of CM1 of the ATOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or

Service Name:	Gtm_GetAtomCompareValCm1
	supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.18 Gtm_SetTomShadowValSr0

Table 484 Specification of API Gtm_SetTomShadowValSr0

Service Name:	Gtm_SetTomShadowValSr0
Syntax:	<pre>inline void Gtm_SetTomShadowValSr0 (uint8 ModuleNo, uint8 ChannelNo, uint16 Sr0Value)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TOM Module Number ChannelNo – TOM Channel Number Sr0Value – SR0 Value
Parameters (out):	None

Service Name:	Gtm_SetTomShadowValSr0
Return Value:	None
Description:	This inline function sets the value of SR0 of the TOM channel with the value given by Sr0Value. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.19 Gtm_SetAtomShadowValSr0

Table 485 Specification of API Gtm_SetAtomShadowValSr0

Service Name:	Gtm_SetAtomShadowValSr0
Syntax:	<pre>inline void Gtm_SetAtomShadowValSr0 (uint8 ModuleNo, uint8 ChannelNo, uint32 Sr0Value)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous

Service Name:	Gtm_SetAtomShadowValSr0
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –ATOM Module Number ChannelNo – ATOM Channel Number Sr0Value – SR0 Value
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of SR0 of the ATOM channel with the value given by Sr0Value. Sr0Value must be max 24 bits If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters)..
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.20 Gtm_GetTomShadowValSr0

Table 486 Specification of API Gtm_GetTomShadowValSr0

Service Name:	Gtm_GetTomShadowValSr0
Syntax:	<pre>inline uint16 Gtm_GetTomShadowValSr0 (uint8 ModuleNo, uint8 ChannelNo</pre>

Service Name:	Gtm_GetTomShadowValSr0
	)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None
Return Value:	SR0 Value
Description:	This inline function returns the value of SR0 of the TOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.21 Gtm_GetAtomShadowValSr0

Table 487 Specification of API Gtm_GetAtomShadowValSr0

Service Name:	Gtm_GetAtomShadowValSr0
Syntax:	inline uint32 Gtm_GetAtomShadowValSr0

Service Name:	Gtm_GetAtomShadowValSr0
	<pre>(uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –ATOM Module Number ChannelNo – ATOM Channel Number
Parameters (out):	None
Return Value:	SR0 Value
Description:	This inline function returns the value of SR0 of the ATOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.22 Gtm_SetTomShadowValSr1

Table 488 Specification of API Gtm_SetTomShadowValSr1

Service Name:	Gtm_SetTomShadowValSr1
Syntax:	<pre> inline void Gtm_SetTomShadowValSr1 (uint8 ModuleNo, uint8 ChannelNo, uint16 Sr1Value) </pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TOM Module Number ChannelNo – TOM Channel Number Sr1Value – SR1 value
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of SR1 of the TOM channel with the value given by Sr1Value. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.23 Gtm_SetAtomShadowValSr1

Table 489 Specification of API Gtm_SetAtomShadowValSr1

Service Name:	Gtm_SetAtomShadowValSr1
Syntax:	<pre> inline void Gtm_SetAtomShadowValSr1 (uint8 ModuleNo, uint8 ChannelNo, uint32 Sr1Value) </pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –ATOM Module Number ChannelNo – ATOM Channel Number Sr1Value – SR1 value
Parameters (out):	None
Return Value:	None
Description:	This inline function sets the value of SR1 of the ATOM channel with the value given by Sr1Value. Sr1Value must be limited to 24 bits. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'.

Service Name:	Gtm_SetAtomShadowValSr1
	This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.24 Gtm_GetTomShadowValSr1

Table 490 Specification of API Gtm_GetTomShadowValSr1

Service Name:	Gtm_GetTomShadowValSr1
Syntax:	<pre>inline uint16 Gtm_GetTomShadowValSr1 (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo –TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None
Return Value:	SR1 value
Description:	This inline function returns the value of SR1 of the TOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'.

Service Name:	Gtm_GetTomShadowValSr1
	This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.25 Gtm_GetAtomShadowValSr1

Table 491 Specification of API Gtm_GetAtomShadowValSr1

Service Name:	Gtm_GetAtomShadowValSr1
Syntax:	<pre>inline uint32 Gtm_GetAtomShadowValSr1 (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – ATOM Module Number ChannelNo – ATOM Channel Number
Parameters (out):	None
Return Value:	SR1 value
Description:	This inline function returns the value of SR1 of the ATOM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided -

Service Name:	Gtm_GetAtomShadowValSr1
	configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'.
	This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'.
	This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.26 Gtm_SetTimCounterShadowValCnts

Table 492 Specification of API Gtm_SetTimCounterShadowValCnts

Service Name:	Gtm_SetTimCounterShadowValCnts
Syntax:	<pre>inline void Gtm_SetTimCounterShadowValCnts (uint8 ModuleNo, uint8 ChannelNo, uint32 CntsValue)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number CntsValue – CNTS value to be written
Parameters (out):	None
Return Value:	None
Description:	Inline function to set CNTS value of the corresponding TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--

Service Name:	Gtm_SetTimCounterShadowValCnts
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	The channel should be in TIPM or TBCM mode for CNTS to be editable.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.27 Gtm_GetTimCounterShadowValCnts

Table 493 Specification of API Gtm_GetTimCounterShadowValCnts

Service Name:	Gtm_GetTimCounterShadowValCnts
Syntax:	<pre>inline uint32 Gtm_GetTimCounterShadowValCnts (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number
Parameters (out):	None
Return Value:	CNTS value
Description:	Inline function to return CNTS value of the corresponding TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true').

Service Name:	Gtm_GetTimCounterShadowValCnts
	(Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.28 Gtm_GetTimCounterValCnt

Table 494 Specification of API Gtm_GetTimCounterValCnt

Service Name:	Gtm_GetTimCounterValCnt
Syntax:	<pre>inline uint32 Gtm_GetTimCounterValCnt (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number
Parameters (out):	None
Return Value:	CNT value
Description:	Inline function to return CNT value of the corresponding TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false').

Service Name:	Gtm_GetTimCounterValCnt
	McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.29 Gtm_GetTimGeneralPurposeRegGpr0

Table 495 Specification of API Gtm_GetTimGeneralPurposeRegGpr0

Service Name:	Gtm_GetTimGeneralPurposeRegGpr0
Syntax:	<pre>inline uint32 Gtm_GetTimGeneralPurposeRegGpr0 (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number
Parameters (out):	None
Return Value:	GPR0 value
Description:	Inline function to return GPR0 value of the corresponding TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable.

Service Name:	Gtm_GetTimGeneralPurposeRegGpr0
	McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.30 Gtm_GetTimGeneralPurposeRegGpr1

Table 496 Specification of API Gtm_GetTimGeneralPurposeRegGpr1

Service Name:	Gtm_GetTimGeneralPurposeRegGpr1
Syntax:	<pre>inline uint32 Gtm_GetTimGeneralPurposeRegGpr1 (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number
Parameters (out):	None
Return Value:	GPR1 value
Description:	Inline function to return GPR1 value of the corresponding TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode

Service Name:	Gtm_GetTimGeneralPurposeRegGpr1
	is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.31 Gtm_SetTimTimeoutVal

Table 497 Specification of API Gtm_SetTimTimeoutVal

Service Name:	Gtm_SetTimTimeoutVal
Syntax:	<pre>inline void Gtm_SetTimTimeoutVal (uint8 ModuleNo, uint8 ChannelNo, uint8 TimeOutDuration)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number TimeOutDuration – Timeout value to set

Service Name:	Gtm_SetTimTimeoutVal
Parameters (out):	None
Return Value:	None
Description:	Inline function to set the Timeout duration for the TIM Channel to the parameter TimeOutDuration. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.32 Gtm_EnableTimTimeoutDetection

Table 498 Specification of API Gtm_EnableTimTimeoutDetection

Service Name:	Gtm_EnableTimTimeoutDetection
Syntax:	<pre>inline void Gtm_EnableTimTimeoutDetection (uint8 ModuleNo, uint8 ChannelNo,)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous

Service Name:	Gtm_EnableTimTimeoutDetection
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number
Parameters (out):	None
Return Value:	None
Description:	Inline function to enable Timeout detection for the TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.33 Gtm_DisableTimTimeoutDetection

Table 499 Specification of API Gtm_DisableTimTimeoutDetection

Service Name:	Gtm_DisableTimTimeoutDetection
Syntax:	<pre>inline void Gtm_DisableTimTimeoutDetection (uint8 ModuleNo, uint8 ChannelNo,)</pre>

Service Name:	Gtm_DisableTimTimeoutDetection
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number
Parameters (out):	None
Return Value:	None
Description:	Inline function to disable Timeout detection for the TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.34 Gtm_SelectTimTimeoutClock

Table 500 Specification of API Gtm_SelectTimTimeoutClock

Service Name:	Gtm_SelectTimTimeoutClock
Syntax:	<pre>inline void Gtm_SelectTimTimeoutClock (</pre>

Service Name:	Gtm_SelectTimTimeoutClock
	uint8 ModuleNo, uint8 ChannelNo, Gtm_ConfigurableClockType ClockSource)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number ClockSource - Clock Source for Timeout detection
Parameters (out):	None
Return Value:	None
Description:	Inline function to select the clock source used for Timeout detection for the TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.35 Gtm_GetTimCurrentTimeoutVal

Table 501 Specification of API Gtm_GetTimCurrentTimeoutVal

Service Name:	Gtm_GetTimCurrentTimeoutVal
Syntax:	<pre>inline Gtm_ConfigurableClockType Gtm_GetTimCurrentTimeoutVal (uint8 ModuleNo, uint8 ChannelNo,)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – TIM Channel Number
Parameters (out):	None
Return Value:	Current Timeout value
Description:	Inline function to return the current Timeout value for Timeout detection for the TIM Channel If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.36 Gtm_EnableTbuChannel

Table 502 Specification of API Gtm_EnableTbuChannel

Service Name:	Gtm_EnableTbuChannel
Syntax:	inline void Gtm_EnableTbuChannel (uint8 TbuChannelNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	TbuChannelNo – TBU Channel to be enabled
Parameters (out):	None
Return Value:	None
Description:	Inline function to enable Tbu Channel given by the parameter TbuChannelNo If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	TbuChannelNo must be 0,1 or 2 only.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.37 Gtm_DisableTbuChannel

Table 503 Specification of API Gtm_DisableTbuChannel

Service Name:	Gtm_DisableTbuChannel
Syntax:	inline void Gtm_DisableTbuChannel (uint8 TbuChannelNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	TbuChannelNo – TBU Channel to be disabled
Parameters (out):	None
Return Value:	None
Description:	Inline function to disable Tbu Channel given by the parameter TbuChannelNo If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	TbuChannelNo must be 0,1 or 2 only.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.38 Gtm_EnableTomChannel

Table 504 Specification of API Gtm_EnableTomChannel

Service Name:	Gtm_EnableTomChannel
Syntax:	inline void Gtm_EnableTomChannel (uint8 ModuleNo, uint8 ChannelNo)
Source:	This is a vendor defined API.

Service Name:	Gtm_EnableTomChannel
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None
Return Value:	None
Description:	Inline function to enable Tom Channel given by ModuleNo and ChannelNo. This function determines the corresponding TGC's and enables the channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.39 Gtm_DisableTomChannel

Table 505 Specification of API Gtm_DisableTomChannel

Service Name:	Gtm_DisableTomChannel
Syntax:	inline void Gtm_DisableTomChannel (uint8 ModuleNo, uint8 ChannelNo)
Source:	This is a vendor defined API.
Service ID:	--

Service Name:	Gtm_DisableTomChannel
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None
Return Value:	None
Description:	Inline function to disable Tom Channel given by ModuleNo and ChannelNo. This function determines the corresponding TGC's and disables the channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.40 Gtm_EnableAtomChannel

Table 506 Specification of API Gtm_EnableAtomChannel

Service Name:	Gtm_EnableAtomChannel
Syntax:	inline void Gtm_EnableAtomChannel (uint8 ModuleNo, uint8 ChannelNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous

Service Name:	Gtm_EnableAtomChannel
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – ATOM Module Number ChannelNo – ATOM Channel Number
Parameters (out):	None
Return Value:	None
Description:	Inline function to enable Atom Channel given by ModuleNo and ChannelNo. This function determines the corresponding AGC's and enables the channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.41 Gtm_DisableAtomChannel

Table 507 Specification of API Gtm_DisableAtomChannel

Service Name:	Gtm_DisableAtomChannel
Syntax:	inline void Gtm_DisableAtomChannel (uint8 ModuleNo, uint8 ChannelNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant

Service Name:	Gtm_DisableAtomChannel
Parameters (in):	ModuleNo – ATOM Module Number ChannelNo – ATOM Channel Number
Parameters (out):	None
Return Value:	None
Description:	Inline function to disable Atom Channel given by ModuleNo and ChannelNo. This function determines the corresponding AGC's and disables the channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.42 Gtm_AddFifoEntry

Table 508 Specification of API Gtm_AddFifoEntry

Service Name:	Gtm_AddFifoEntry
Syntax:	inline void Gtm_AddFifoEntry (uint8 FifoChannel, uint32 Entry)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	FifoChannel – FIFO Channel Number

Service Name:	Gtm_AddFifoEntry
	Entry – Entry to be written
Parameters (out):	None
Return Value:	None
Description:	Inline function to add an entry given by “Entry” to the fifo channel given by “FifoChannel”. This information is added to the register GTM_AFD[i]_CHx_BUF_ACC where i is the AFD module number. If configuration parameter McuRunningInUser0Mode is ‘True’ then user0 mode is supported for GTM. If this switch is ‘false’ then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is ‘true’ then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is ‘false’). McuUserModeRuntimeApiEnable is ‘true’ then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is ‘true’). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	“Entry” is 29 bit information. If Fifo is full, then the information will be lost.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is ‘True’ and McuUserModeRuntimeApiEnable is ‘True’. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is ‘False’ and McuUserModeRuntimeApiEnable is ‘True’. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is ‘False’ and McuUserModeRuntimeApiEnable is ‘False’.

5.2.43 Gtm_ReadFifoEntry

Table 509 Specification of API Gtm_ReadFifoEntry

Service Name:	Gtm_ReadFifoEntry
Syntax:	inline uint32 Gtm_ReadFifoEntry (uint8 FifoChannel)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	FifoChannel – FIFO Channel Number

Service Name:	Gtm_ReadFifoEntry
Parameters (out):	None
Return Value:	Fifo topmost Entry
Description:	Inline function to reads topmost entry in the fifo channel given by "FifoChannel". This information is read from the register GTM_AFD[i]_CHx_BUF_ACC where i is AFD module number. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	"Entry" is 29 bit information. If Fifo is full, then the information will be lost.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.44 Gtm_EnableDpll

Table 510 Specification of API Gtm_EnableDpll

Service Name:	Gtm_EnableDpll
Syntax:	inline void Gtm_EnableDpll ()
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	None
Parameters (out):	None

Service Name:	Gtm_EnableDpll
Return Value:	None
Description:	Inline function to enable DPLL. This Api, sets DEN bit in GTM_DPLL_CTRL_1. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.45 Gtm_DisableDpll

Table 511 Specification of API Gtm_DisableDpll

Service Name:	Gtm_DisableDpll
Syntax:	inline void Gtm_DisableDpll ()
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	None
Parameters (out):	None
Return Value:	None
Description:	Inline function to disable DPLL. This Api, resets DEN bit in GTM_DPLL_CTRL_1.

Service Name:	Gtm_DisableDpll
	If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.46 Gtm_GetTbuTimerValue

Table 512 Specification of API Gtm_GetTbuTimerValue

Service Name:	Gtm_GetTbuTimerValue
Syntax:	inline uint32 Gtm_GetTbuTimerValue(uint8 TbuChannelNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	TbuChannelNo – TBU channel number (0-2)
Parameters (out):	None
Return Value:	BASE register content
Description:	Inline function to get the GTM_TBU_CHx_BASE register content of the Tbu Channel number given by TbuChannelNo

Service Name:	Gtm_GetTbuTimerValue
	If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.47 Gtm_GetTomOutputState

Table 513 Specification of API Gtm_GetTomOutputState

Service Name:	Gtm_GetTomOutputState
Syntax:	<pre>inline Gtm_OutputLevelType Gtm_GetTomOutputState(uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo - TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None
Return Value:	Current Output State
Description:	Inline function to get the Output state of the TOM Channel given by TOM<ModuleNo>.Channel<ChannelNo>.

Service Name:	Gtm_GetTomOutputState
	If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.48 Gtm_GetAtomOutputState

Table 514 Specification of API Gtm_GetAtomOutputState

Service Name:	Gtm_GetAtomOutputState
Syntax:	<pre>inline Gtm_OutputLevelType Gtm_GetAtomOutputState(uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo - ATOM Module Number ChannelNo – ATOM Channel Number
Parameters (out):	None
Return Value:	Current Output State

Service Name:	Gtm_GetAtomOutputState
Description:	Inline function to get the Output state of the ATOM Channel given by ATOM<ModuleNo>.Channel<ChannelNo>. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.49 Gtm_EnableTomCoherentUpdate

Table 515 Specification of API Gtm_EnableTomCoherentUpdate

Service Name:	Gtm_EnableTomCoherentUpdate
Syntax:	<pre>inline void Gtm_EnableTomCoherentUpdate (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo - TOM Module Number ChannelNo – TOM Channel Number
Parameters (out):	None

Service Name:	Gtm_EnableTomCoherentUpdate
Return Value:	None
Description:	Inline function to enable Coherent Update for the mentioned TOM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.50 Gtm_EnableAtomCoherentUpdate

Table 516 Specification of API Gtm_EnableAtomCoherentUpdate

Service Name:	Gtm_EnableAtomCoherentUpdate
Syntax:	<pre>inline void Gtm_EnableAtomCoherentUpdate (uint8 ModuleNo, uint8 ChannelNo)</pre>
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo - ATOM Module Number ChannelNo – ATOM Channel Number

Service Name:	Gtm_EnableAtomCoherentUpdate
Parameters (out):	None
Return Value:	None
Description:	Inline function to enable Coherent Update for the mentioned ATOM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.51 Gtm_ModifyOutputCtrl

Table 517 Specification of API Gtm_ModifyOutputCtrl

Service Name:	Gtm_ModifyOutputCtrl
Syntax:	inline void Gtm_ModifyOutputCtrl (uint8 TimerModule, uint8 ModuleNo, uint8 ChannelNo, boolean ToEnable)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	TimerModule - 0 – TOM 1 - ATOM

	ModuleNo – TOM/ATOM Module Number ChannelNo – TOM/ATOM Channel Number ToEnable – TRUE -> Enable TOM/ATOM Output FALSE -> Disable TOM/ATOM Output
Parameters (out):	None
Return Value:	None
Description:	This service enables/disables the TOM/ATOM Channel's output. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.52 Gtm_TrigActTimebase

Table 518 Specification of API Gtm_TrigActTimebase

Service Name:	Gtm_TrigActTimebase
Syntax:	Inline void Gtm_TrigActTimebase (uint8 TimerModule, uint8 ModuleNo, uint8 GroupNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant

Parameters (in):	TimerModule – 0 – TOM 1 - ATOM ModuleNo – TOM/ATOM Module Number GroupNo – TGC/AGC Number
Parameters (out):	None
Return Value:	None
Description:	This service sets Action Timebase trigger request for the corresponding TGC/AGC of the TOM/ATOM module. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	Group number should be 0 or 1 in case of TOM. It must be only 0 in case of ATOM. Module Number must be within desired limits.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.53 Gtm_SetTimFilterThreshold

Table 519 Specification of API Gtm_SetTimFilterThreshold

Service Name:	Gtm_SetTimFilterThreshold
Syntax:	Inline void Gtm_SetTimFilterThreshold (uint8 ModuleNo,uint8 ChannelNo, uint32 FltRisingEdge, uint32 FltFallingEdge)
Source:	This is a vendor defined API.
Service ID:	--

Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – TIM Module Number ChannelNo – Channel Number FltRisingEdge – Filter Threshold for Rising Edge FltFallingEdge – Filter Threshold for Falling Edge
Parameters (out):	None
Return Value:	None
Description:	This service sets the Filter Threshold value for rising and falling edge for the TIM Channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.54 Gtm_FlushFifo

Table 520 Specification of API Gtm_FlushFifo

Service Name:	Gtm_FlushFifo
Syntax:	Inline void Gtm_FlushFifo (uint8 FifoModuleNo, uint8 FifoChannelNo)
Source:	This is a vendor defined API.
Service ID:	--

Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	FifoModuleNo – FIFO module Number FifoChannelNo – FIFO channel Number
Parameters (out):	None
Return Value:	None
Description:	This service flushes the FIFO channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	This service shall be called by user after Gtm_Init and before writing to FIFO RAM.
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	This API will not wait until the Flush operation is completed. This will only trigger a FLUSH process.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.55 Gtm_CtrlFifoStream

Table 521 Specification of API Gtm_CtrlFifoStream

Service Name:	Gtm_CtrlFifoStream
Syntax:	Inline void Gtm_CtrlFifoStream (uint8 FifoModuleNo, uint8 FifoChannelNo, boolean ToStart)
Source:	This is a vendor defined API.
Service ID:	--

Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	FifoModuleNo – FIFO module Number FifoChannelNo – FIFO channel Number ToStart – TRUE – Start FIFO Stream FALSE – Stop FIFO Stream
Parameters (out):	None
Return Value:	None
Description:	This service starts/stops a FIFO stream. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.56 Gtm_ClearMcsRam

Table 522 Specification of API Gtm_ClearMcsRam

Service Name:	Gtm_ClearMcsRam
Syntax:	Inline void Gtm_ClearMcsRam (uint8 McsModule)
Source:	This is a vendor defined API.
Service ID:	--

Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModule – MCS module Number
Parameters (out):	None
Return Value:	None
Description:	This service resets the RAM corresponding to the MCS module number. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	This service shall be called by user after Gtm_Init and before writing to MCS RAM.
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	This API only issues a RAM Reset but does not wait for its completion.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.57 Gtm_CtrlMcsTrigger

Table 523 Specification of API Gtm_CtrlMcsTrigger

Service Name:	Gtm_CtrlMcsTrigger
Syntax:	Inline void Gtm_CtrlMcsTrigger (uint8 McsModule, uint8 TriggerNo, boolean TriggerCtrl)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant

Parameters (in):	McsModule – MCS module number TriggerNo – Trigger Number TriggerCtrl – FALSE – Clear Trigger TRUE - Set Trigger
Parameters (out):	None
Return Value:	None
Description:	This service clears/sets MCS trigger. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.58 Gtm_TrigDpIISwRst

Table 524 Specification of API Gtm_TrigDpIISwRst

Service Name:	Gtm_TrigDpIISwRst
Syntax:	Inline void Gtm_TrigDpIISwRst ()
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant

Parameters (in):	None
Parameters (out):	None
Return Value:	None
Description:	This service resets all register and internal states of the DPLL. If configuration parameter <code>McuRunningInUser0Mode</code> is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on <code>McuUserModeRuntimeApiEnable</code>. <code>McuUserModeRuntimeApiEnable</code> is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided <code>McuRunningInUser0Mode</code> is 'false'). <code>McuUserModeRuntimeApiEnable</code> is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided <code>McuRunningInUser0Mode</code> is 'true'). (Refer Mcu user manual for <code>McuUserModeRuntimeApiEnable</code> and <code>McuUserModeRuntimeApiEnable</code> configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'True' and <code>McuUserModeRuntimeApiEnable</code> is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'False' and <code>McuUserModeRuntimeApiEnable</code> is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'False' and <code>McuUserModeRuntimeApiEnable</code> is 'False'.

5.2.59 Gtm_GetDpllStatus

Table 525 Specification of API Gtm_GetDpllStatus

Service Name:	Gtm_GetDpllStatus
Syntax:	Inline Gtm_DpllStatType Gtm_GetDpllStatus()
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	None
Parameters (out):	None
Return Value:	Status :-

	DpllStatus – Provides the DPLL status IntStatus – Provides the DPLL interrupt Status
Description:	This service provides DPLL module and DPLL interrupt status If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.60 Gtm_SetDpllRefMode

Table 526 Specification of API Gtm_SetDpllRefMode

Service Name:	Gtm_SetDpllRefMode
Syntax:	Inline void Gtm_SetDpllRefMode(uint8 RefMode)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	RefMode – 0 -> Normal Mode 1 -> Emergency Mode
Parameters (out):	None
Return Value:	None

Description:	This service sets the Reference mode of the DPLL module If configuration parameter <code>McuRunningInUser0Mode</code> is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on <code>McuUserModeRuntimeApiEnable</code>. <code>McuUserModeRuntimeApiEnable</code> is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided <code>McuRunningInUser0Mode</code> is 'false'). <code>McuUserModeRuntimeApiEnable</code> is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided <code>McuRunningInUser0Mode</code> is 'true'). (Refer Mcu user manual for <code>McuUserModeRuntimeApiEnable</code> and <code>McuUserModeRuntimeApiEnable</code> configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'True' and <code>McuUserModeRuntimeApiEnable</code> is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'False' and <code>McuUserModeRuntimeApiEnable</code> is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'False' and <code>McuUserModeRuntimeApiEnable</code> is 'False'.

5.2.61 Gtm_CtrlSpeModule

Table 527 Specification of API Gtm_CtrlSpeModule

Service Name:	Gtm_CtrlSpeModule
Syntax:	Inline void Gtm_CtrlSpeModule(uint8 ModuleNo, uint8 ToEnable)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – SPE Module number ToEnable – 1 -> Enable SPE Module 0 -> Disable SPE Module
Parameters (out):	None
Return Value:	None
Description:	This service enables/disables SPE module.

	If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.62 Gtm_GetSpelInputPat

Table 528 Specification of API Gtm_GetSpelInputPat

Service Name:	Gtm_GetSpelInputPat
Syntax:	Inline uint16 Gtm_GetSpelInputPat(uint8 ModuleNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	ModuleNo – SPE Module number
Parameters (out):	None
Return Value:	uint16 InputPat -> Bit 0 to 2 : AIP (Actual Input Pattern) Bit 3 -> 0 Bit 4 to 6 : PIP (Previous Input Pattern) Bit 7 -> 0

	Bit 8 to 10 : NIP (New Input Pattern)
Description:	This service provides the Input Pattern history of SPE. If configuration parameter <code>McuRunningInUser0Mode</code> is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on <code>McuUserModeRuntimeApiEnable</code>. <code>McuUserModeRuntimeApiEnable</code> is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided <code>McuRunningInUser0Mode</code> is 'false'). <code>McuUserModeRuntimeApiEnable</code> is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided <code>McuRunningInUser0Mode</code> is 'true'). (Refer Mcu user manual for <code>McuUserModeRuntimeApiEnable</code> and <code>McuUserModeRuntimeApiEnable</code> configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'True' and <code>McuUserModeRuntimeApiEnable</code> is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'False' and <code>McuUserModeRuntimeApiEnable</code> is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'False' and <code>McuUserModeRuntimeApiEnable</code> is 'False'.

5.2.63 Gtm_CtrlMapInput

Table 529 Specification of API Gtm_CtrlMapInput

Service Name:	Gtm_CtrlMapInput
Syntax:	Inline void Gtm_CtrlMapInput(uint32 MapInput)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	MapInput – Same format as Gtm_Map_Ctrl register
Parameters (out):	None
Return Value:	None
Description:	This service provides the Input selection for MAP module.

	If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.64 Gtm_SelectBrcDest

Table 530 Specification of API Gtm_SelectBrcDest

Service Name:	Gtm_SelectBrcDest
Syntax:	Inline void Gtm_SelectBrcDest(uint8 SourceNo, uint32 SelDest)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	SourceNo – BRC source number SelDest – Each bit field represents the Destination to be selected for the BRC Source. If all destination need to be cleared then bit 22 should be set to activate the Trash bin functionality.
Parameters (out):	None
Return Value:	None
Description:	This service provides Trash bin functionality and Destination select for BRC Source. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode

	is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.65 Gtm_CtrlAruFrame

Table 531 Specification of API Gtm_CtrlAruFrame

Service Name:	Gtm_CtrlAruFrame
Syntax:	Gtm_AruReqStatus Gtm_CtrlAruFrame(uint16 AruAddress, uint8 ReqType, uint32* AruFrame, uint8 ReadReq)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	AruAddress – ARU Address frame to be acted upon ReqType – 0 – Request for Writing ARU frame 1 – Request for Reading ARU frame AruFrame – Pointer to the location where - In case of Write request, 2 uint32 ARU frame content will be fetched from. - In case of read request, 2 uint32 ARU frame will be written to. Location pointed by AruFrame address will contain the Upper word of ARU frame. Location pointer by AruFrame + 4 will contain the Lower word of ARU

	frame. ReadReq – This is applicable only if ReqType is 1 0 – Only Issue a Read Request. ARU frame shall be reported only if the read is completed. 1 – Do not issue a Read Request. Only ARU frame content should be reported.
Parameters (out):	None
Return Value:	Gtm_CtrlAruFrame – GTM_INCORRECT_REQUEST – Incorrect parameter passed to the function. When ReqType = 0, GTM_ARU_WRITE_SUCCESSFUL – Write request issued successfully. GTM_ARU_WRITE_FAILED – Write could not be issued because of a previously pending request When ReqType = 1, ReadReq = 0, GTM_ARU_RD_ISSUED_AND_COMPLETED – Read request issued successfully. ARU Frame also read. Content pointed by AruFrame will be valid GTM_ARU_RD_ISSUED_BUT_NOT_CMPLT - Read request issued successfully but ARU frame reading is not completed. GTM_ARU_READ_NOT_ISSUED - Read request not issued successfully because of previously pending request. When ReqType = 1, ReadReq = 1, GTM_ARU_READ_FAILED – ARU Frame not read as ARU reading is not complete. Content pointed by AruFrame will be invalid GTM_ARU_READ_SUCCESSFUL - ARU Frame read. Content pointed by AruFrame will be valid
Description:	This service provides read and write functionality of ARU frames If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	

Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.66 Gtm_GetFifoStatus

Table 532 Specification of API Gtm_GetFifoStatus

Service Name:	Gtm_GetFifoStatus
Syntax:	Inline uint8 Gtm_GetFifoStatus (uint8 FifoModuleNo, uint8 FifoChannelNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	uint8 FifoModuleNo : Fifo module number. uint8 FifoChannelNo : Fifo channel number.
Parameters (out):	None
Return Value:	None
Description:	This service provides the status of the referred FIFO channel number. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None

I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.
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5.2.67 Gtm_GetMcsRamRstStat

Table 533 Specification of API Gtm_GetMcsRamRstStat

Service Name:	Gtm_GetMcsRamRstStat
Syntax:	Inline boolean Gtm_GetMcsRamRstStat (uint8 McsModule)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModule – MCS module number
Parameters (out):	None
Return Value:	Boolean status – TRUE – MCS RAM reset in progress FALSE – no MCS RAM reset active
Description:	This service provides the status of MCS Ram Reset. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and

	McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'..
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5.2.68 Gtm_WriteMcsRam

Table 534 Specification of API Gtm_WriteMcsRam

Service Name:	Gtm_WriteMcsRam
Syntax:	Inline void Gtm_WriteMcsRam (uint8* McsStartAddress, const uint8* const McsCodePtr, uint32 CodeSize)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsStartAddress – Start Address from where the code has to be written to. McsCodePtr – Pointer to the microcode that has to be written into MCS Ram CodeSize – Size of MCS microcode in bytes.
Parameters (out):	None
Return Value:	None
Description:	This service writes microcode into MCS RAM. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'.

	This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.
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5.2.69 Gtm_CtrlMcsChannel

Table 535 Specification of API Gtm_CtrlMcsChannel

Service Name:	Gtm_CtrlMcsChannel
Syntax:	Inline boolean Gtm_CtrlMcsChannel (uint8 McsModNo, uint8 McsChannel, uint8 Activity)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModNo – MCS module number McsChannel – MCS Channel Number Activity – 1 – Start MCS channel 0 – Stop MCS Channel
Parameters (out):	None
Return Value:	TRUE - Function executed successfully FALSE - Function could not be executed since RAM reset is ongoing.
Description:	This service starts or stops an MCS channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided -

	configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.
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5.2.70 Gtm_InitMcsProgCntr

Table 536 Specification of API Gtm_InitMcsProgCntr

Service Name:	Gtm_InitMcsProgCntr
Syntax:	Inline boolean Gtm_InitMcsProgCntr (uint8 McsModNo, uint8 McsChannel, uint32 PcValue)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModNo – MCS module number McsChannel – MCS Channel Number PcValue – Program Counter Value to be initialized
Parameters (out):	None
Return Value:	TRUE: PC written Successfully FALSE: PC not written since Channel is enabled
Description:	This service initializes the program counter of MCS channel with the value given in PcValue. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None

I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.
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5.2.71 Gtm_TrigMcsIntrpt

Table 537 Specification of API Gtm_TrigMcsIntrpt

Service Name:	Gtm_TrigMcsIntrpt
Syntax:	Inline void Gtm_TrigMcsIntrpt (uint8 McsModNo, uint8 McsChannel, uint8 IntrptToTrigger)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModNo – MCS module number McsChannel – MCS Channel Number IntrptToTrigger – 0 – MCS Channel Interrupt 1 – Stack Counter Overflow/Underflow Interrupt 2 – Memory Access out of range interrupt
Parameters (out):	None
Return Value:	None
Description:	This service force triggers interrupt of type specified by IntrptToTrigger. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	Force Interrupt should be enabled globally.
Algorithm Description:	

Design Decisions:	a) Force Interrupt should be enabled globally. b) Interrupt is expected to have been enabled prior.
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.72 Gtm_CtrlMcsIntrpt

Table 538 Specification of API Gtm_CtrlMcsIntrpt

Service Name:	Gtm_CtrlMcsIntrpt
Syntax:	Inline void Gtm_CtrlMcsIntrpt (uint8 McsModNo, uint8 McsChannel, uint8 IntrptToCtrl, uint8 CtrlType)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModNo – MCS module number McsChannel – MCS Channel Number IntrptToCtrl – 0 – MCS Channel Interrupt 1 – Stack Counter Overflow/Underflow Interrupt 2 – Memory Access out of range interrupt CtrlType – 0 – Enable Interrupt 1 – Disable Interrupt
Parameters (out):	None
Return Value:	None
Description:	This service enables or disables interrupt of type specified by IntrptToCtrl. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--

Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.73 Gtm_TrigMcsReset

Table 539 Specification of API Gtm_TrigMcsReset

Service Name:	Gtm_TrigMcsReset
Syntax:	Inline void Gtm_TrigMcsReset (uint8 McsModNo, uint32 ResetVal)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModNo – MCS module number ResetVal – Resetting Channel, Cancelling Aru transfer and WURM instruction in the structure of MCS_RST register.
Parameters (out):	None
Return Value:	None
Description:	This service writes the content of ResetVal into GTM_MCSi_RST register where i depends on MCS module number. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--

DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.74 Gtm_GetMcsErrTrigStatus

Table 540 Specification of API Gtm_GetMcsErrTrigStatus

Service Name:	Gtm_GetMcsErrTrigStatus
Syntax:	Inline uint32 Gtm_GetMcsErrTrigStatus (uint8 McsModNo)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModNo – MCS module number
Parameters (out):	None
Return Value:	ErrTrigStatus – Lower 16 bits holds the Trigger Status of all the channels in the MCS modules. Upper 16 bits holds the error information of all the channels.
Description:	This service returns the error and trigger status of all the channels of the MCS module. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None

Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.

5.2.75 Gtm_ResetMcsErrStatus

Table 541 Specification of API Gtm_ResetMcsErrStatus

Service Name:	Gtm_ResetMcsErrStatus
Syntax:	Inline void Gtm_ResetMcsErrStatus (uint8 McsModNo, uint8 McsChannel)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	McsModNo – MCS module number McsChannel – MCS Channel Number
Parameters (out):	None
Return Value:	None
Description:	This service resets the error information of the requested MCS channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run in user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run in user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run in user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).
Caveats:	--
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	None

I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'True' and <code>McuUserModeRuntimeApiEnable</code> is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'False' and <code>McuUserModeRuntimeApiEnable</code> is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'False' and <code>McuUserModeRuntimeApiEnable</code> is 'False'.
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5.2.76 Gtm_InitDpllRam

Table 542 Specification of API Gtm_InitDpllRam

Service Name:	Gtm_InitDpllRam
Syntax:	Inline void Gtm_InitDpllRam (void)
Source:	This is a vendor defined API.
Service ID:	--
Sync / Async:	Synchronous
Re-entrancy:	Reentrant
Parameters (in):	None
Parameters (out):	None
Return Value:	None
Description:	This service initializes the DPLL RAM. This API does not wait till the completion of RAM Initialization.
Caveats:	This service shall be called by user after Gtm_Init and before writing to DPLL RAM.
Configuration:	--
DET:	None
DEM:	None
Algorithm Description:	
Design Decisions:	DPLL is expected to be in the disabled state before this function is called. If configuration parameter <code>McuRunningInUser0Mode</code> is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on <code>McuUserModeRuntimeApiEnable</code>. <code>McuUserModeRuntimeApiEnable</code> is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided <code>McuRunningInUser0Mode</code> is 'false'). <code>McuUserModeRuntimeApiEnable</code> is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided <code>McuRunningInUser0Mode</code> is 'true'). (Refer Mcu user manual for <code>McuUserModeRuntimeApiEnable</code> and <code>McuUserModeRuntimeApiEnable</code> configuration parameters).
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter <code>McuRunningInUser0Mode</code> is 'True' and

	McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'False'.
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5.2.77 Gtm_GetTimerRegisters

Table 543 Specification of API Gtm_GetTimerRegisters

Service Name:	Gtm_GetTimerRegisters		
Syntax:	uint32 Gtm_GetTimerRegisters(uint8 ModType, uint8 ModuleNo, uint8 GroupNo, Gtm_TimerRegistersType RegName)		
Source:	This is a vendor defined API.		
Service ID:	--		
Sync / Async:	Synchronous		
Re-entrancy:	Reentrant		
Parameters (in):	ModuleNo	Valid TOM/ATOM module number	TOM/ATOM Module Number
	GroupNo	0 - 1	TGC Number in case of TOM. Ignored in case of ATOM.
	ModType	GTM_TOM, GTM_ATOM	Choose the Timer type
	RegName	Refer Gtm_TimerRegistersType	Register index
Parameters (out):	None		
Return Value:	RegVal	Value of the register	
Description:	This service returns the register values of the requested register. If configuration parameter McuRunningInUser0Mode is ‘True’ then user0 mode is supported for GTM. If this switch is ‘false’ then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is ‘true’ then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is ‘false’). McuUserModeRuntimeApiEnable is ‘true’ then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is ‘true’). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).		
Caveats:	--		
Configuration:	--		
DET:	None		
DEM:	None		

Algorithm Description:	
Design Decisions:	None
I/O Mode:	This API can be invoked when the CPU is in either User-1 Mode or Supervisor Mode.

5.2.78 Gtm_GetTimFilterThreshold

Table 544 Specification of API Gtm_GetTimFilterThreshold

Service Name:	Gtm_GetTimFilterThreshold		
Syntax:	inline Gtm_TimFilterType Gtm_GetTimFilterThreshold (uint8 ModuleNo, uint8 ChannelNo)		
Source:	This is a vendor defined API.		
Service ID:	--		
Sync / Async:	Synchronous		
Re-entrancy:	Reentrant		
Parameters (in):	ModuleNo	Valid TIM Module number	TIM Module Number
	ChannelNo	Valid TIM channel number	TIM channel number.
Parameters (out):	None		
Return Value:	FltValue	Filter value is returned in the elements FltRisingEdge and FltFallingEdge.	
Description:	This service returns the filter values of the requested TIM channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).		
Caveats:	--		
Configuration:	--		
DET:	None		
DEM:	None		
Algorithm Description:			
Design Decisions:	None		
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'.		

This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter `McuRunningInUser0Mode` is 'False' and `McuUserModeRuntimeApiEnable` is 'False'.

5.2.79 Gtm_GetTbuChannelStatus

Table 545 Specification of API Gtm_GetTbuChannelStatus

Service Name:	Gtm_GetTbuChannelStatus		
Syntax:	inline boolean Gtm_GetTbuChannelStatus(uint8 TbuChannelNo)		
Source:	This is a vendor defined API.		
Service ID:	--		
Sync / Async:	Synchronous		
Re-entrancy:	Reentrant		
Parameters (in):	TbuChannelNo	Valid TBU channel Number	TBU channel's status that need to be returned
Parameters (out):	None		
Return Value:	ChannelStatus	Status of TBU Channel - if enabled or not	
Description:	Inline function to get the enabled status of the TBU channel. If configuration parameter McuRunningInUser0Mode is 'True' then user0 mode is supported for GTM. If this switch is 'false' then the Api will run is user1 or supervisory mode depending on McuUserModeRuntimeApiEnable. McuUserModeRuntimeApiEnable is 'true' then the API will run is user1 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'false'). McuUserModeRuntimeApiEnable is 'true' then the API will run is user0 mode, else it will run in supervisory mode (provided McuRunningInUser0Mode is 'true'). (Refer Mcu user manual for McuUserModeRuntimeApiEnable and McuUserModeRuntimeApiEnable configuration parameters).		
Caveats:	--		
Configuration:	--		
DET:	None		
DEM:	None		
Algorithm Description:	Checks the bits in GTM_TBU_CHEN corresponding to TbuChannelNo to find if it is 00b or 11b. If 00b, FALSE is returned, else TRUE is returned.		
Design Decisions:	None		
I/O Mode:	This API can be invoked when the CPU is in User-0 Mode provided - configuration parameter McuRunningInUser0Mode is 'True' and McuUserModeRuntimeApiEnable is 'True'. This API can be invoked when the CPU is in User-1 Mode provided - configuration parameter McuRunningInUser0Mode is 'False' and McuUserModeRuntimeApiEnable is 'True'.		

This API can be invoked when the CPU is in Supervisor Mode provided - configuration parameter `McuRunningInUser0Mode` is 'False' and `McuUserModeRuntimeApiEnable` is 'False'.

5.3 Interrupt Handling

GTM Driver is responsible for handling GTM interrupt requests and call the necessary call back functions. On a call of ISR, the configured notification function gets called. If the channel is configured for ICU, then `Icu_17_Isr` Api is called, if the channel is configured for PWM, then `Pwm_17_Isr` is called and if the channel is configured for Gpt then `Gpt_Isr` is called. In case of TOM, ATOM or TIM sub-modules, If the channel is not configured for any of these drivers then the notification field in the channel configured has to be provided the Callback function name. Please note, the callback functions/notifications configured should be unique for different channels. For other Sub-Modules, the interrupt frames have to be updated manually to mention the function to be called on interrupt request.

5.3.1 Gtm_IsrTomModule

Table 546 Specification of API Gtm_IsrTomModule

Service Name:	Gtm_IsrTomModule	
Syntax:	<pre>void Gtm_IsrTomModule (uint8 ModuleNo, uint8 ChannelNo)</pre>	
Service ID:	90	
Sync / Async:	Synchronous	
Re-entrancy:	Reentrant	
	ModuleNo	TOM/ATOM module number of the Pwm channel hardware
	ChannelNo	TOM/ATOM channel number of the Pwm channel hardware
Parameters (out):	None	
Return Value:	None	
Description:	One Interrupt node is allotted for every 2 TOM Channels. And each TOM channel can have 2 Interrupt sources – CCU0 and CCU1. GTM complex driver notifications are of link time and the channels for which GPT//PWM is used it will call a function with single parameter i.e. corresponding modules channel number.	
Caveats:	None	
Configuration:	None	
DET:	None	
DEM:	None	
SAFE MCAL ERROR	GTM_E_SPURIOUS_INTRPT – Spurious/Erroneous interrupt triggered from the Interrupt Router (IR).	
Implementation Comments:	None	

5.3.2 Gtm_IsrTimModule

Table 547 Specification of API Gtm_IsrTimModule

Service Name:	Gtm_IsrTimModule	
Syntax:	<pre>void Gtm_IsrTimModule (uint8 ModuleNo, uint8 ChannelNo)</pre>	
Service ID:	91	
Sync / Async:	Synchronous	
Re-entrancy:	Reentrant	
	ModuleNo	TOM/ATOM module number of the Pwm channel hardware
	ChannelNo	TOM/ATOM channel number of the Pwm channel hardware
Parameters (out):	None	
Return Value:	None	
Description:	Each TIM channel can have 6 Interrupt sources – GTM complex driver notifications are of link time and the channels for which ICU is used it will call a function with single parameter i.e. corresponding modules channel number.	
Caveats:	None	
Configuration:	None	
DET:	None	
DEM:	None	
SAFE MCAL ERROR	GTM_E_SPURIOUS_INTRPT – Spurious/Erroneous interrupt triggered from the Interrupt Router (IR).	
Implementation Comments:	None	

5.3.3 Gtm_IsrAtomModule

Table 548 Specification of API Gtm_IsrAtomModule

Service Name:	Gtm_IsrAtomModule	
Syntax:	<pre>void Gtm_IsrAtomModule (uint8 ModuleNo, uint8 ChannelNo)</pre>	
Service ID:	92	
Sync / Async:	Synchronous	
Re-entrancy:	Reentrant	
	ModuleNo	TOM/ATOM module number of the Pwm channel hardware
	ChannelNo	TOM/ATOM channel number of the Pwm channel hardware

Parameters (out):	None
Return Value:	None
Description:	One Interrupt node is allotted for every 2 ATOM Channels. And each ATOM channel can have 2 Interrupt sources – CCU0 and CCU1. GTM complex driver notifications are of link time and the channels for which GPT//PWM is used it will call a function with single parameter i.e. corresponding modules channel number. Separate notification configurability is added for CCU0 and CCU1. This feature is optional and can be disabled by keeping AtomGlobalCcuSel false. Note: AtomGlobalCcuSel to be made true only if CCU0 and CCU1 require separate configurability for the corresponding channel.
Caveats:	None
Configuration:	None
DET:	None
DEM:	None
SAFE MCAL ERROR	GTM_E_SPURIOUS_INTRPT – Spurious/Erroneous interrupt triggered from the Interrupt Router (IR).
Implementation Comments:	None

6 Data Description

The below table list all the global variables used in the GTM driver

Table 549 Global Data Description

Parameter	Type/Range	Description
Gtm_kConfigPtr	const static Gtm_ConfigType*	To store pointer to the GTM configuration structure.

7 Example Usage

7.1 Configuration of Driver

GTM driver configuration (CLOCK and MUX programming)

1. MCU Driver: Configure the system clock and GTM prescaler.
2. GTM Driver: Configure GTM Module Clock and CMU settings. Configure TBU's with corresponding Clock signal. Get the list of resources required in ICU / PWM / GPT Driver. Same resource configured for different drivers will be identified by the configuration and an assert will be raised. For specific GTM configuration the user should be well acquainted with GTM resources and can make use of the basic Interfaces that are provided with Driver
3. PORT Driver: For all the port pins that were used in GTM Driver as input / output pins, configure the same in PORT Driver. For output pins set the direction to output and port function as ALT1. This can be easily identified in GTM User manual/TS. For input pins set the port direction as input.
4. Interrupt priority initialization: The priority of interrupts should be configured in IRQ Driver and appropriate priorities should be given.

7.1.1 Sample Configuration of Driver

7.1.1.1 PWM Generation using TOM Sub-Module.

This example configuration generates PWM signal from TOM0 Channel 0. This will be visible in Port0 Pin10.

MCU Driver:

Configure the MCU Clock dividers and the GTM Prescaler.

GTM Driver:

Let the Initial value of all the GTM Configuration Parameters be default value. Apart from that, perform the following

Table 550 GTM Driver configuration

Container Name	Parameter Name	Value
Cmu /CmuFixedClock	CmuEnableAllFixedClocks	True
TOM_0 / TomChannel_0/TomChannelUsage	TomChannelUsage	USED_BY_GTM_DRIVER
TOM_0 /TomChannel_0/ TomChannelOutput	TomChannelOutputControl	ENABLE_OUTPUT_AT_START
TOM_0 /TomChannel_0/ TomChannelOutput	TomChannelPortPinSelect	TOUT19_SELA_PORT0_PIN10
TOM_0 /TomChannel_0/ TomChannelTimerParameters	TomChannelPeriodCompareValCm0	1000
TOM_0 /TomChannel_0/ TomChannelTimerParameters	TomChanelDutyCycleCompareValCm1	500

Port Driver:

Table 551 Port Driver Configuration

Container Name	Parameter Name	Value
PortConfigSet/*[]/PortContainer_0/PortPin_10	PortPinDirection	PORT_PIN_OUT
PortConfigSet/*[]/PortContainer_0/PortPin_10	PortPinInitialMode	PORT_PIN_MODE_ALT1

8 Limitations and Assumptions

8.1 Assumptions and Deviations from Software Specification

The GTM complex driver notifications are of link time. However when a GTM channel is configured for GPT/ICU/PWM, the call to this function is postbuild. This implies that if GTM complex driver notifications are configured then in case of selectable configuration, all the sets should use the same GTM module and channel number for the same module. The first configuration set should be used as the super set configuration (in case of difference in channel across configuration). As an example, in the first configuration set if there are 5 channels configured for TOM module and 3 channels in the second TOM module then same channels should be present in the second configuration set also. If 3 TOM channels are configured in

the first config set and 4 TOM channels are configured in second configset then notifications for 4th TOM channel will not be available.

8.2 Limitations

The Container names within GtmConfiguration container cannot be changed by the user. This limitation is imposed because of the dependencies of other modules with GTM.

The MCFG configurations generated in GTM is not updated in the GTM_MCFG_CTRL register in the cases when McfgMemConfigForMcs_3 is not set to BORROW. The values are not changed by the driver in these cases.

8.3 Hardware features not supported

GTM AEI bridge configuration is not supported.

GTM DTM featured in TC23x device is not supported.

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